

Geothermal Wireline Logging Services

NESR

NESR Environment & Decarbonization Applications

Presented by Kadek Agus Dedyantara

Jakarta Drilling Society

Prepared for JDS



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EDUCATION

Bachelor at Petroleum Engineer
UPN Veteran Yogyakarta (2003 – 2008)

WORK EXPERIENCE

2018 – PRESENT	**NESR INDONESIA**	**SERVICE LEADER**
2010 – 2018	**EXSPAN PETROGAS INTRANUSA**	**FIELD WIRELINE ENGINEER**
2008 – 2010	**PATRA NUSA DATA**	**PETROLEUM ENGINEER**

INDONESIA OPERATION



No.	Location	Client	Project
1	Sorik - Sumut	KS ORKA	Geothermal - Wireline Services
2	Langgak - Riau	SPR Langgak	O&G - Wireline, Cementing, Fracturing
3	Jambi	Pertamina Z1	O&G - Cementing
4	Palembang	Tropik Energi	O&G - Wireline, Slickline, Mini CTU, Stimulation
5	Bengkulu , lampung	PGE	Geothermal - Wireline Services, Capillary String
6	Jawa Barat	PGE, Geodipa	Geothermal - Wireline Services, Capillary String, Hydrofracturing
7	Cirebon	Pertamina Z7	O&G - CTU, Stimulation, CO2 Huff and Puff
8	Jawa Timur	Pertamina Z11	O&G - CTU, Stimulation, Cementing
9	Kalimantan	Pertamina Z9, PHSS	O&G - Capillary String, Mini CTU, Stimulation, Cementing
10	Flores	KS Orka, PLN	Geothermal - Wireline Services, Stimulation, Well Test , Mini CTU



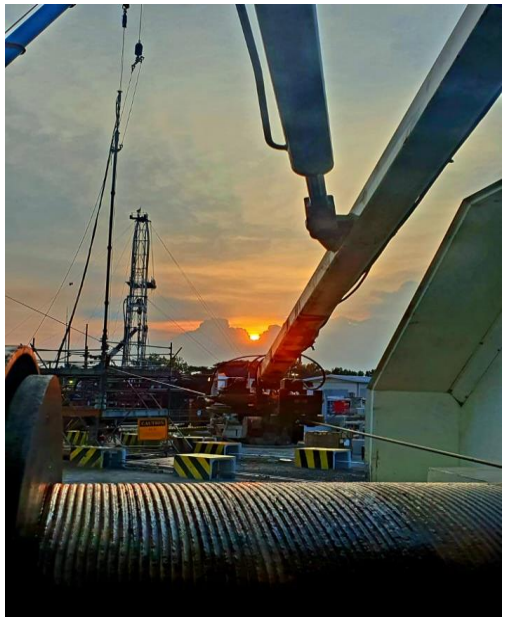




Geothermal Operation











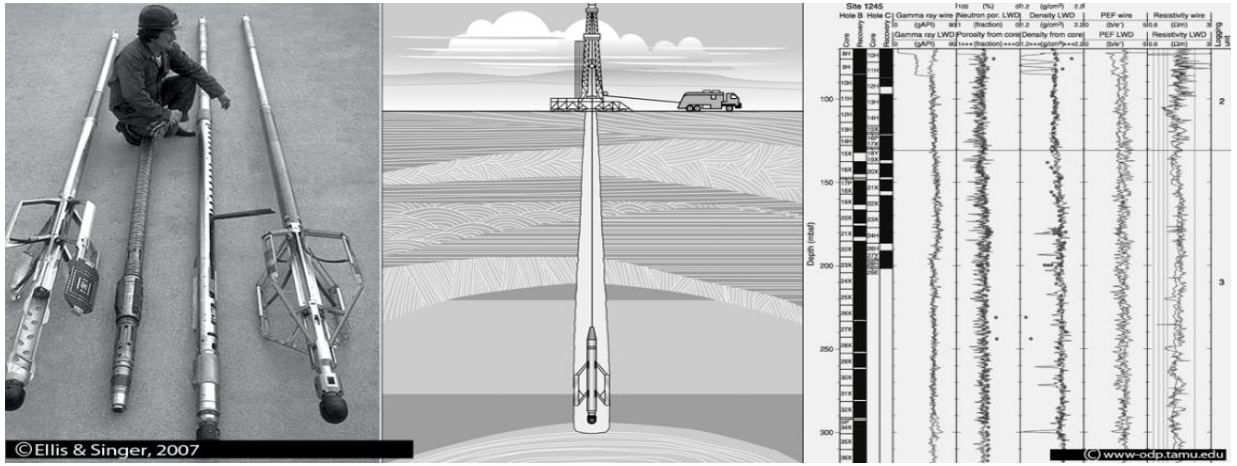




Outline

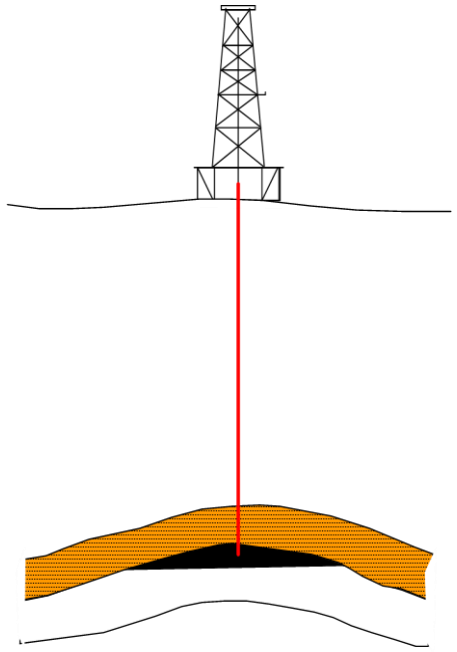
1. Introduction
2. Well Completions
3. Wireline Logging
4. Type of Wireline Services
5. Well Integrity



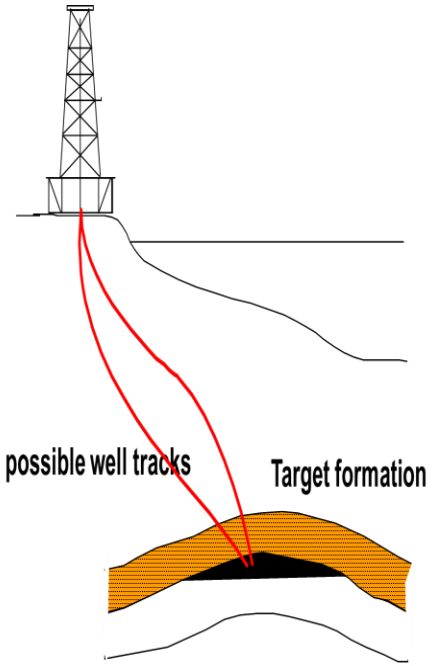


Well Categories

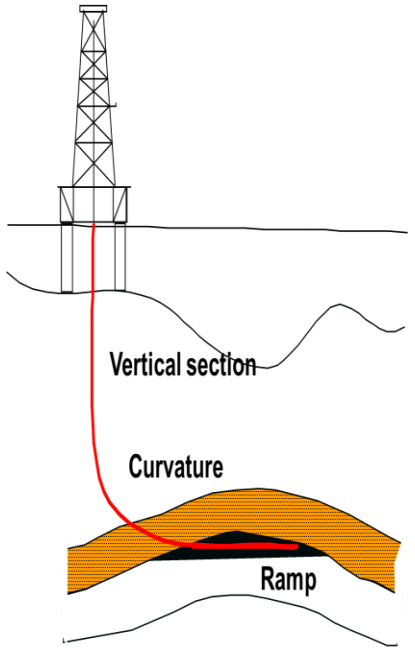
Vertical Wells



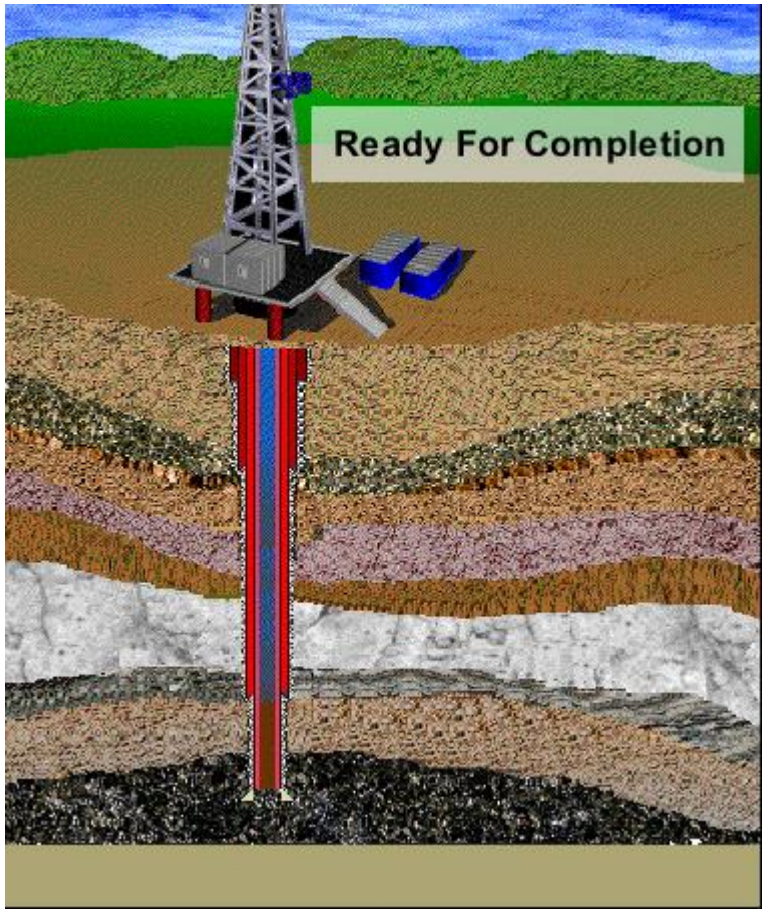
Deviated well



Horizontal well

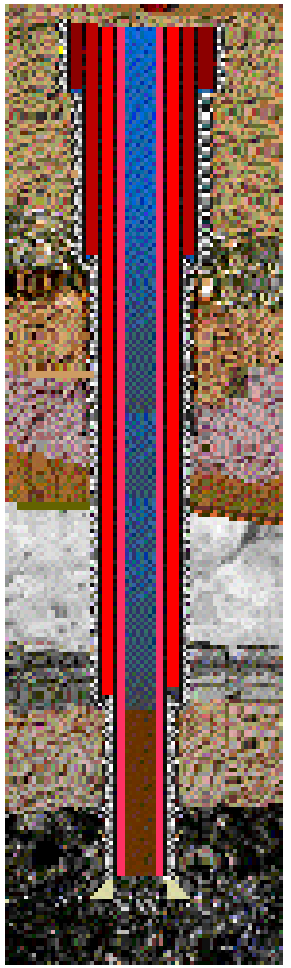


Well Completions

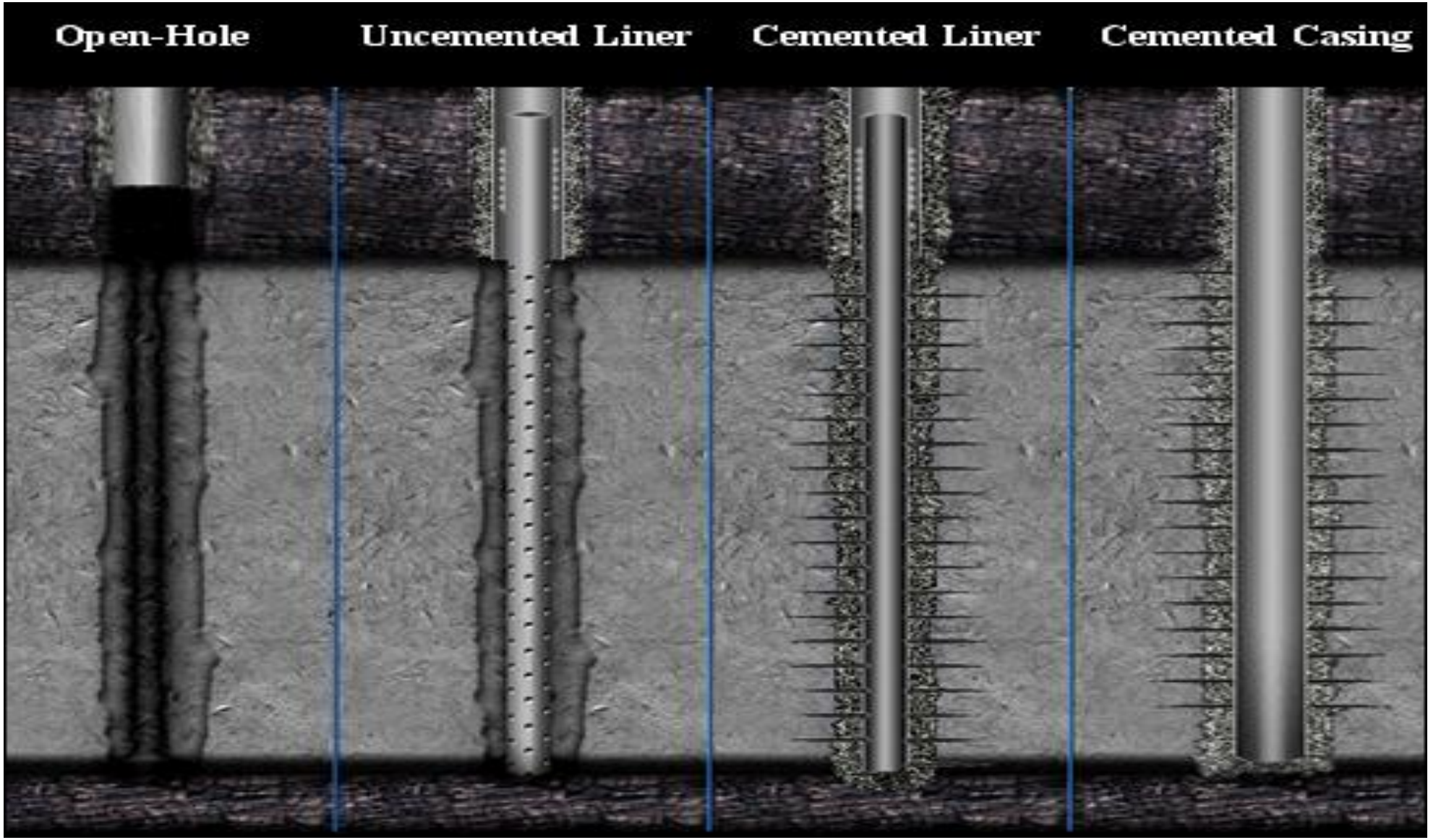


The hole is drilled in stages and at the end of every stage, casing is put into the hole and cemented into place.

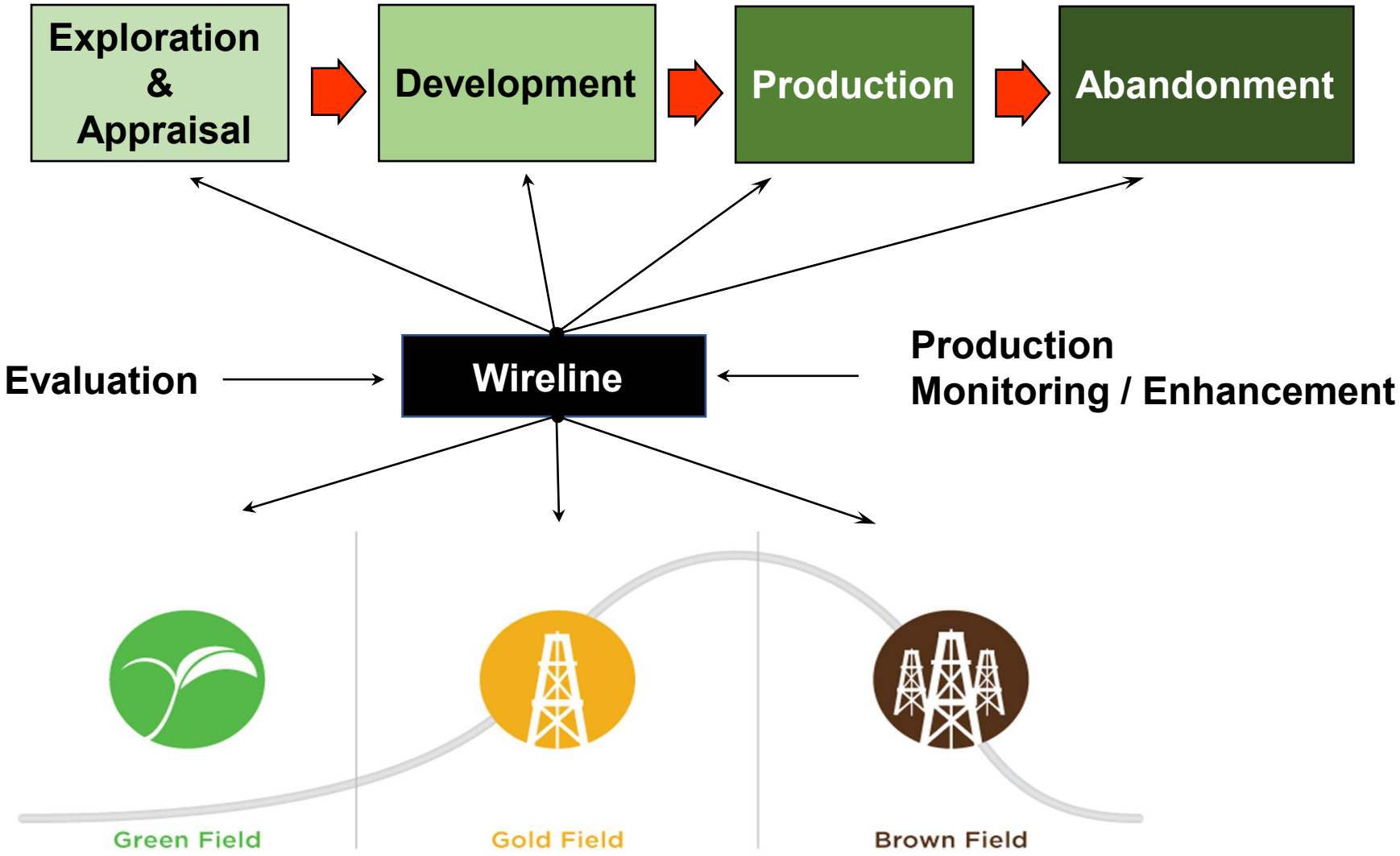
After the well has been drilled to final depth and all of the casing has been cemented into place, the well is completed.



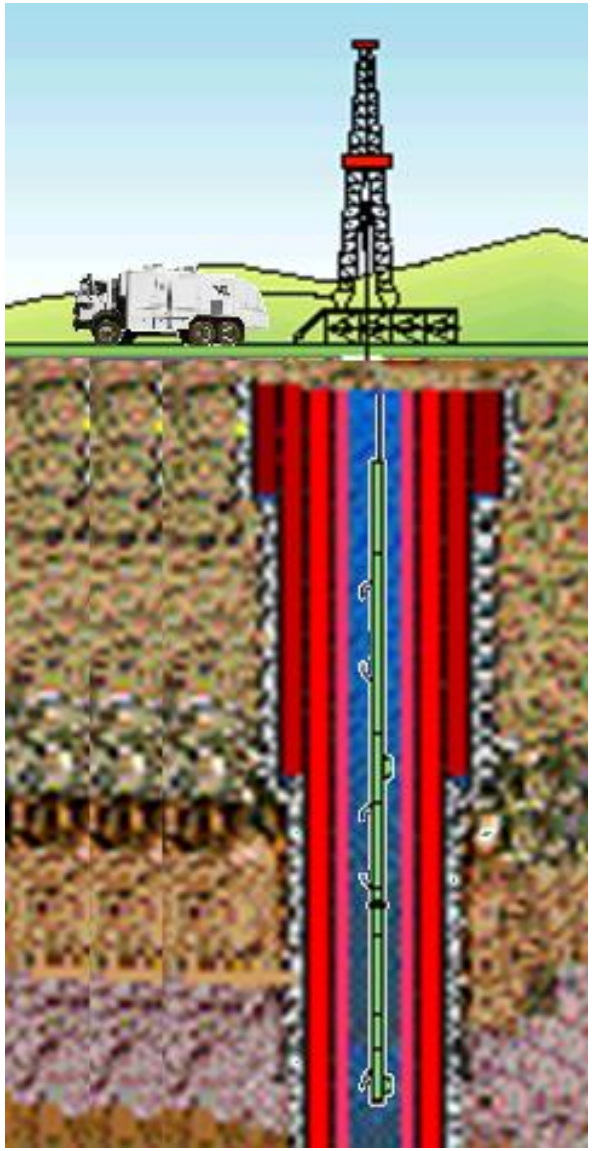
Well Completions Configurations



Wireline: Life of Reservoir / Field Stages



What is Wireline Logging?



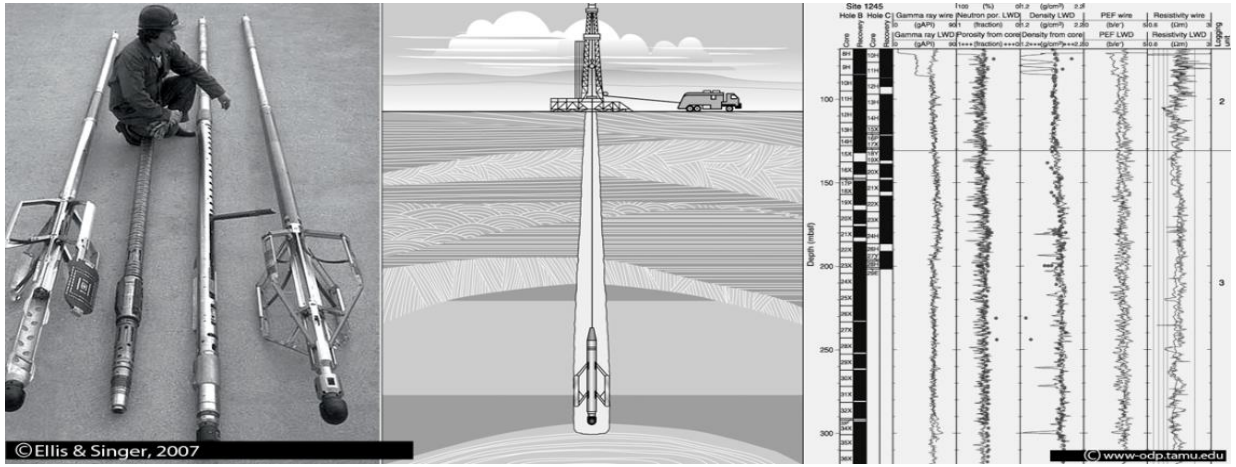
Lowering of sensors into the (open/cased) wellbore using a wireline cable to provide the customer with answers such as:

- Geological
- Petrophysical
- Borehole Seismic Measurements
- Pressure Surveys and Sampling
- Cement and Corrosion evaluation
- Perforating
- Formation evaluation through casing
- Production Logging and Production Monitoring
- Remedial Workover

What is Wireline Tools?

- Wireline Tools are certain category of tools which are used when any well drilling, workover and intervention related activities needs to be performed in an already producing oil or gas well, and also geothermal well.
- These tools are designed in such a way that they can be easily lowered down into the well bore.







Type of Wireline Services



OPEN HOLE

Reservoir Properties

Fluid/Rock Sampling

Borehole Seismic

Borehole Images



CASED HOLE / PRODUCTION

Cement / Corrosion

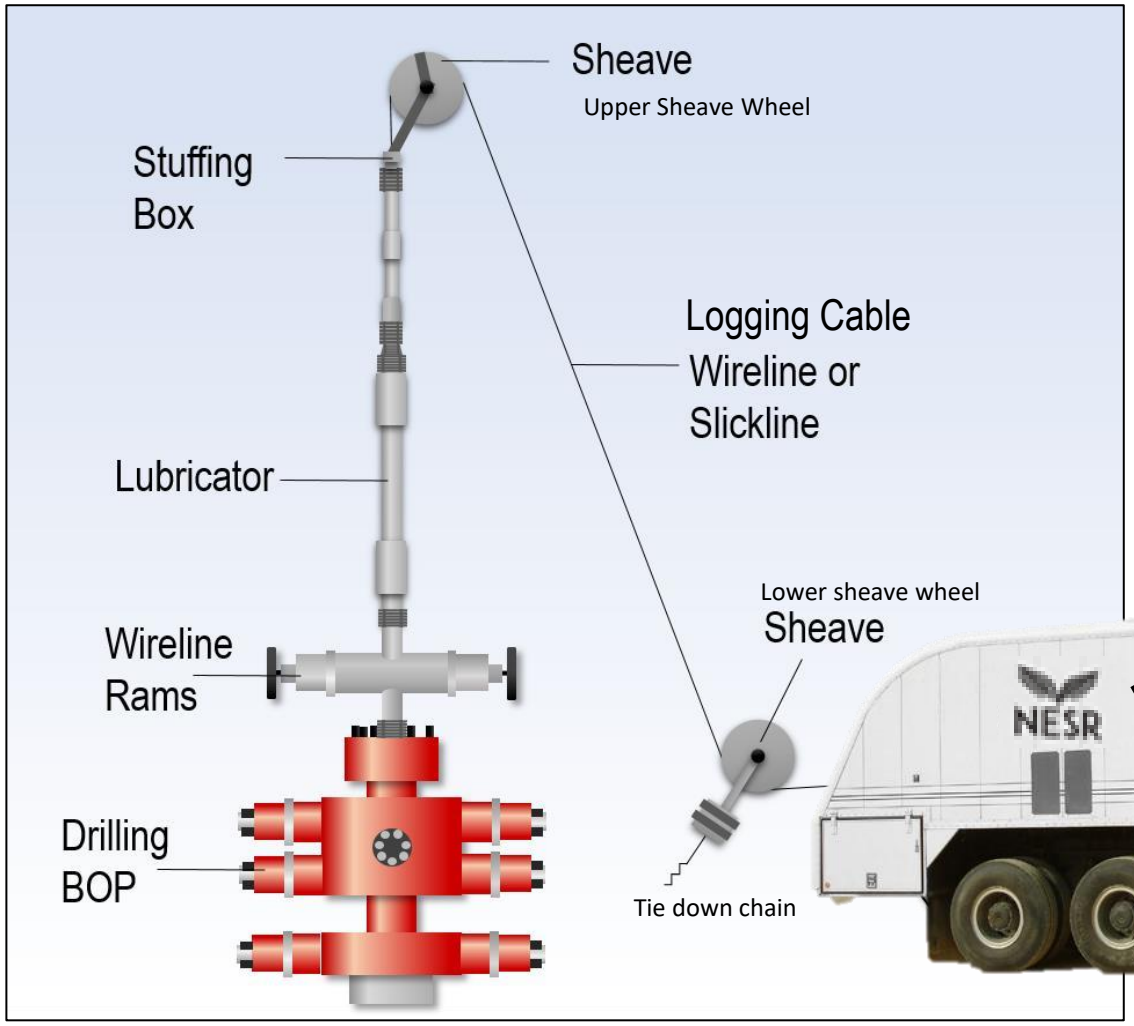
Diagnosis / Monitoring (PL)

Perforation

Repair / Remedial Work

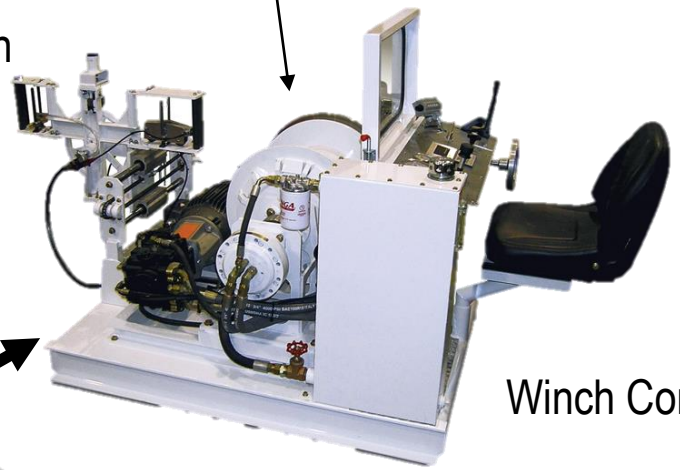
Re-evaluation

Wireline Setup



Cable Drum

Depth / Tension Device

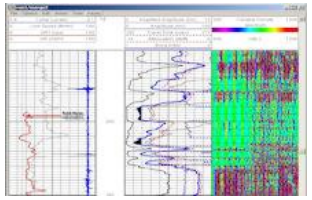


Winch Control





Acquisition System



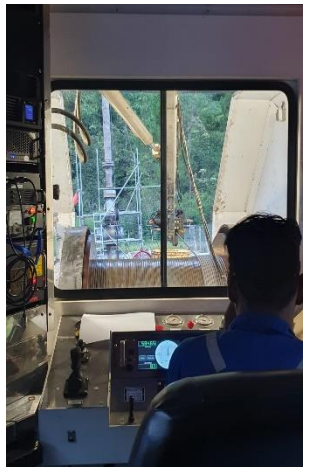
Winch Control

This allows the control of the winch, having:

- Display monitor
- Direction handle
- Winch Brake
- Throttle
- Monitoring Gauges
- Emergency Shutdown



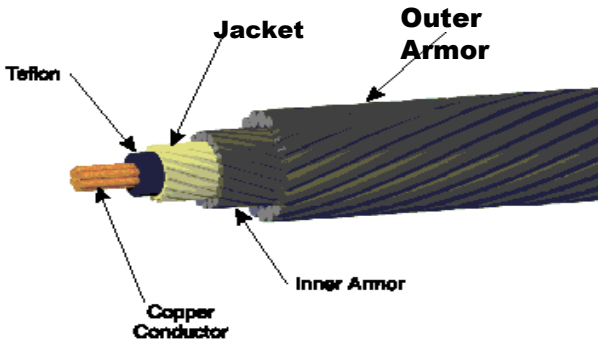






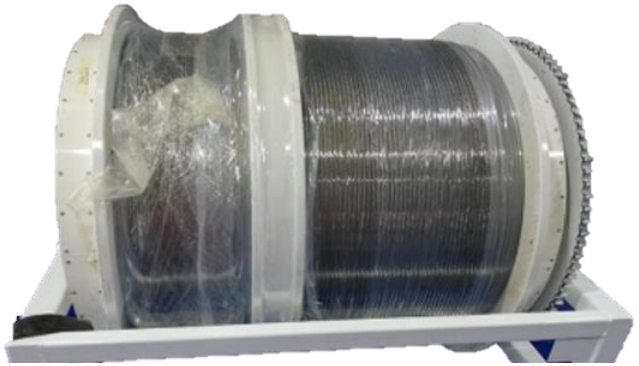
Wireline Cable

Monocable

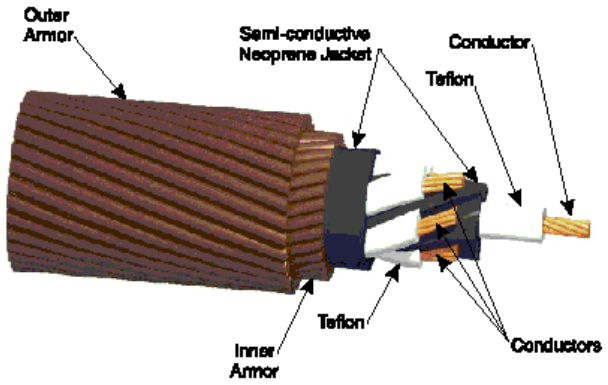


Purpose:

- ✓ Electrical Connection
- ✓ Mechanical Connection
- ✓ Depth Measurement

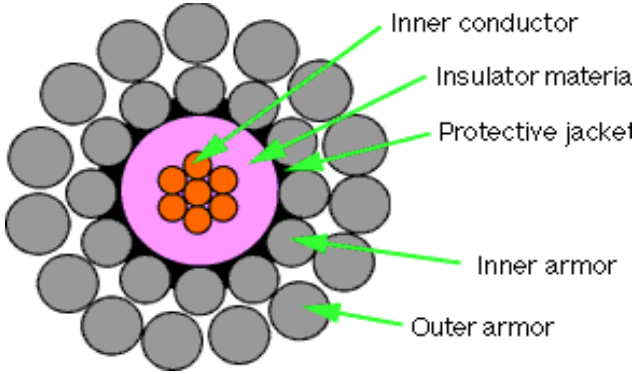


Heptacable



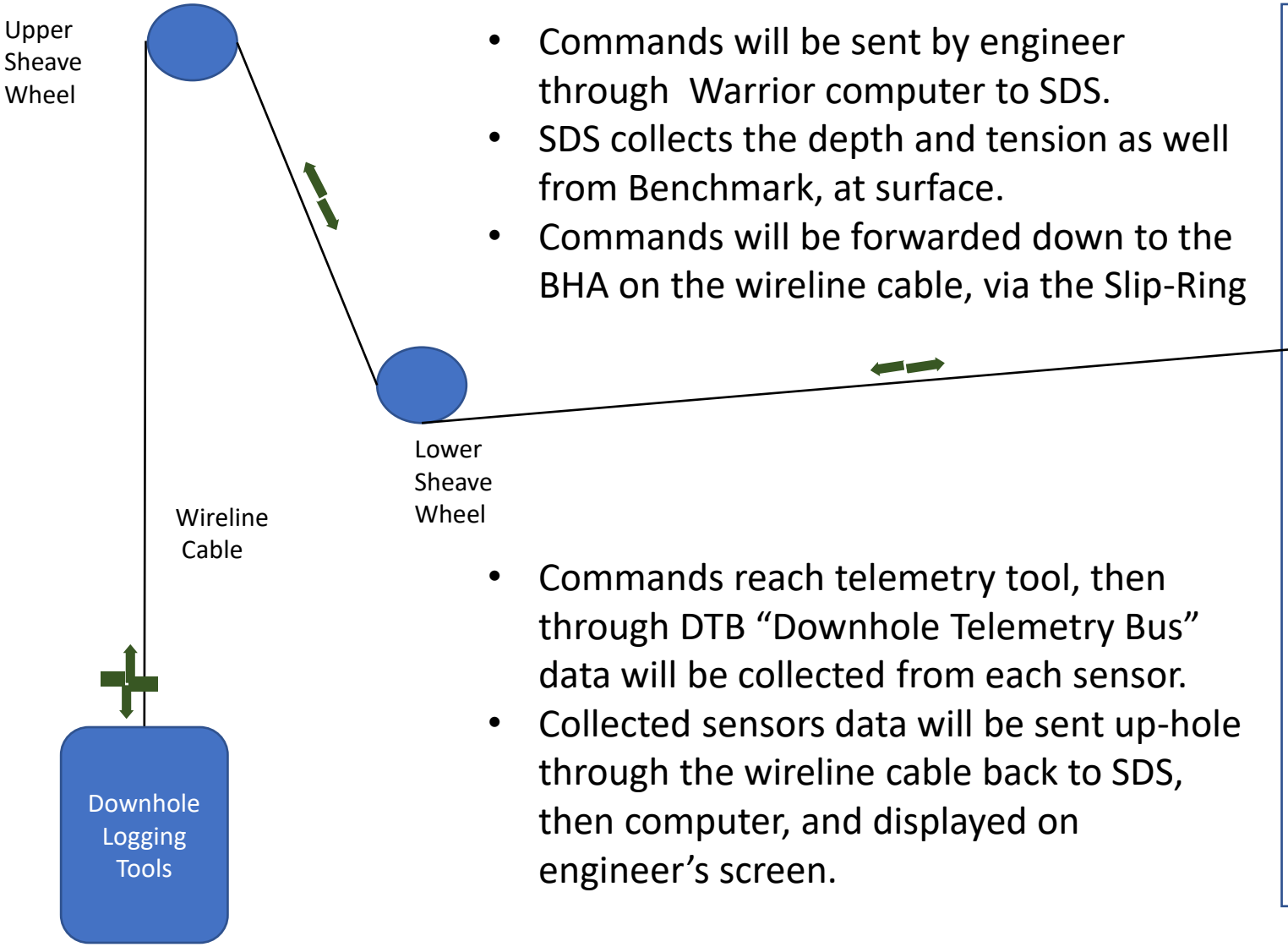


Heptacable



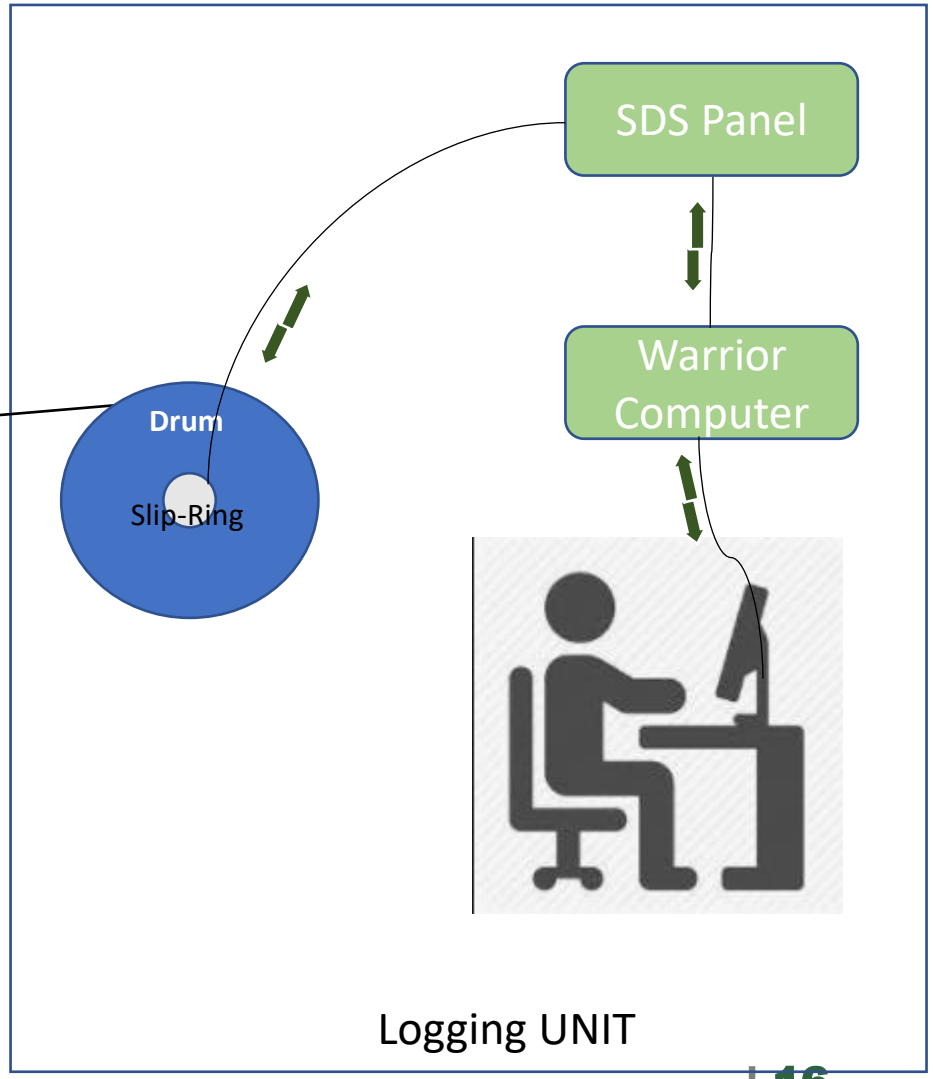
Monocable

Data Transmission system



- Commands will be sent by engineer through Warrior computer to SDS.
- SDS collects the depth and tension as well from Benchmark, at surface.
- Commands will be forwarded down to the BHA on the wireline cable, via the Slip-Ring

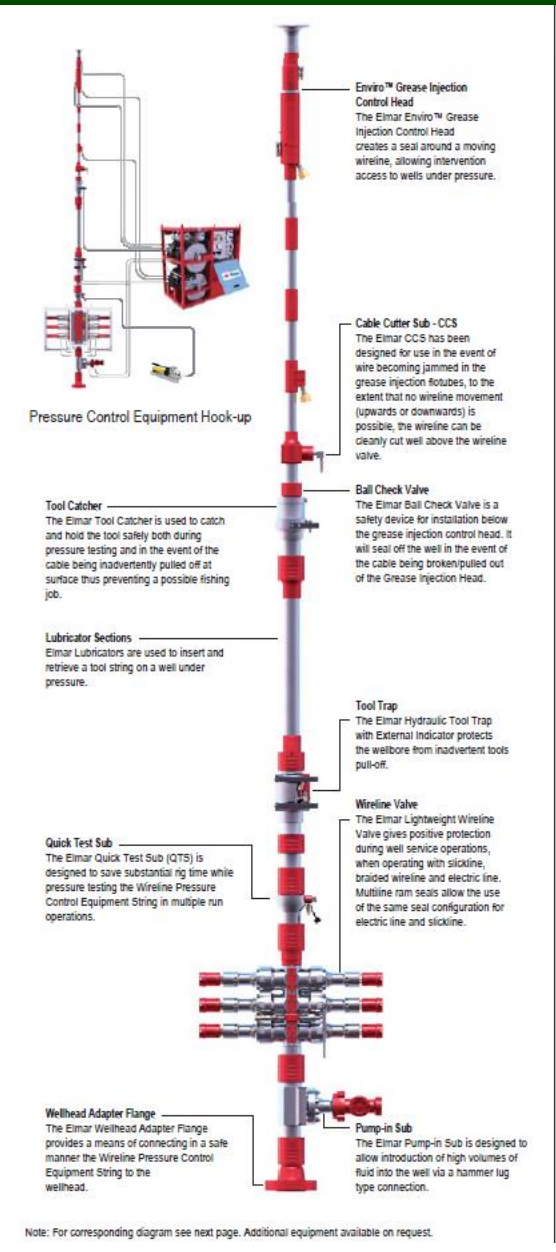
- Commands reach telemetry tool, then through DTB “Downhole Telemetry Bus” data will be collected from each sensor.
- Collected sensors data will be sent up-hole through the wireline cable back to SDS, then computer, and displayed on engineer’s screen.



Pressure Control Equipment

PCE use is required to have a transition environment for the toolstring “BHA”, to get in/out of a Live/Pressurized well, safely:

- Blow Out Preventer “BOP” or Wireline valve
- Lubricators (BHA temporary home)
- Tool Catcher / Ball check valve
- Tool Trap
- Quick-Test Sub “QTS”
- Pump-in Sub
- Wellhead adaptor /Flange
- Grease Injection Head



Rig up Equipment

- **Rig up requires:**

- Sheaves wheels
- Sheave Hanger
- Sheave Hanger Adapter
- Lower sheave wheel stand
- Vertical Make-Up Plate
- Gooseneck
- Tie-Down Chain / Sheave Anchor Chain
- D-Shackle
- Dual Hook Device
- Thread Protector
- Lifting Caps
- Wire Sling

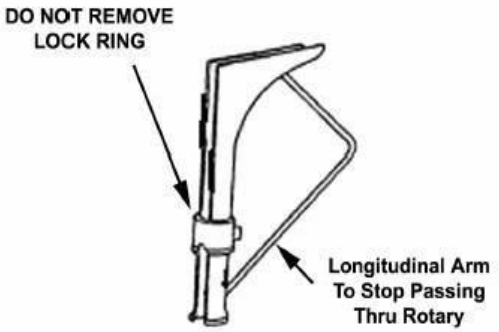




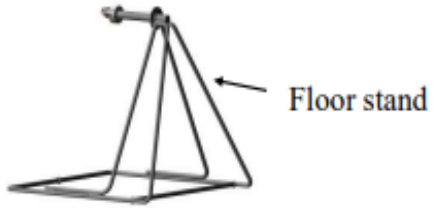






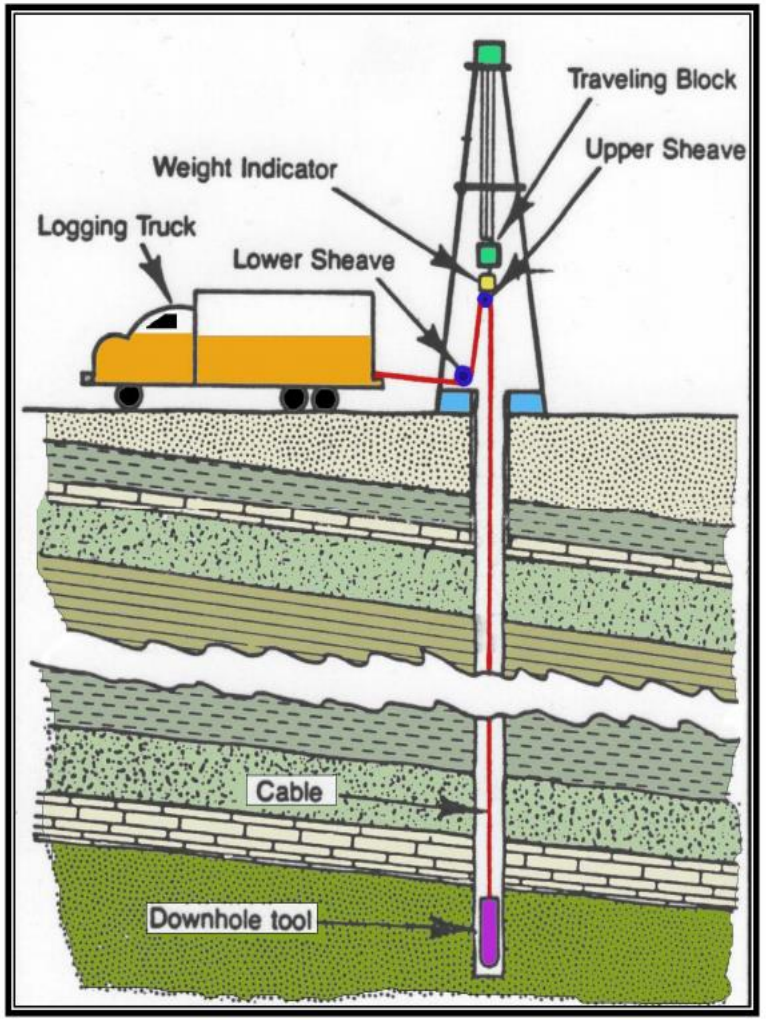






Wireline Logging Procedure

1. Rig-up
   - Place sonde(s) on cat-walk
   - Erect sheaves
   - Thread cable through sheaves
   - Connect head to sonde
   - Using cable, lift sonde to rig floor
   - Set 0 (zero) depth reference
2. Tool to TD
3. Repeat section 300 ft / 100 m
4. Tool to TD
5. Full survey
6. Pull out of hole
7. Rig down



Safety Risk Analysis / Mitigation and Prevention

Pinch Points

- Benchmark, Sheaves, Drum, Snatch Pulls.

High Tension

- Great amount of energy.
- Barrier areas- near cable drum, rig floor,
- No personnel walking under cable.

Electrical Hazards

- Insulation checks- always short conductors with mass.

Heavy Rotating Equipment

- Use rated lifting beams
- Strict compliance with Mechanical Lifting Standard







Type of Wireline Services

Open Hole

- Caliper Log
- Borehole Imager
- Density/Neutron Log
- Sonic Log

Well Completion

- PTS Survey
- Gross Permeability Test
- Pressure Fall of Test
- Heating-Up

Well Production Test

- Static Well Test
- Flowing Well Test
- Injection Test

Well Integrity

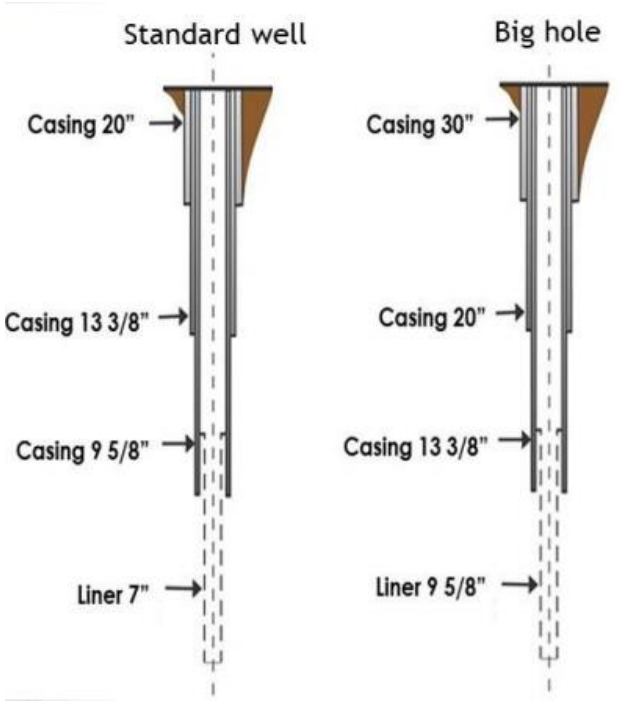
- Multi Finger Caliper
- Magnetic Thickness Detector
- Annular Leak Flow Analyzer
- Downhole Video



Geothermal Well

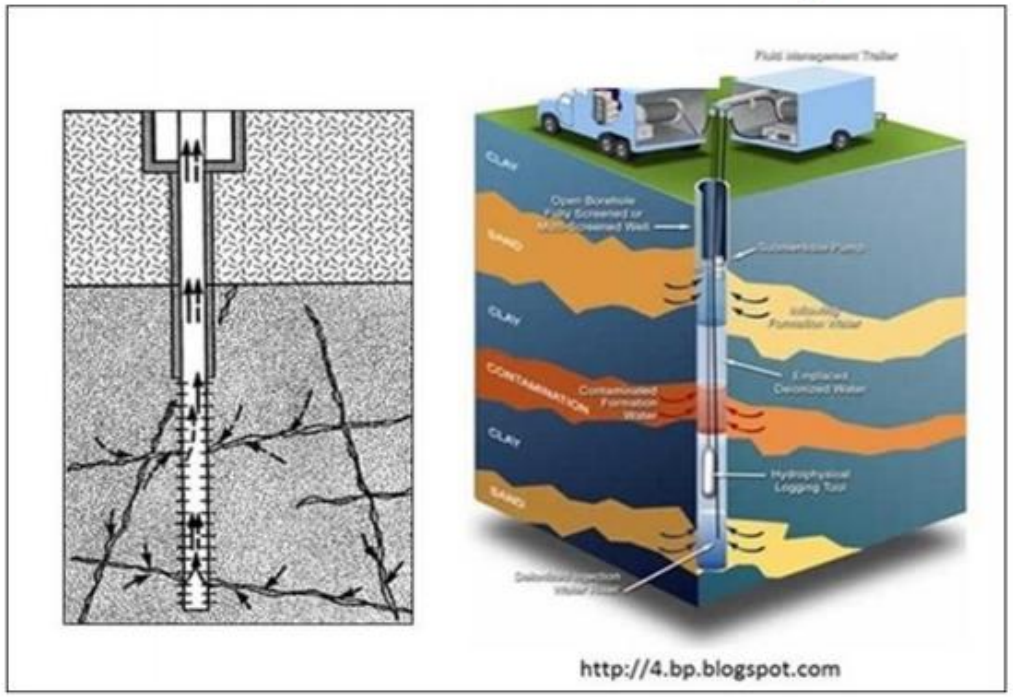


- Drilling Target : High Temperature Zone & high fracture (good permeability)
- The target depth is still economies to drill. Usually in Indonesia range of depth 1-3 km.



Geothermal Well





- Open Hole Logging
- Completion Test:
  a. Injection Test (Water Loss Test)
  b. Gross Permeability Test (Multi-rate Injection Test)
  c. Fall Off Test

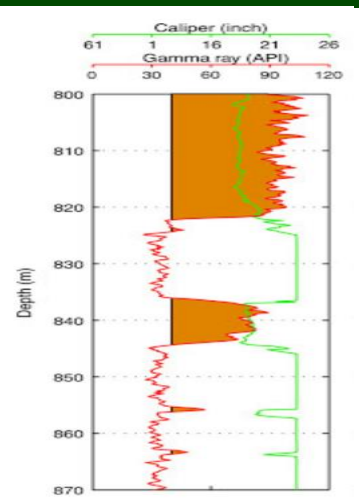
Open Hole Logging

➢ Caliper Log

Function:

- Identification of washouts
- Identification of fractured zones/weak formations
- Cement volume calculation
- Correlation with lost circulation zones



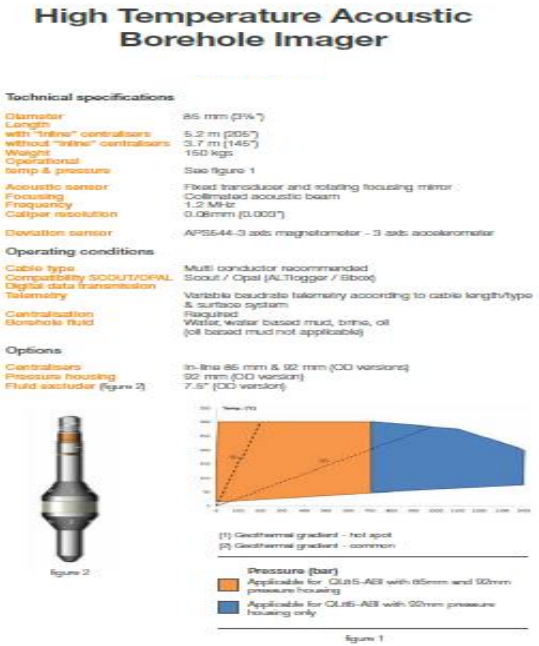


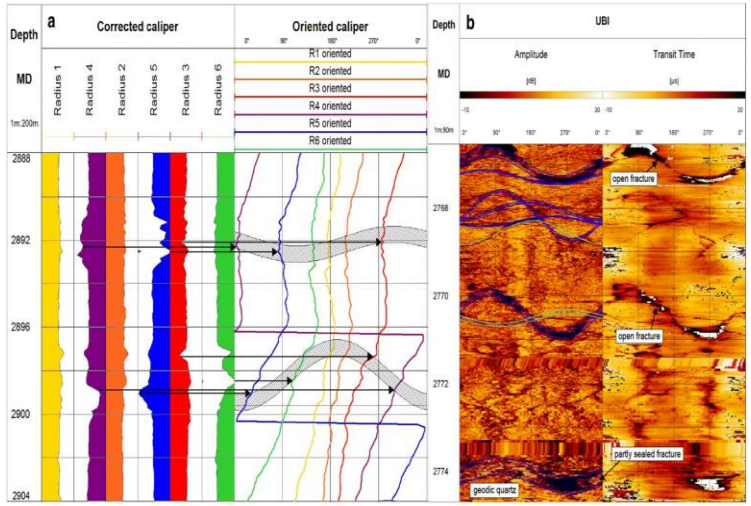
➢ Borehole Image Log

Function:

- Fracture identification
- Determining fracture orientation
- Breakouts and drilling-induced fractures
- Structural interpretation
- Assisting in identifying fractures that likely contribute to permeability







Open Hole Logging

➢Density / Neutron Log

Function:

- Formation Density
- Porosity indication
- Lithology
- Fracture/secondary porosity indication

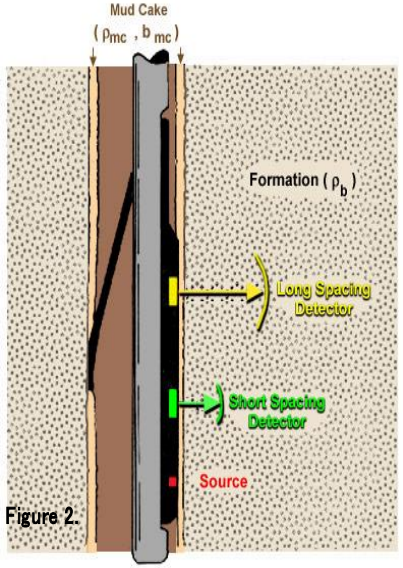


Figure 2.

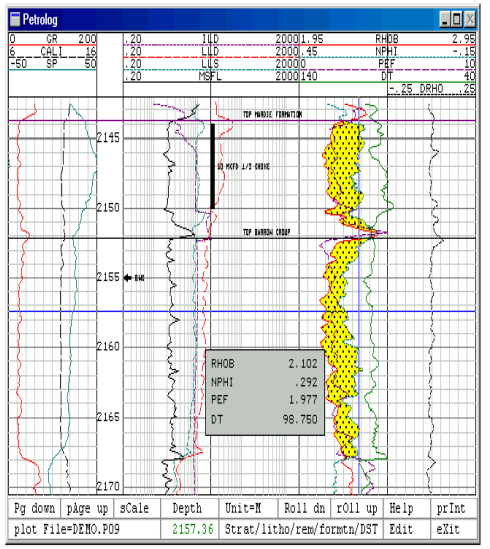


Figure 6: Typical Density/Neutron Log Presentation

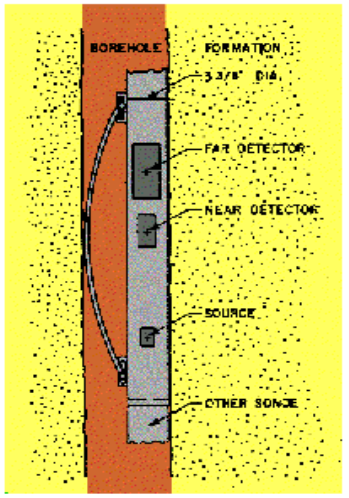
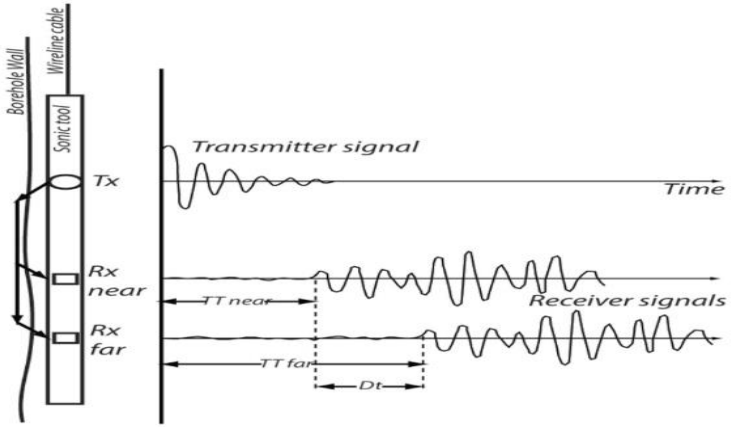


Figure 3: Neutron tool schematic

➢Sonic / Acoustic Log

Function:

- Formation velocity
- Fracture identification
- Rock strength indication
- Wellbore stability analysis



Geothermal Well Completion Test

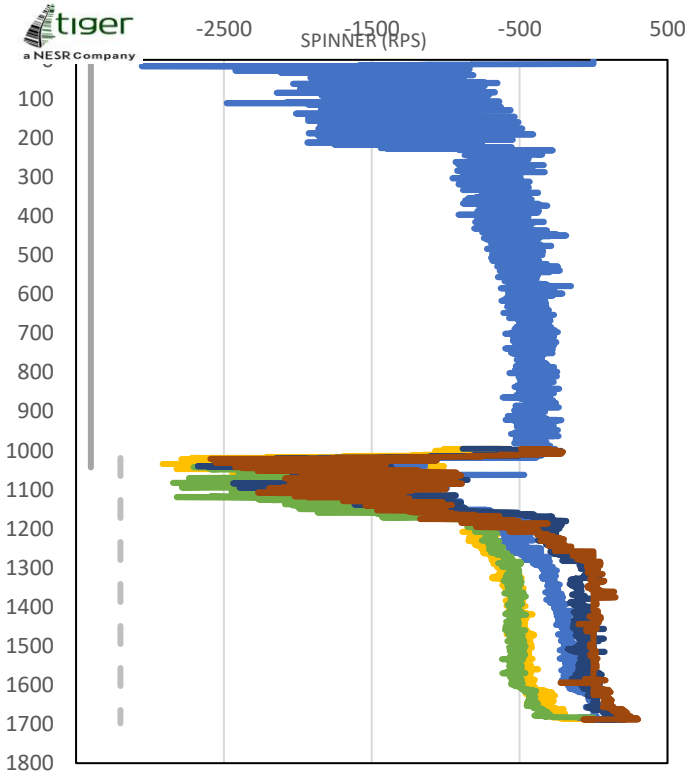
Completion Test

- When: immediately after the well is completed & the rig is still on the well, before installing production wellhead.
- Duration: 1-2 days.

A. Pressure Temperature Spinner (PTS) survey: to identify initial conditions inside the well / along the wellbore:

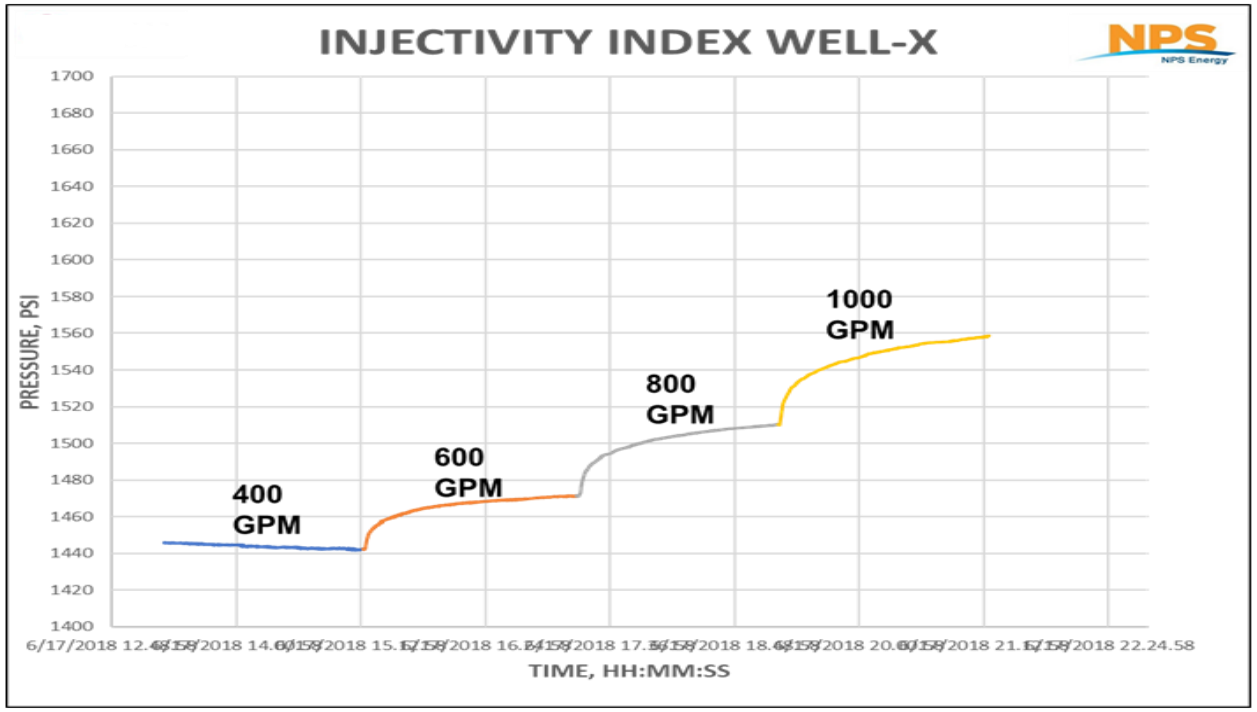
1. Temperature profile
2. Pressure profile and fluid level
3. Entry / feed zones

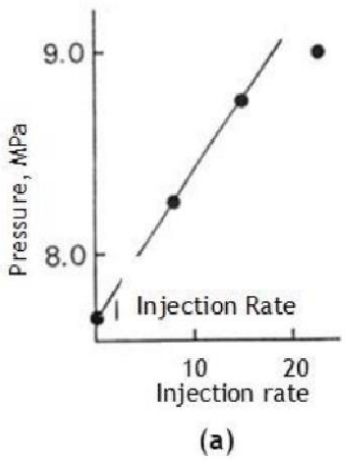


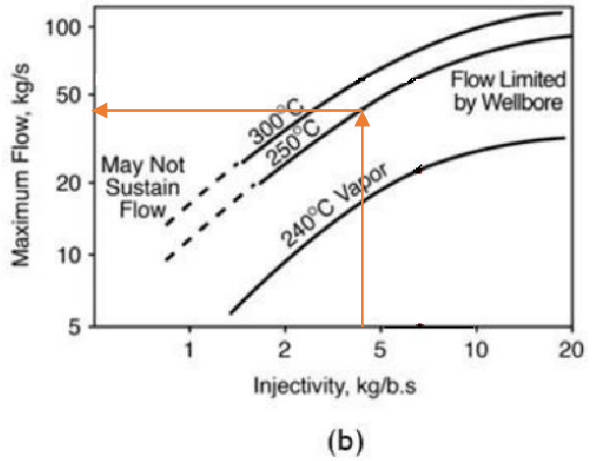


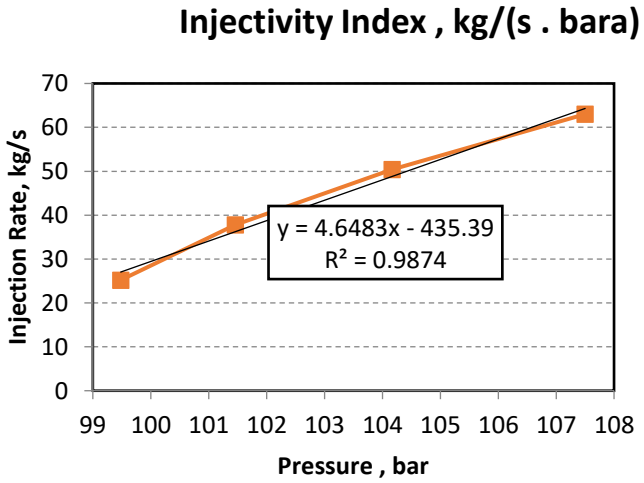
B. Gross Permeability Test (GPT)

Objective: To estimate the injectivity and mass flow rate of the well (well test design)





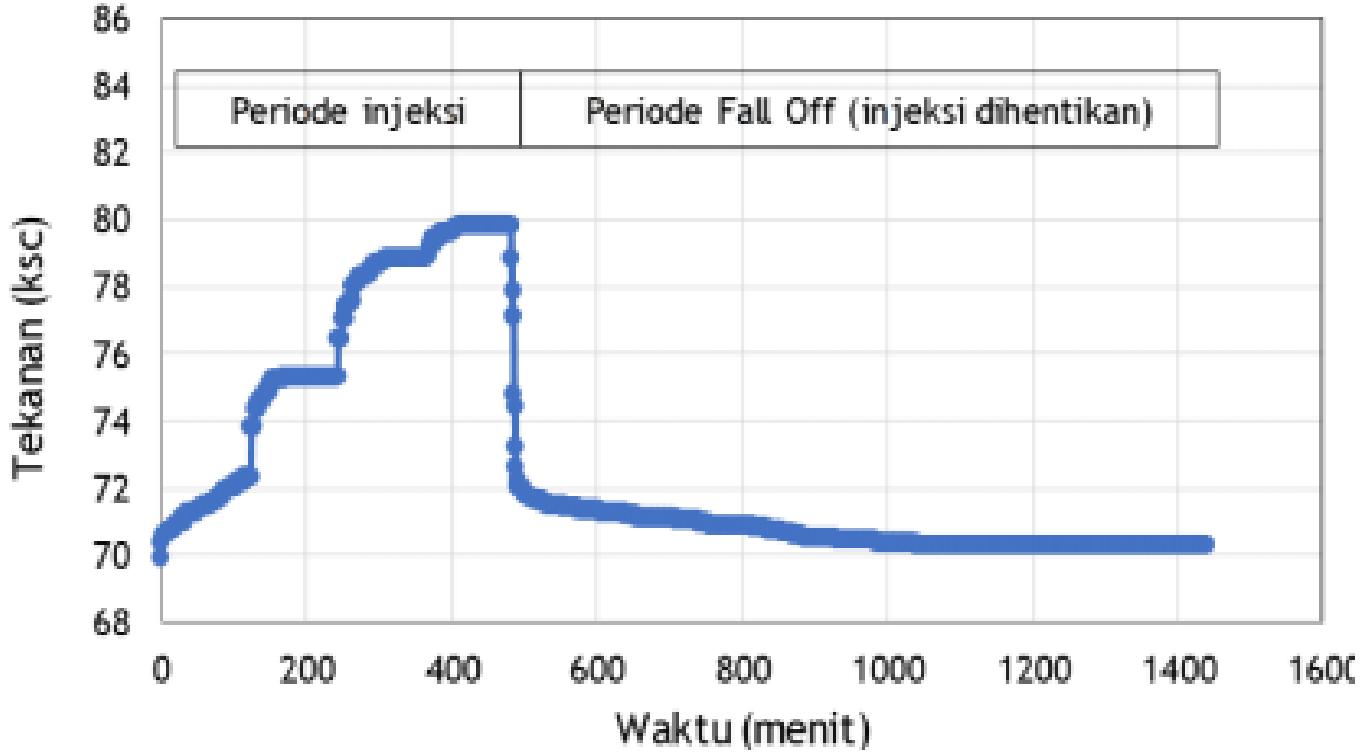


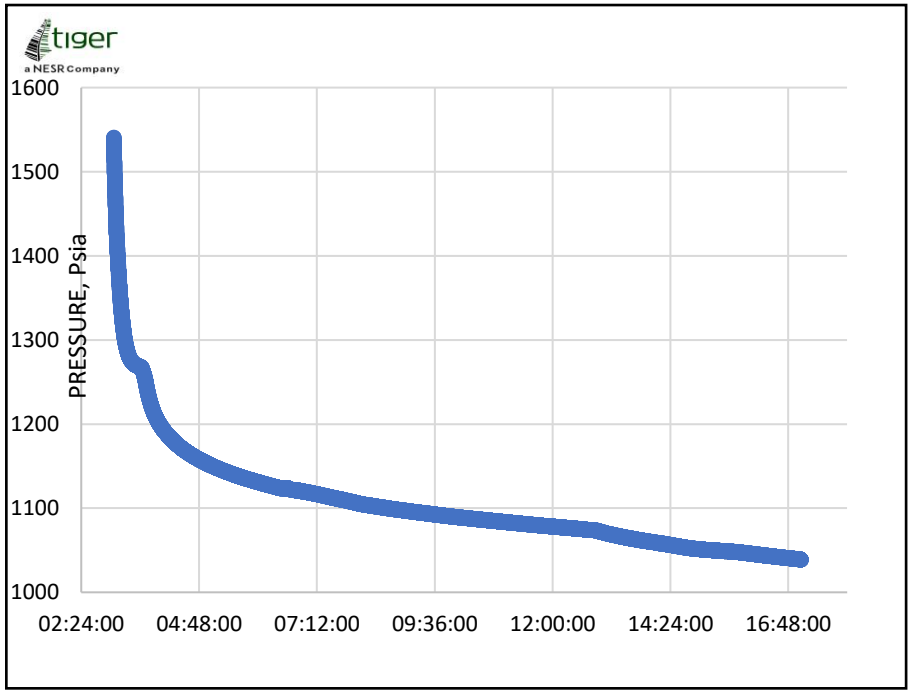


Well Completion

C. Fall Off Test (FOT)

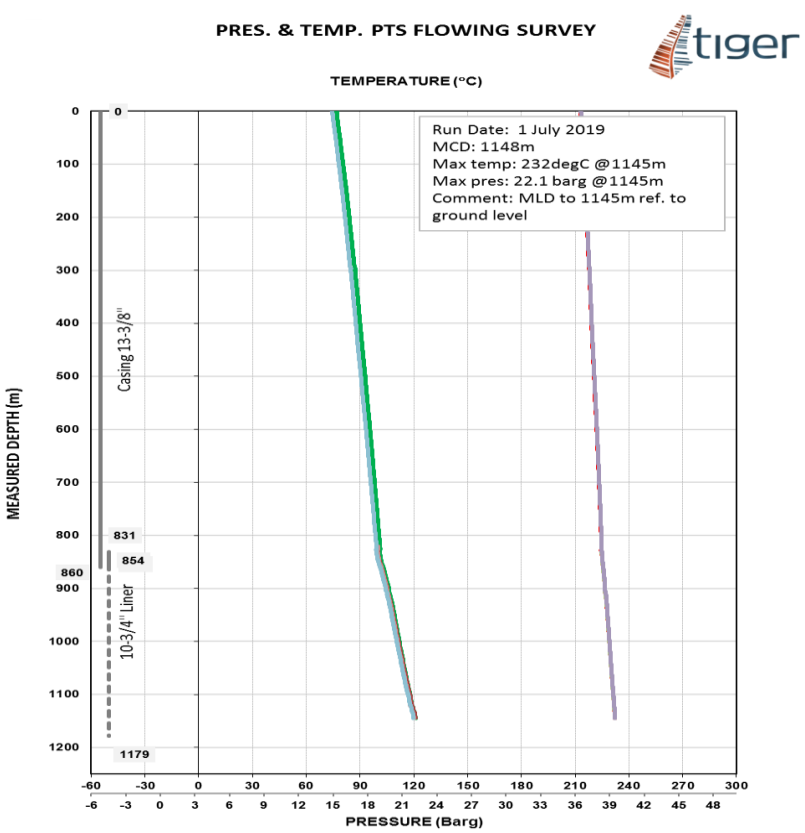
Objective: To determine the transmissivity and whether there is damage around the wellbore (Skin)

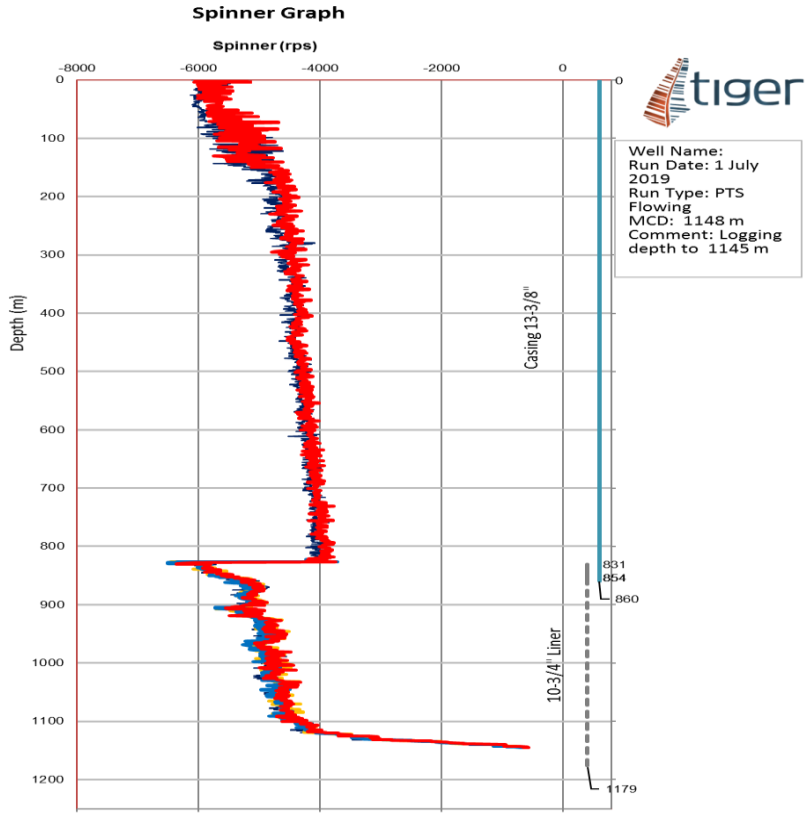




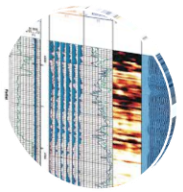
Well Production Test

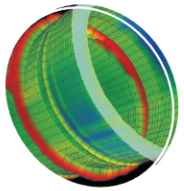
- PT Shut-in: to monitor PT profile of well
- PTS Flowing: to determine contribution of each feed zone











Well Integrity Overview

Well Integrity Diagnostic

Well Integrity Solution

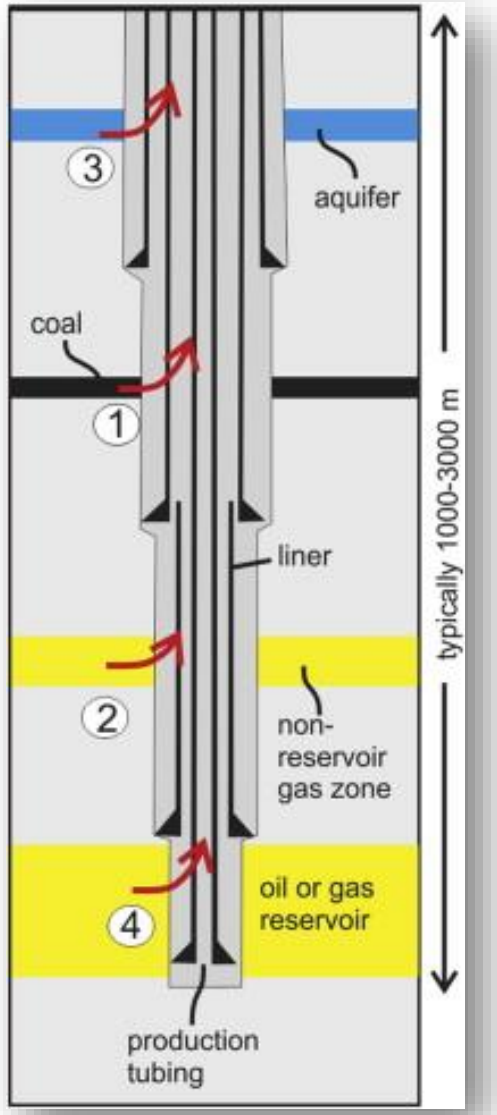
Problem - Gas Bubble on cellar



Problem – Overflow from conductor casing



Overview



Well Integrity

Barrier Inspection

Diagnostic Services

Cement Evaluation

Corrosion Inspection

Annular Flow Detection

Transient Monitoring

Cement Bond

Impedance Map

Mechanical

Ultrasonic

Magnetic

Electro-Magnetic

Electrical

Noise Logging

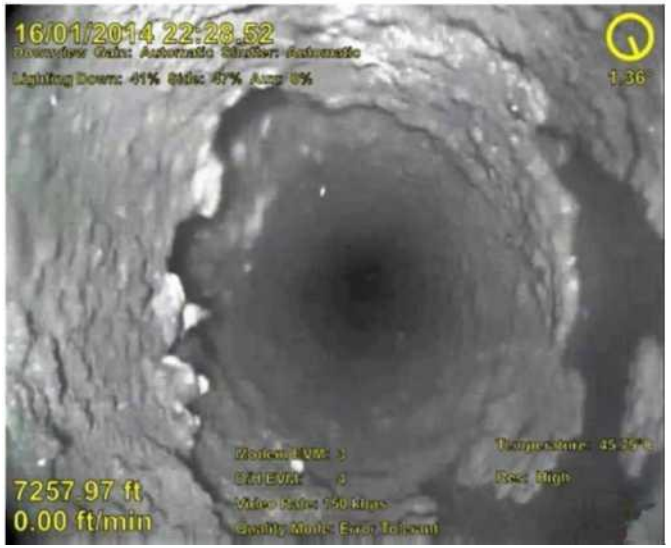
Pulsed Neutron

Temp. Survey

Press. Survey

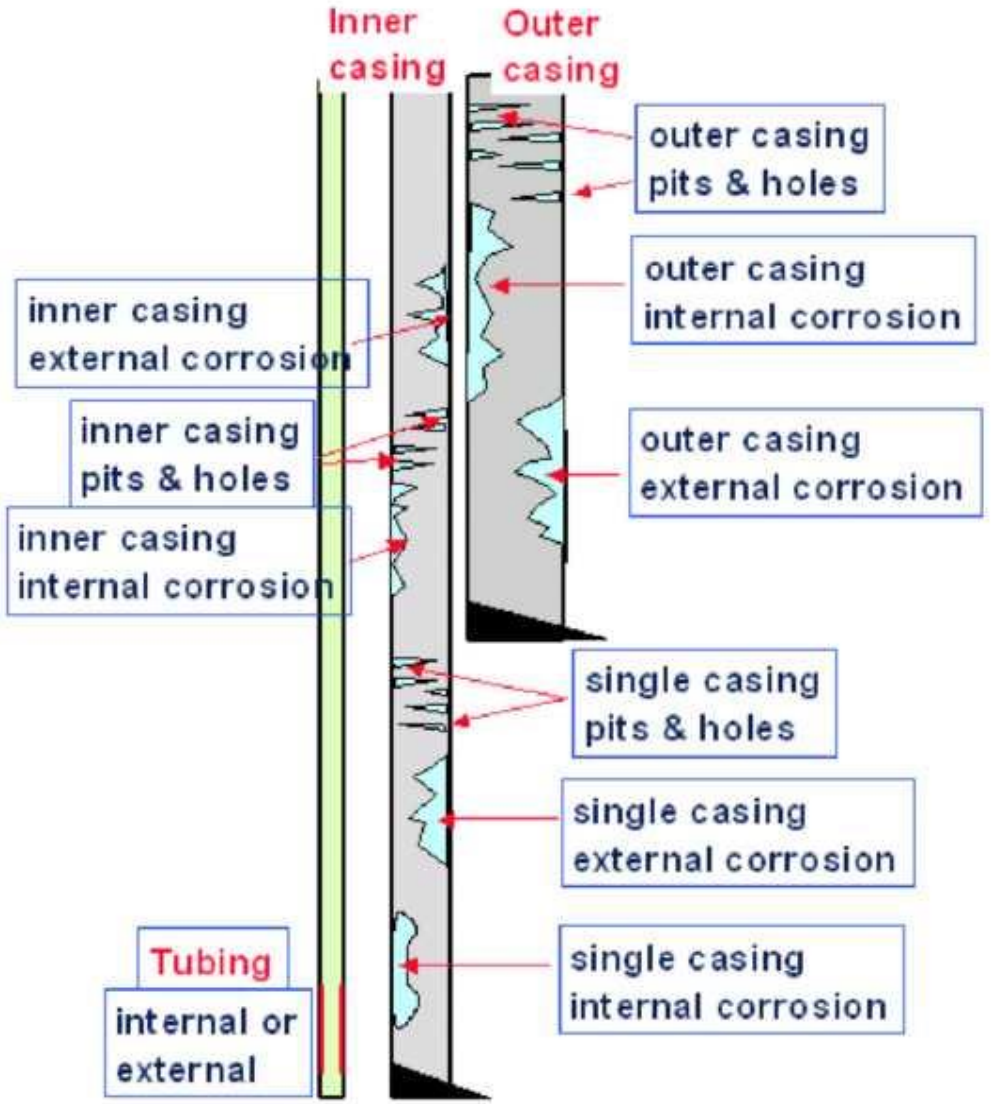
Why Evaluate Casing Integrity?





- Implementing Preventive measures.
- Precise identification of failure can minimize repair expense.
- Remedial action can be taken before the well is seriously damaged.
- Optimization of Workover Programs for maximum efficiency.

Location of Pipe Integrity Problems



Cause of Well Integrity Problems

- **Effect of corrosive fluids ie acids, saline waters etc**
- **Poor cementing. This leads to:**

- Increased stress

- allows corrosive formation fluids to reach casing

- **Stress**
- **Metal anomalies especially at joints**
- **Abrasion due to**:

- drill pipe

- fluid (and accompanying solids) flow especially at high rates

Purpose of Cement

- **Cement is used for:**
  - **structural support to the casing to reduce the risk of a casing failure**
  - providing hydraulic & gas isolation
  - preventing production of unwanted formation fluids
  - isolating intervals to ground water
  - providing structural strength and isolation during fracturing



Basic CBL Principal

No Cement



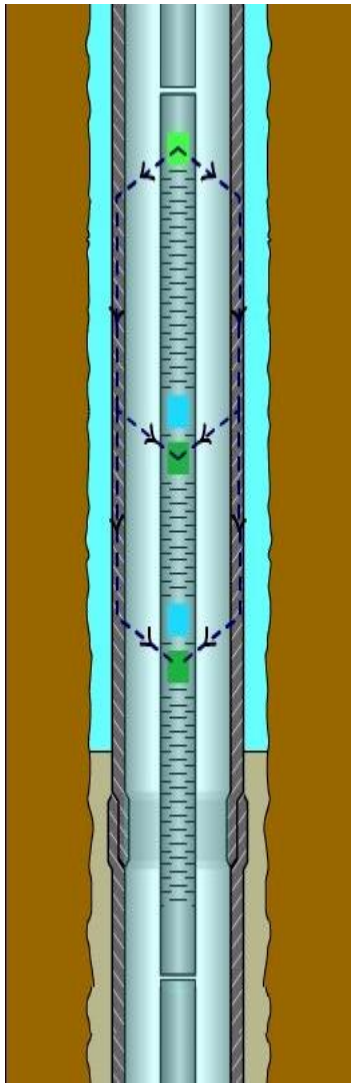
Similar to a Ringing Bell

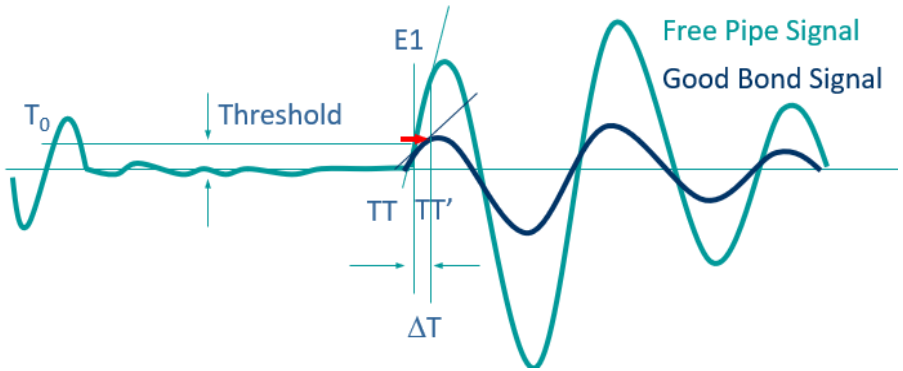
When Fluid is behind Casing, pipe is free to vibrate [loud sound]

Good Bond

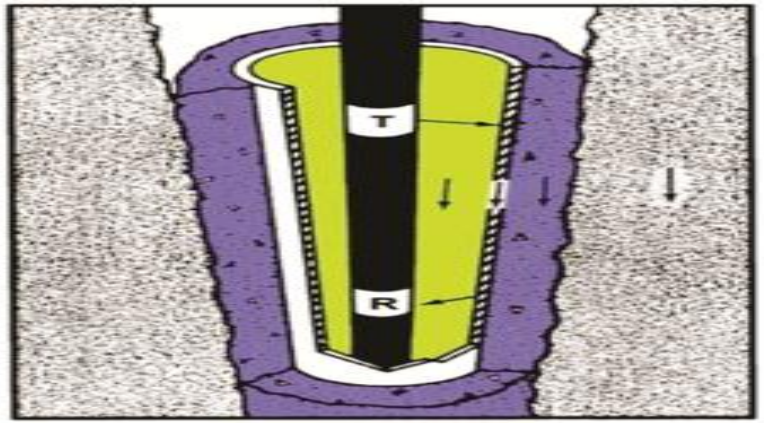


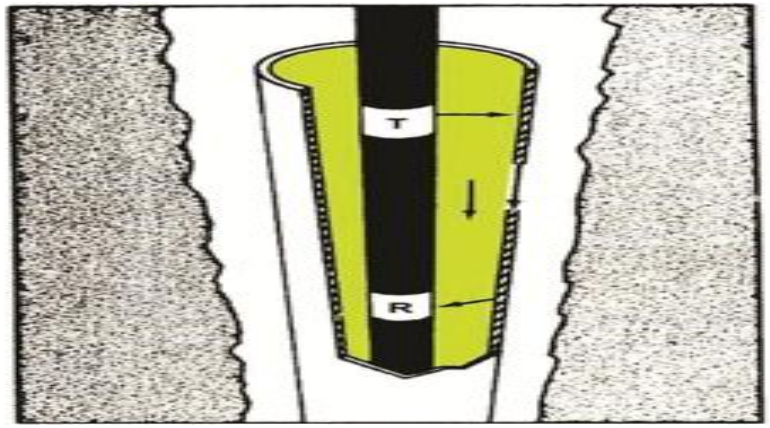
When the casing is bonded to hard cement, casing vibrations are attenuated proportionally to bonded surface



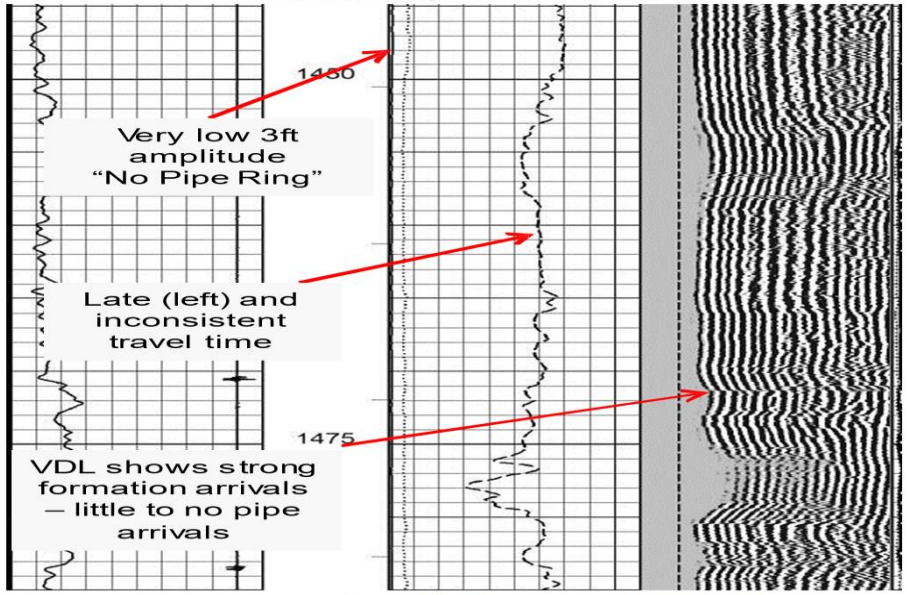


Amplitude, Travel Time & VDL – Example Extremes

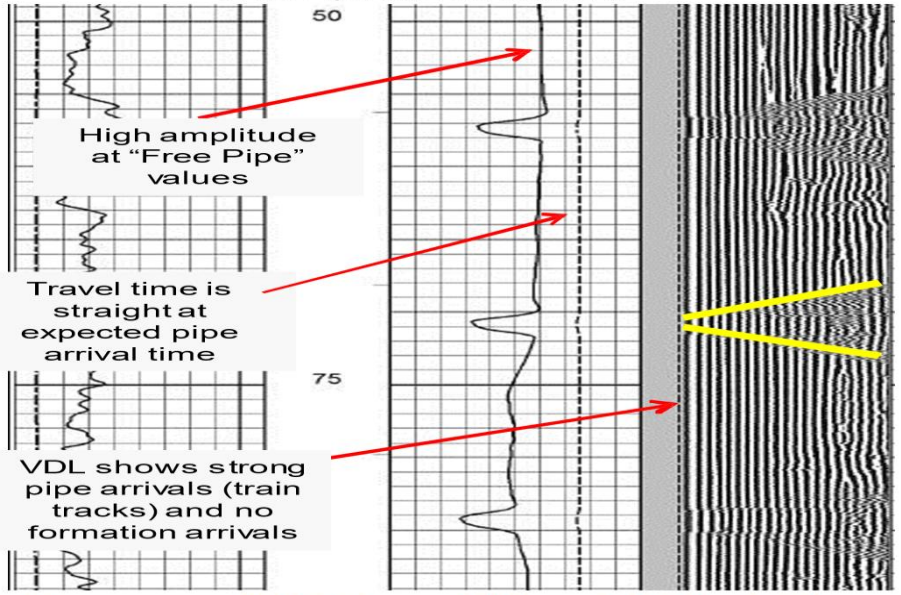




Good Bond to Pipe and Formation



Free Pipe with no cement

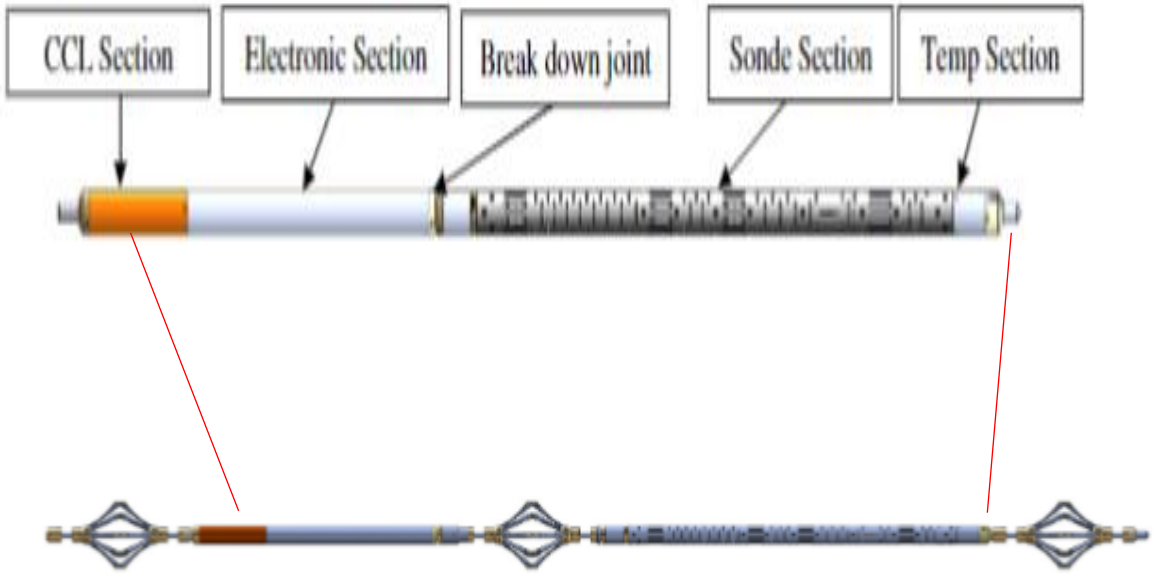


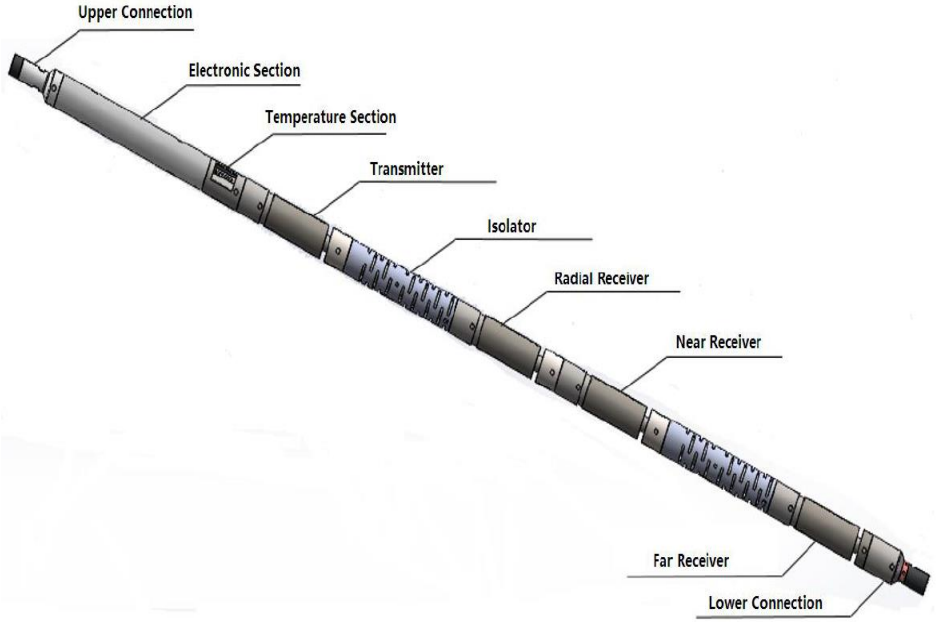
Note: L to R: standard *Amplitude* scaled *0-100 mV*; standard *Travel Time* scaled *650-150 µsec, VDL scaled 200 – 1200 µsec*

Radial Bond Log

Applications :

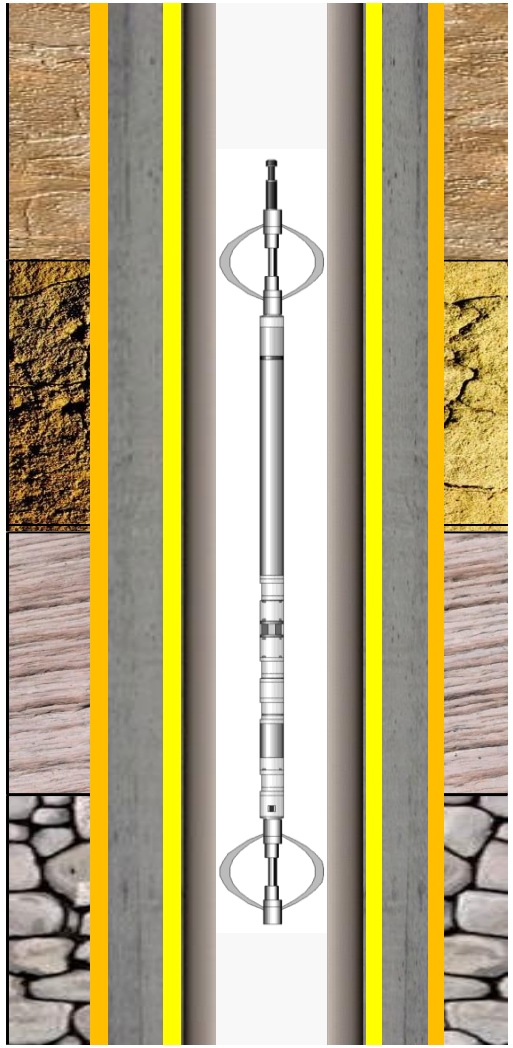
- Provides a 360deg cement map
- Indicates channels and intervals using radial receivers





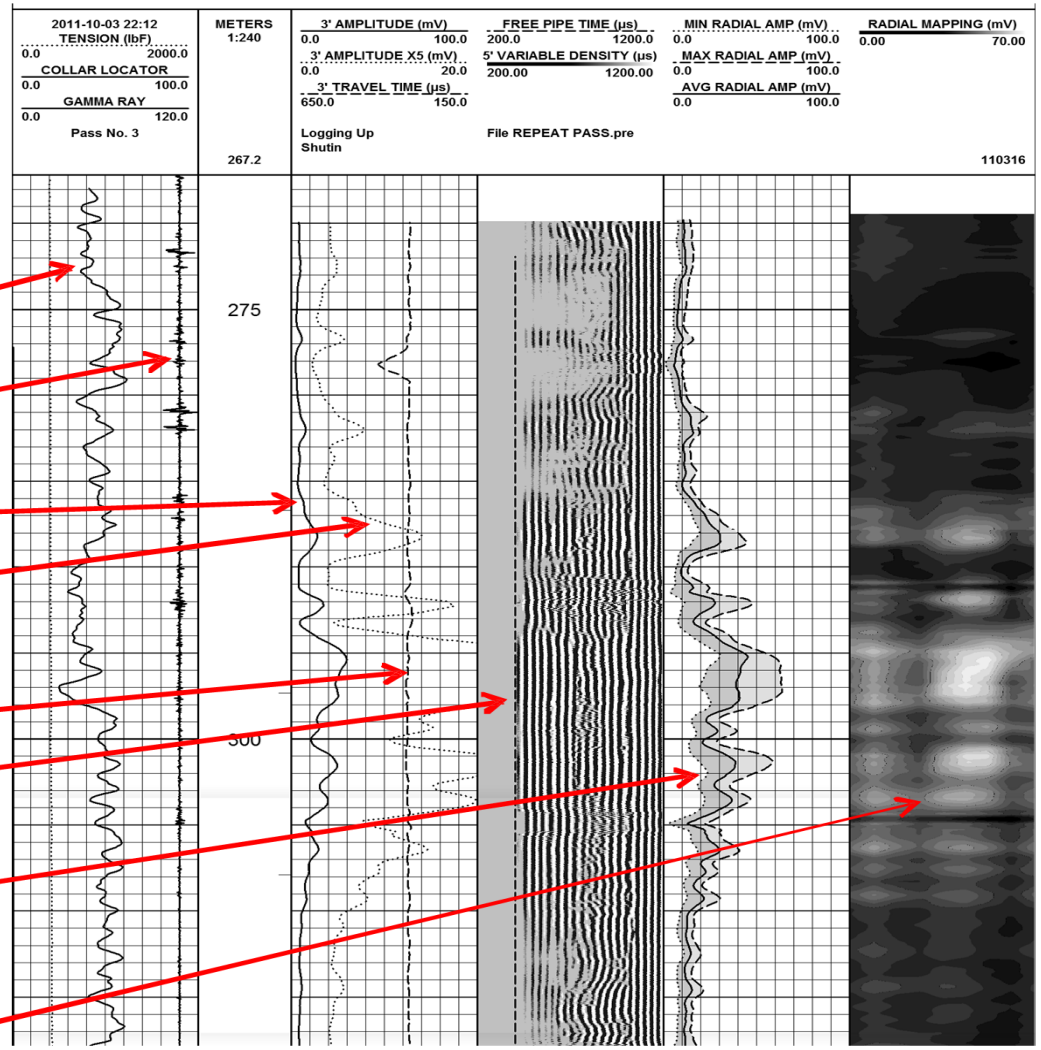
Reasons for Radial Bond Log

- **Radial Bond logs are run to determine:**
  - Cement to casing relationship
  - Cement to formation relationship
  - Evaluate cement conditions:
    - **Channeling**
    - Compromised cement (i.e. gas cut, dehydrated, etc.)
    - Cement stages
    - Cement top
    - Micro annulus



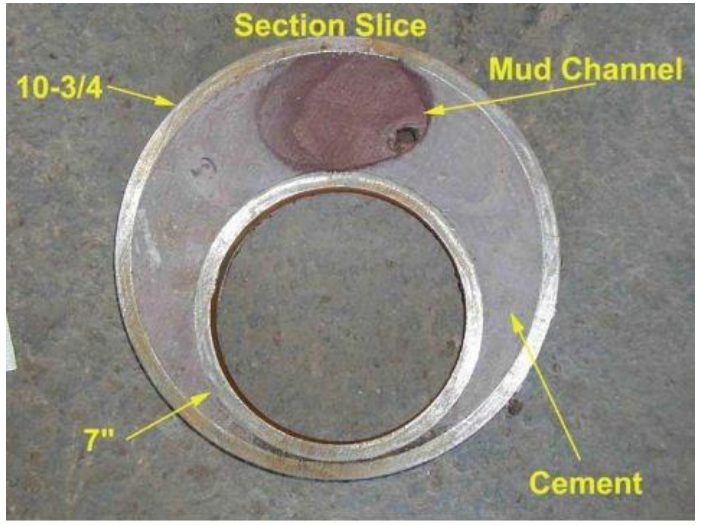
Radial Cement Bond Log Tool (RCBL)

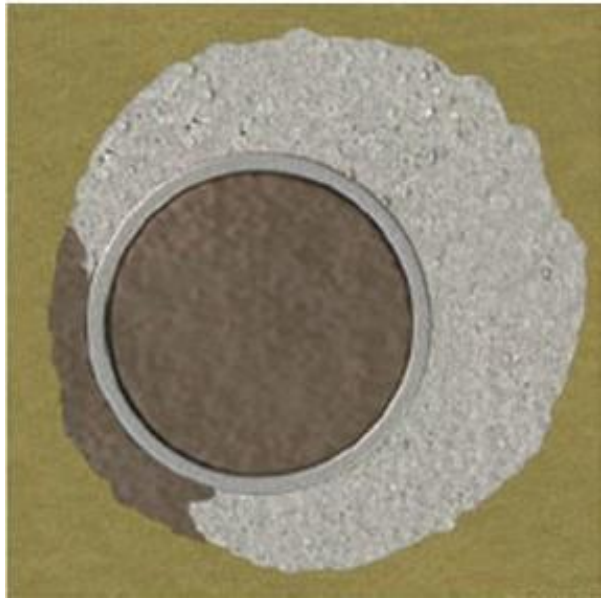
- Presented on the log:
  - Natural Gamma Ray
  - Casing Collar Locator
  - Amplitude from the 3ft
  - Amplitude x 5
  - Travel Time from the 3ft
  - VDL from the 5ft
  - Amplitudes from the eight / six radials (if RBL)
  - Min, Max and Average of the Radials (if RBL)



Bond log considerations

- **Poor Primary Cementing/Hole conditioning**
  - Mud Contaminated cement
  - Gas Cut Cement
  - Channeled cement
- Decentralized casing string
  - Vertical
  - Slant
  - Horizontal







Corrosion Evaluation Techniques

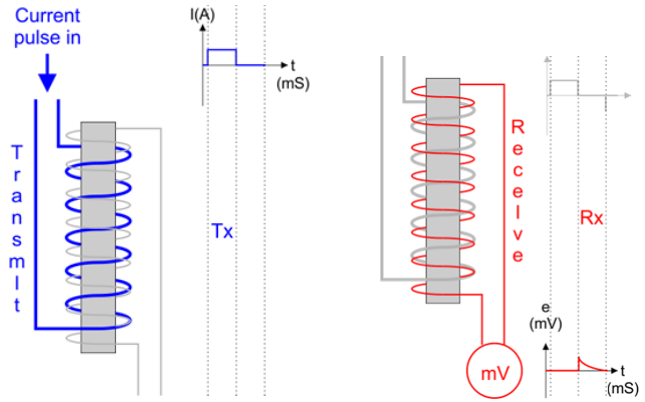
Mechanical (e.g. MIT/MFC)

- Good resolution/accuracy
- Inner surface only
- Restriction/Scale/Solid accumulation

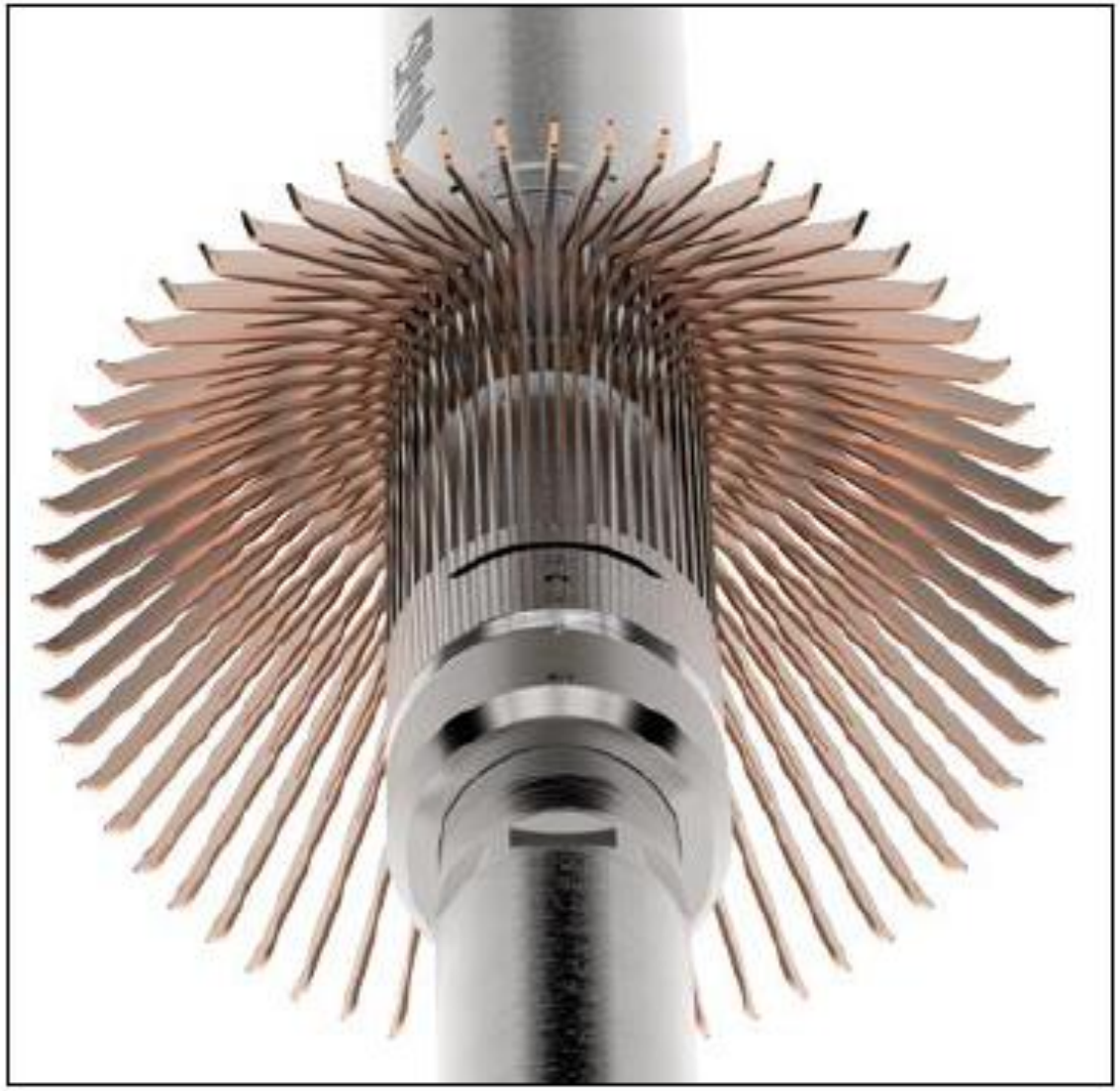


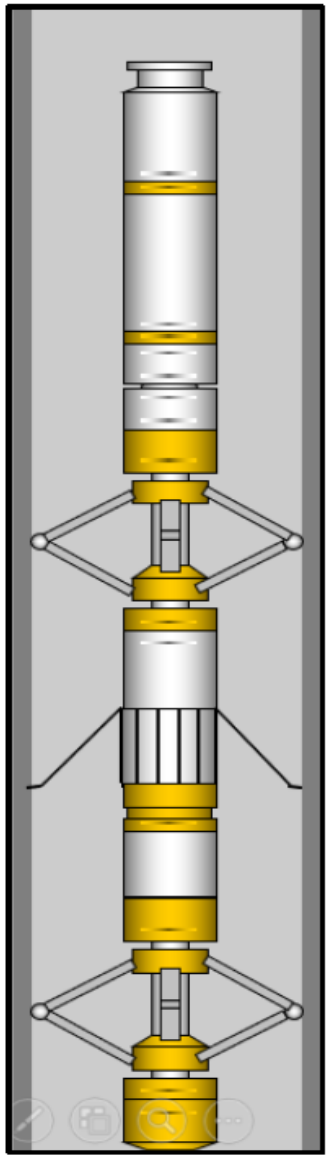
Magnetic (e.g. MTD, ePDT)

- Multiple pipe evaluation
- No need retrieve the tubing
- Quantitative assessment



Multi Finger Caliper (MFC)

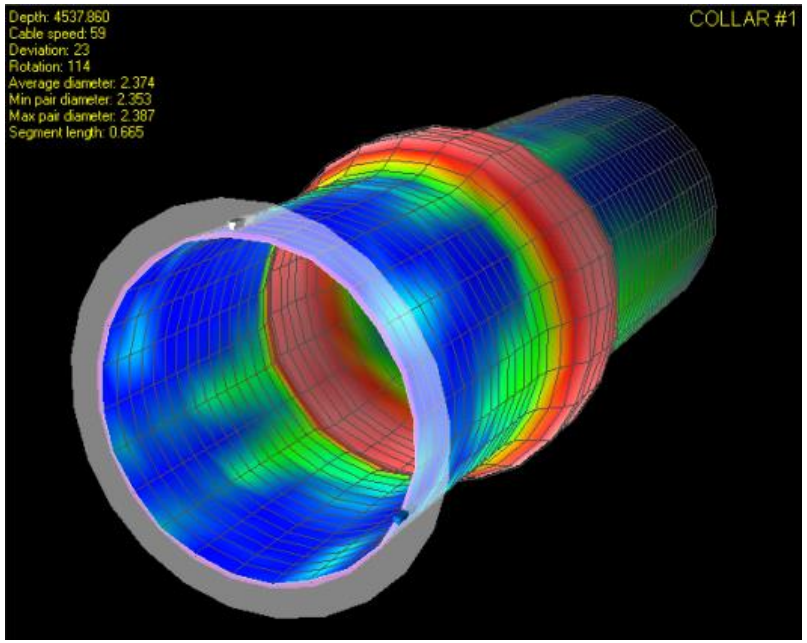


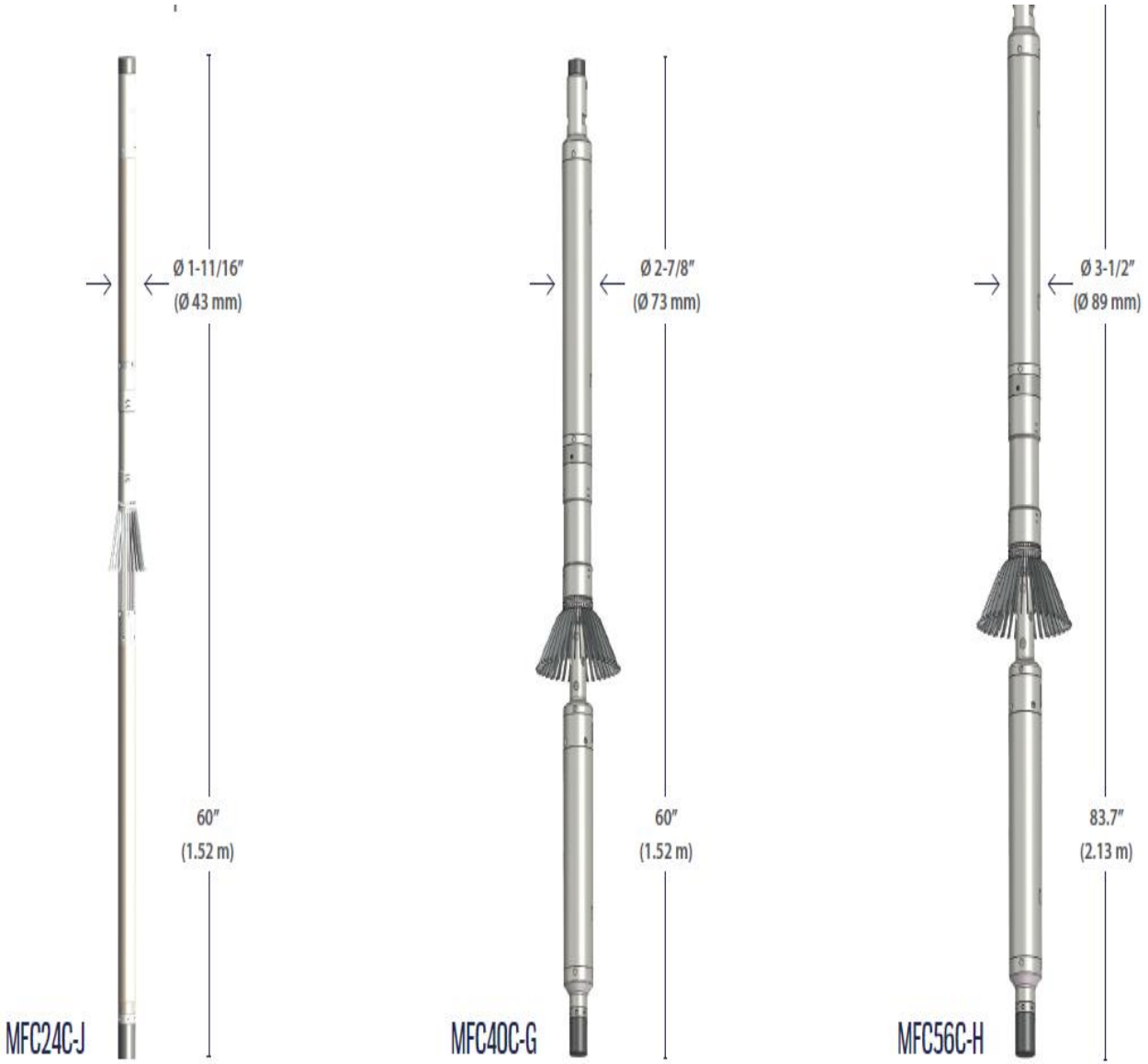


Multi Finger Caliper (MFC)

Multi-fingers tools are used to detect very small changes to the internal surface condition of tubing or casing, with a very high degree of accuracy.

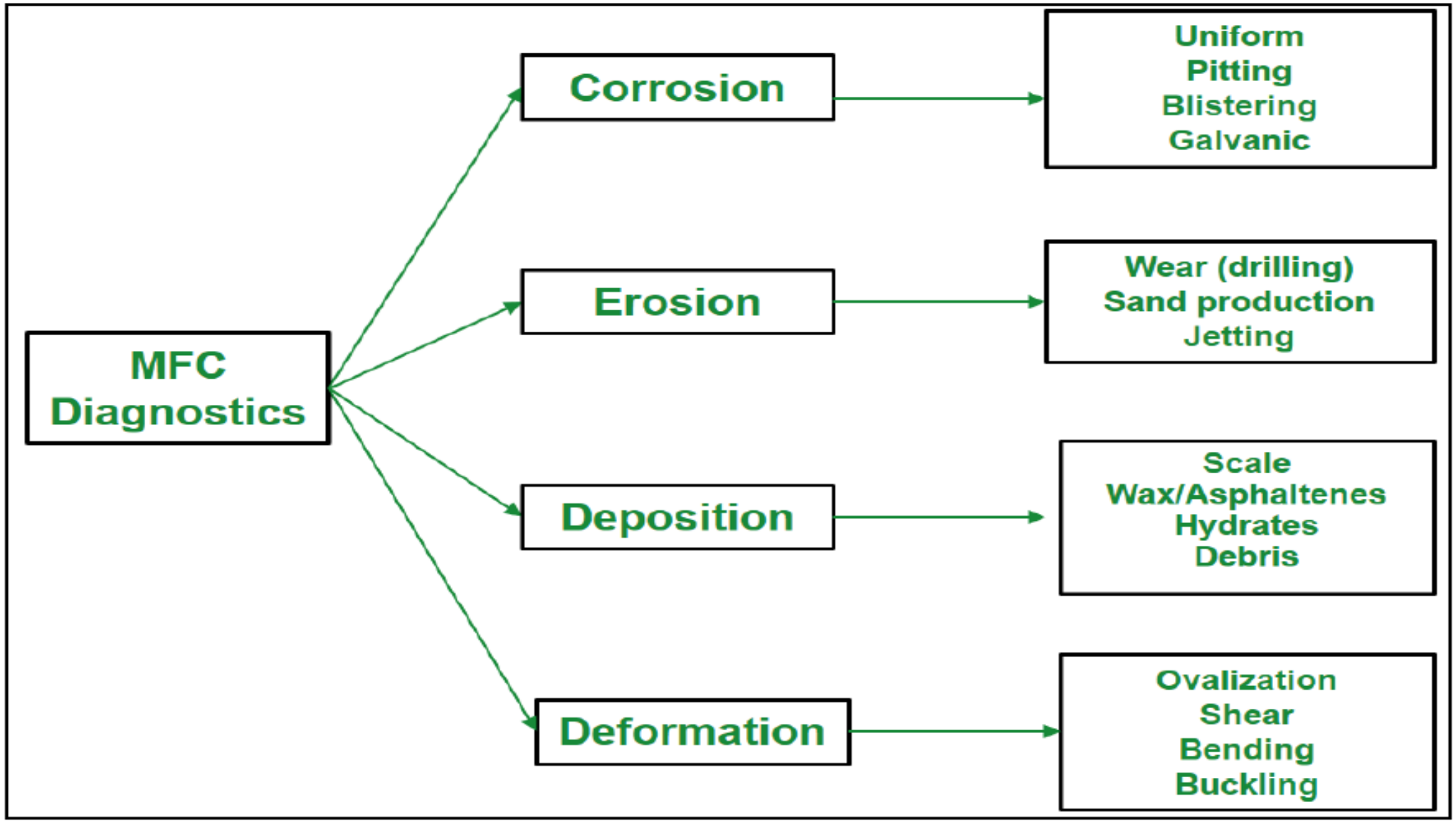




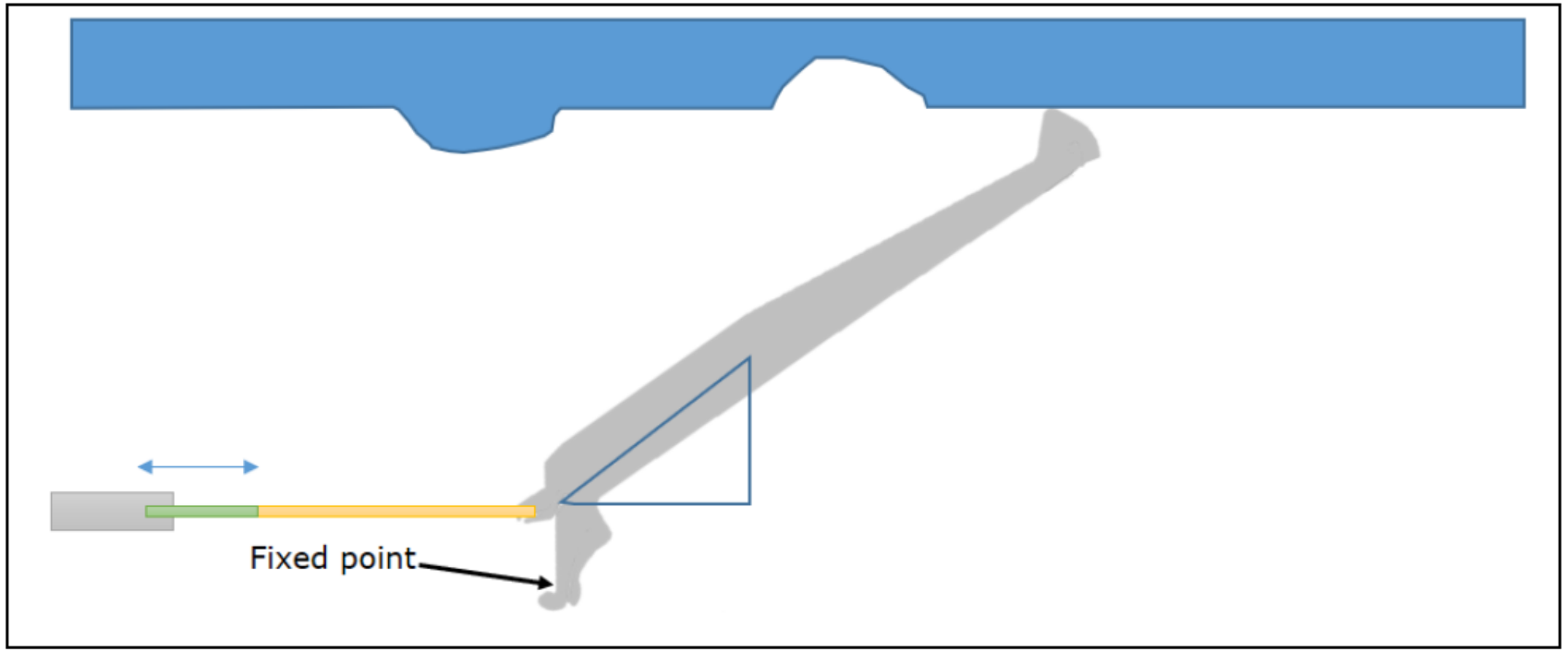


		MFC24C-J	MFC40C-G	MFC56C-H
GENERAL SPECS		**P/N 100511763**	**P/N 100511762**	**P/N 100512537**
Maximum Pressure		15000 PSI (103 Mpa)		
Maximum Temperature		350 °F (175 °C)		
Diameter		1-11/16 in. (43 mm)	2-7/8 in. (73 mm)	3-1/2 in. (89 mm)
Length		59.3in. (1.51 m)	60 in. (1.52 m)	66.7 in (1.69 m)
Caliper Measure Point		28.7in (728 mm)	21.9in. (556 mm)	26.3 in (669 mm)
Weight		28.7 lbs (13 kg)	81.6 lbs (37.0 kg)	138.9 lbs (63.0 kg)
Max Tensile Strength		10000lbf		
Material		17-4 SST		
CALIPER MEASUREMENT				
Number of arms		24 arms	40 arms	56 arms
STD	Minimum, Diameter Maximum, Diameter Accuracy, Radial Resolution	2 in. (51 mm) 7 in. (178 mm) ±0.02 in. (0.5 mm) 0.0039 in (0.1mm)	3.5 in. (89 mm) 8-1/4 in. (209.6 mm) ±0.02 in. (0.5 mm) 0.0039 in (0.1mm)	4 in. (101.6 mm) 9-5/8 in. (244.5 mm) ±0.02 in. (0.5 mm) 0.0039 in (0.1mm)
EF	Minimum, Diameter Maximum, Diameter Accuracy, Radial Resolution	2 in. (51mm) 9-5/8 in. (244.5 mm) ±0.035 in. (0.89 mm) 0.005 in (0.13mm)	3.5 in. (89 mm) 9-5/8 in. (244.5 mm) ±0.035 in. (0.89 mm) 0.005 in (0.13mm)	4 in. (101.6 mm) 13-3/8 in. (340 mm) ±0.035 in. (0.89 mm) 0.005 in (0.13mm)
EXF	Minimum, Diameter Maximum, Diameter Accuracy, Radial Resolution	N/A		4 in. (101.6 mm) 16 in. (406.4 mm) ±0.045 in. (1.14 mm) 0.005 in (0.13mm)
Sensor type		Linear Displacement sensor		
INCLINATION MEASUREMENT				
Minimum		0°		
Maximum		180°		
Accuracy		±5.0°		
RELATIVE BEARING MEASUREMENT				
Minimum		0°		
Maximum		360°		
Accuracy		±5.0° (Dev≥5.0°)		
VERTICAL RESOLUTION				
Typical Logging Speed		30 ft/min (9.14 m/min)		
Vertical Resolution @100 samples/ ft		0.12 in. (3.05mm)		
BUILDIN CENTRALIZER				
Buildin Centralizer		NONE		

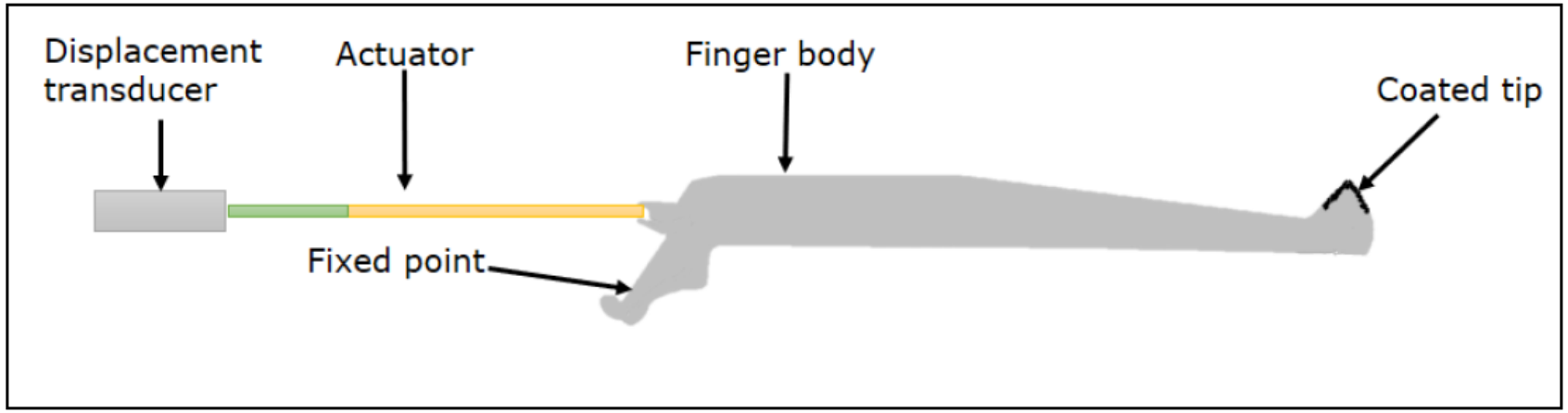
MFC Applications



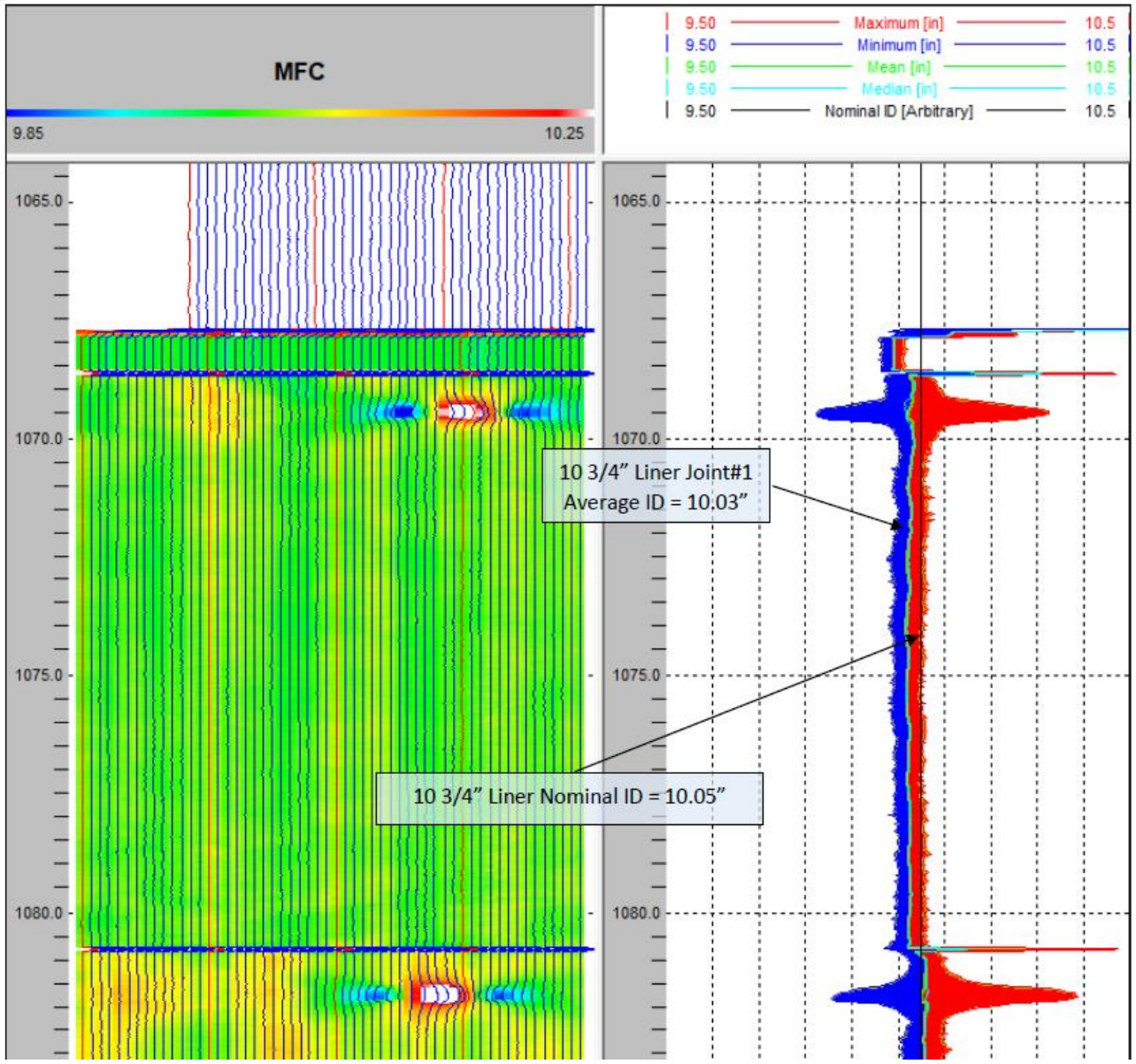
Finger Assembly







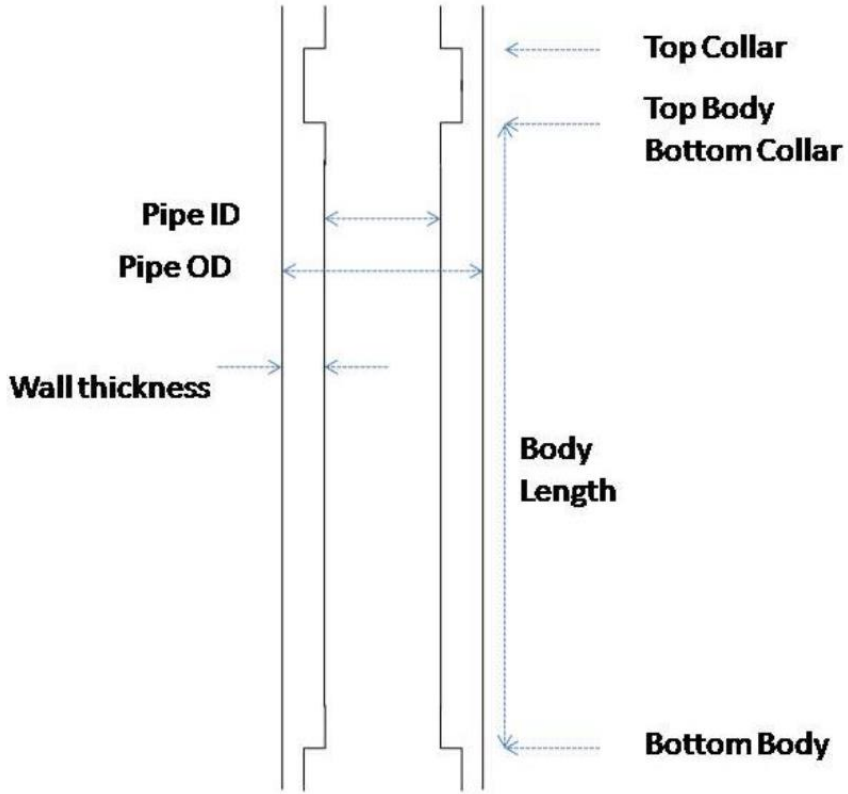
Definitions in Caliper Log



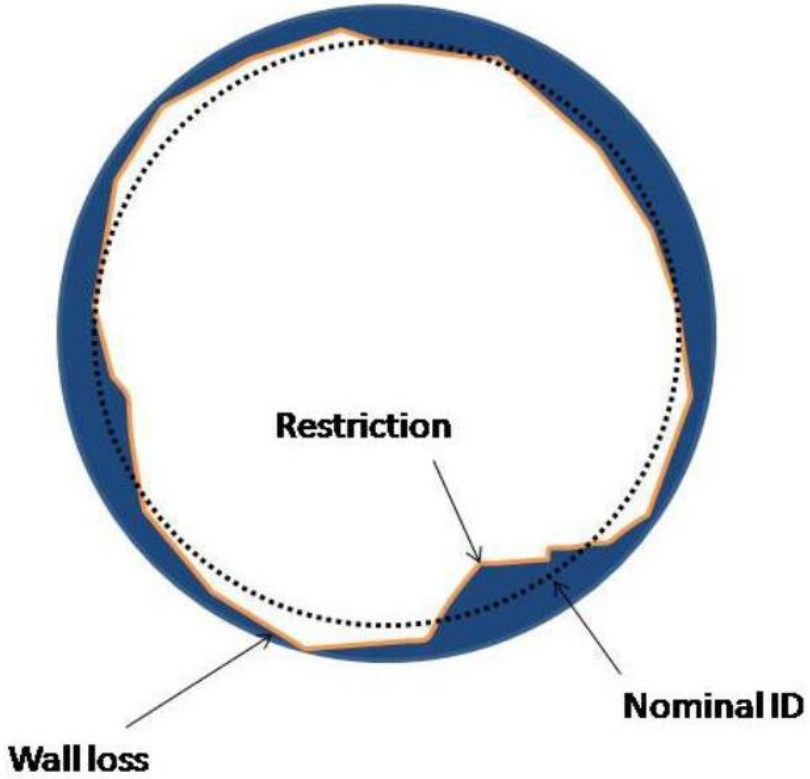
Minimum, minimum of 56 arms

Maximum, maximum of 56 arms

Average, average of 56 arms



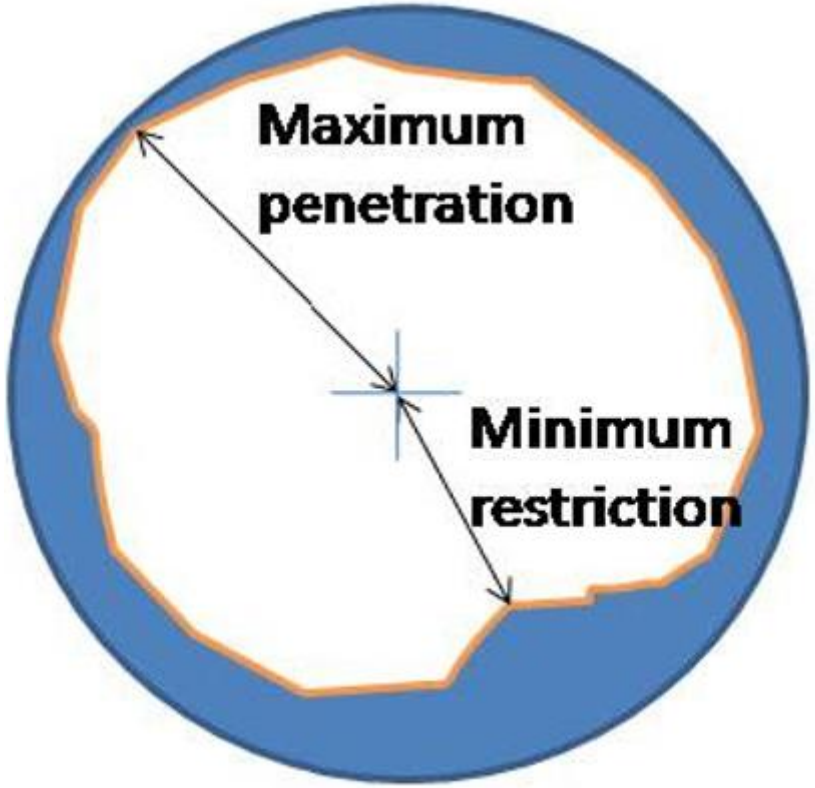
Definition of Metal Loss and Scale



Metal Loss=
100* ((Max. Radius)-(Nom, Radius))/
Nominal Thickness

Scale=
100* ((Min. Radius)-(Nom, Radius))/
Nominal Thickness

Definition



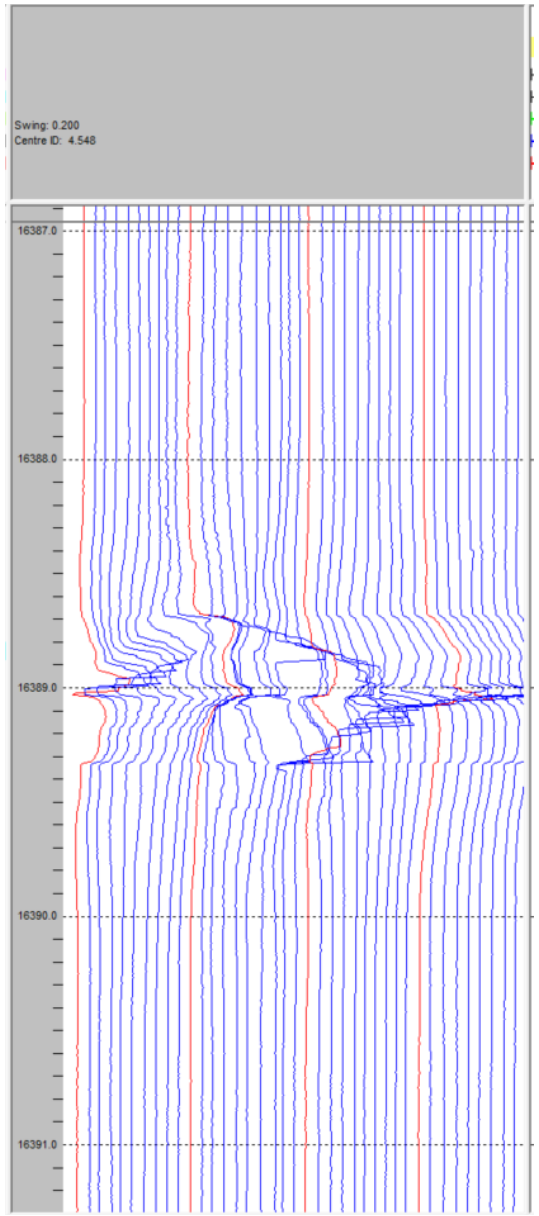
Maximum penetration % [Max.Pen (%)]
Maximum penetration of the wall in the pipe section, expressed as a percentage relative to the difference between Nominal thickness at the maximum penetration point.

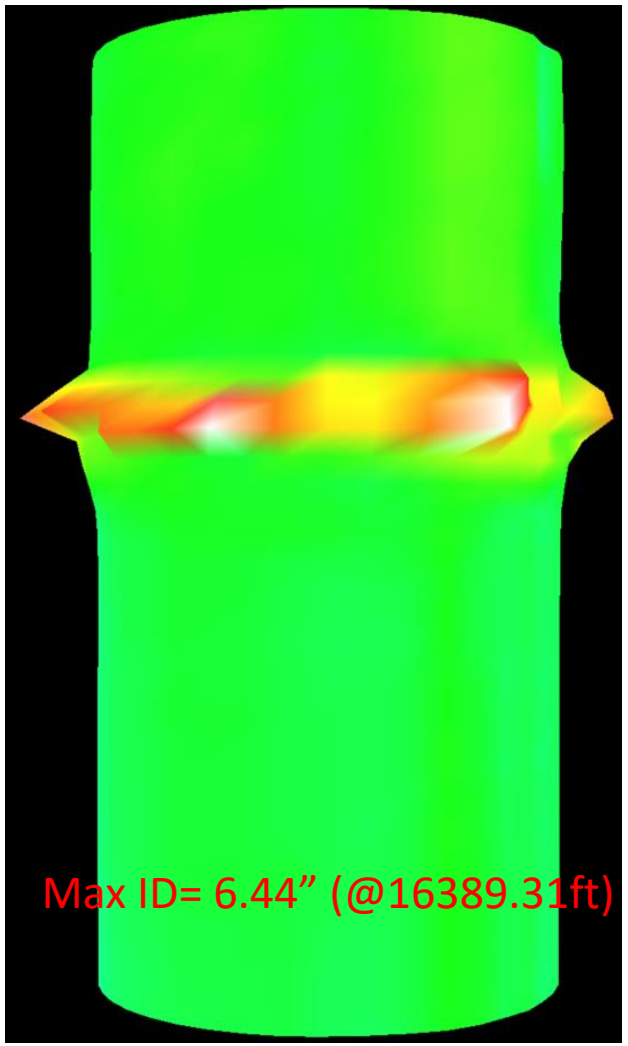
Minimum radial restriction % [Min Res (%)]
Smallest arm reading in the pipe section, expressed as a percentage relative to the Nominal ID and Outer diameters. This is reported as percentage since minimum radius is lower than the Nominal Inside Radius of the pipe.

- **100 % Metal Loss means penetration or hole in the pipe**

- **100 % Scale does not mean that there is complete blockage in the pipe. It means that restriction in the pipe as the same quantity of metal thickness**

Penetration Example

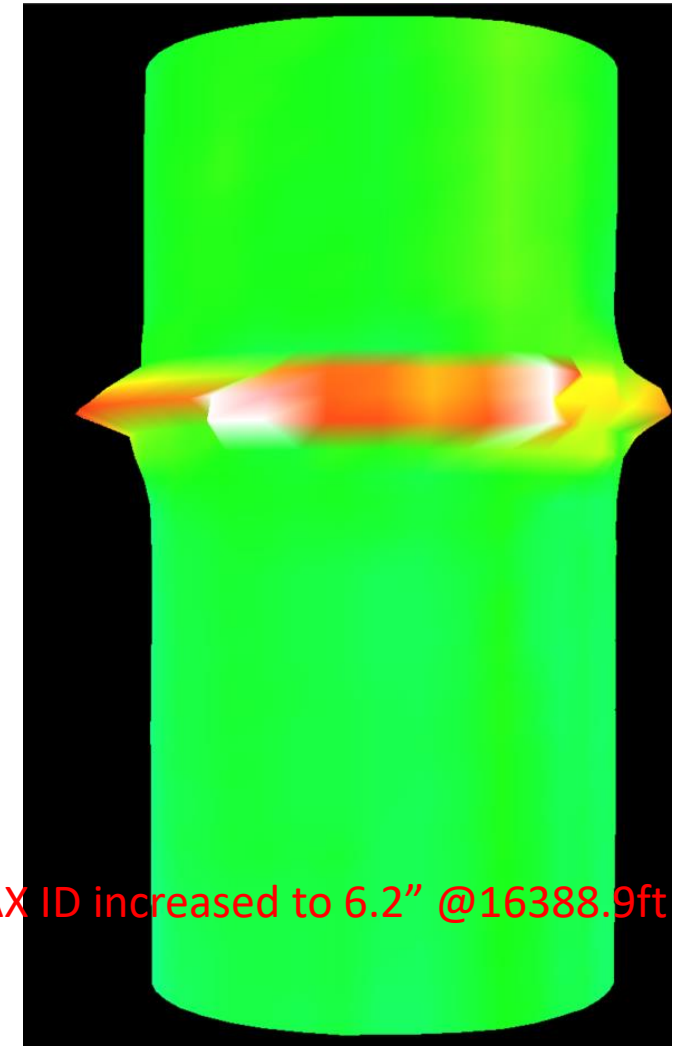




16385ft

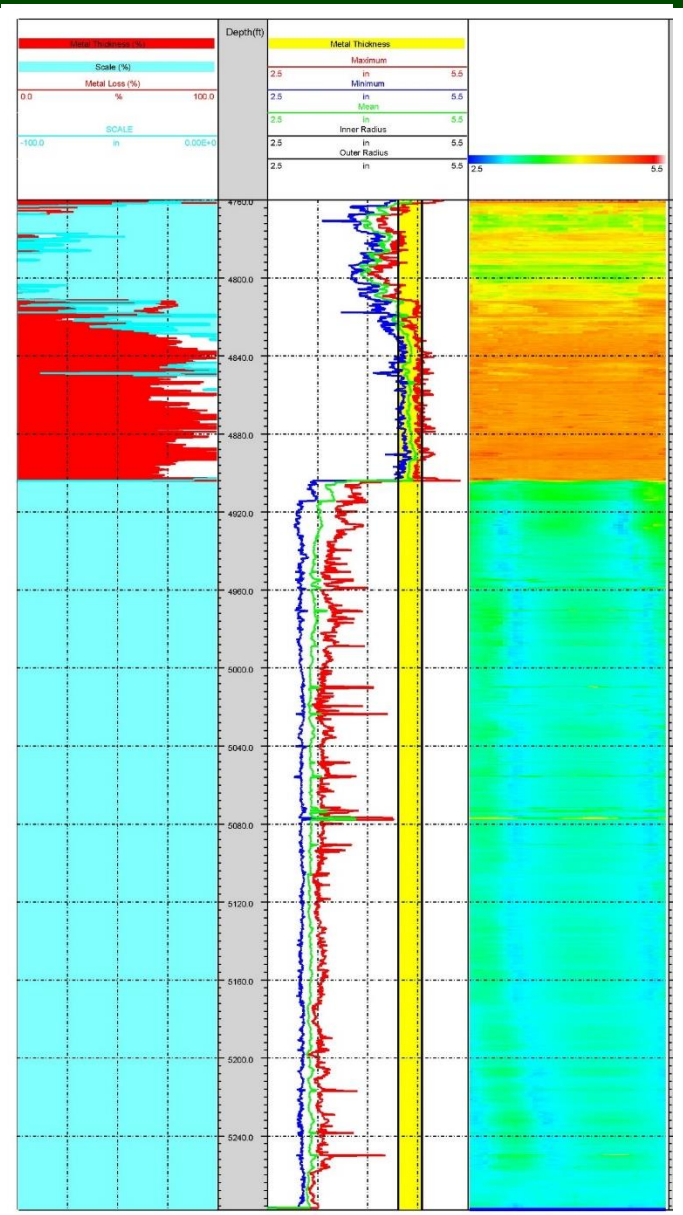
16389ft

16395ft

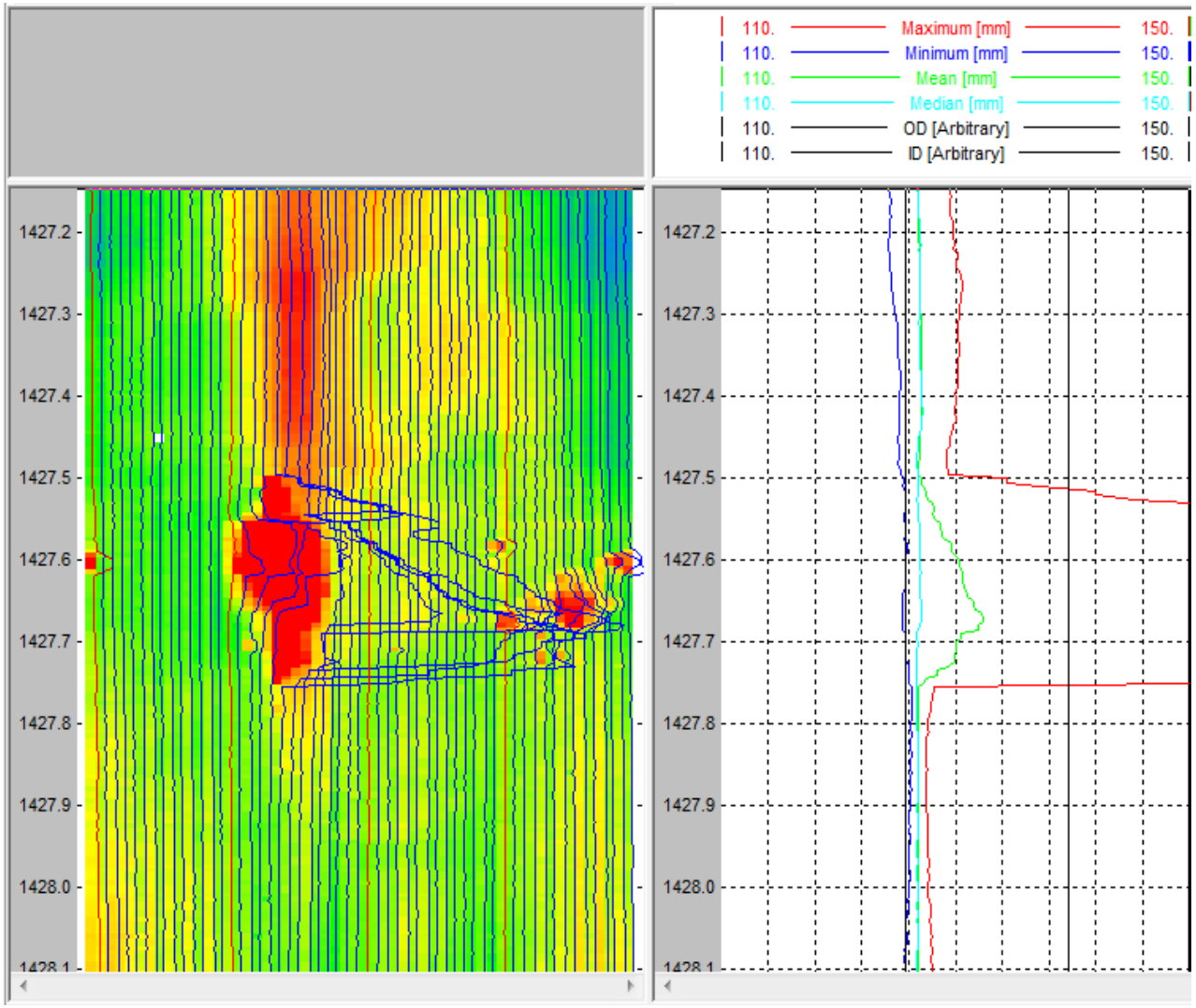


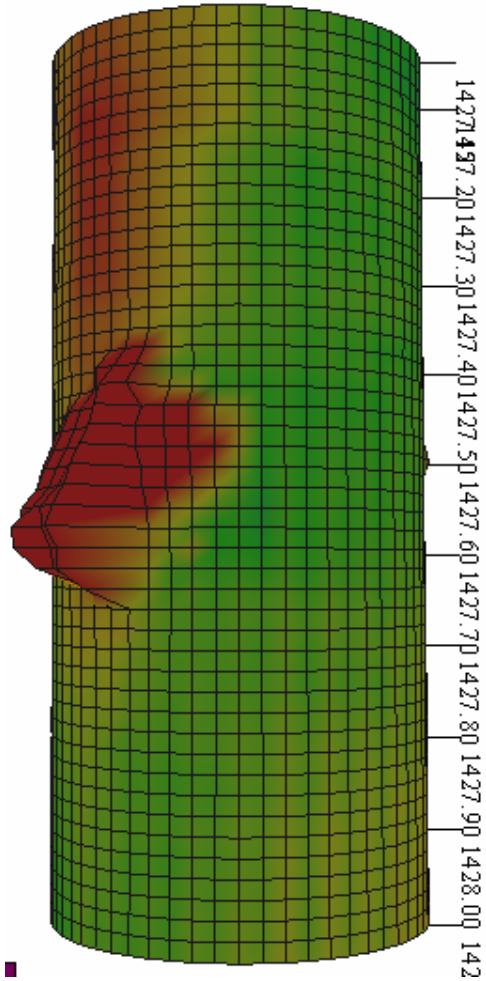
Scale Example

- 100% scale exists at the bottom section till to end of the logging interval
- Significant reduction in ID
- Most likely some perforations are plugged
- Productivity significantly reduced



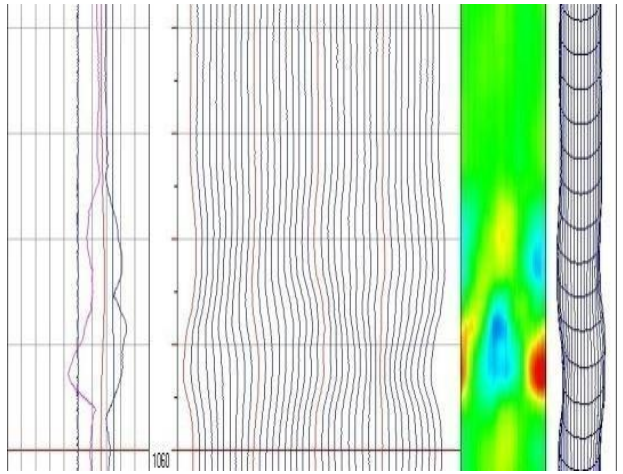
Casing Break/Damage



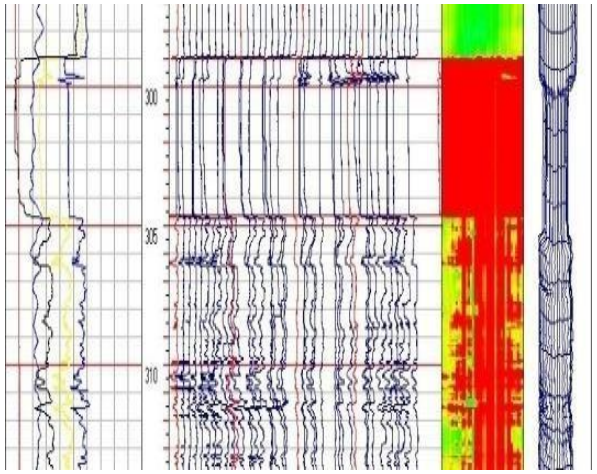


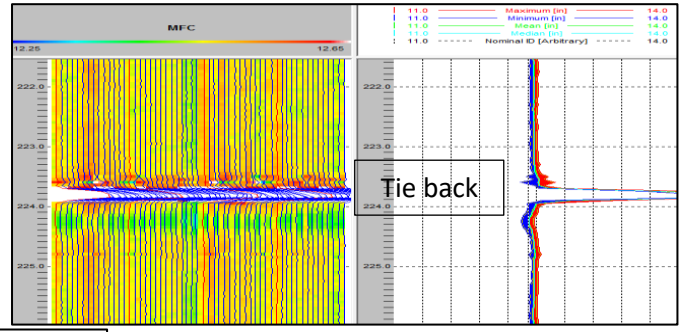
Multi Finger Caliper

Bending

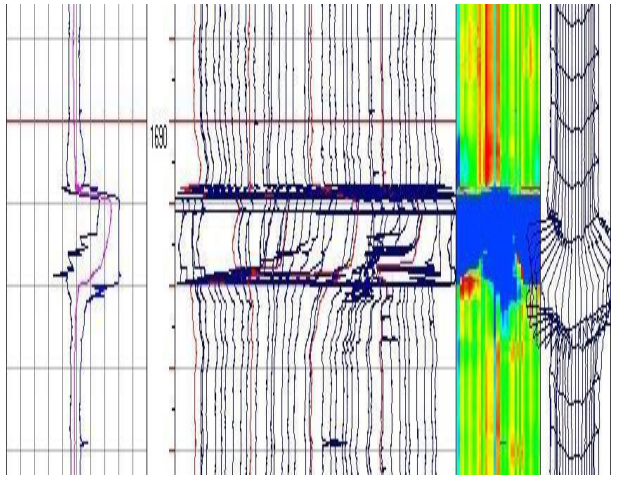


Shrinking

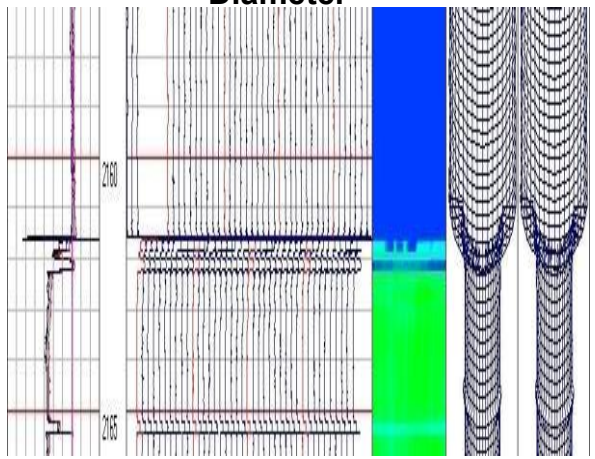


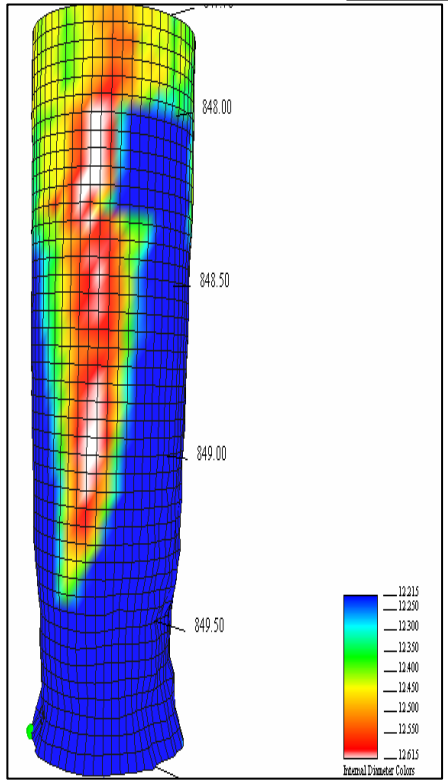


Breakage

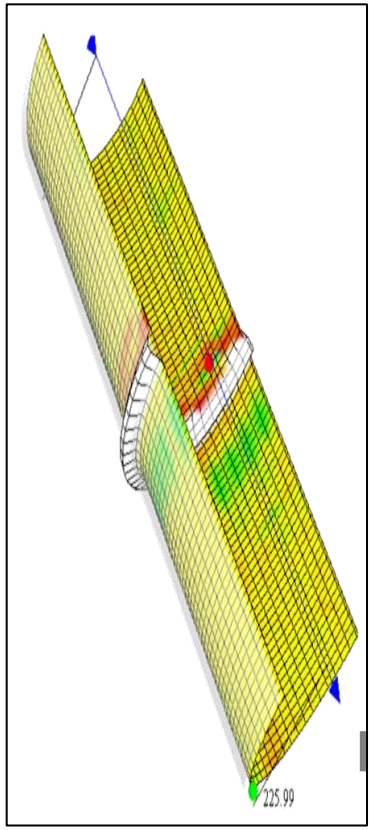


Change of Diameter

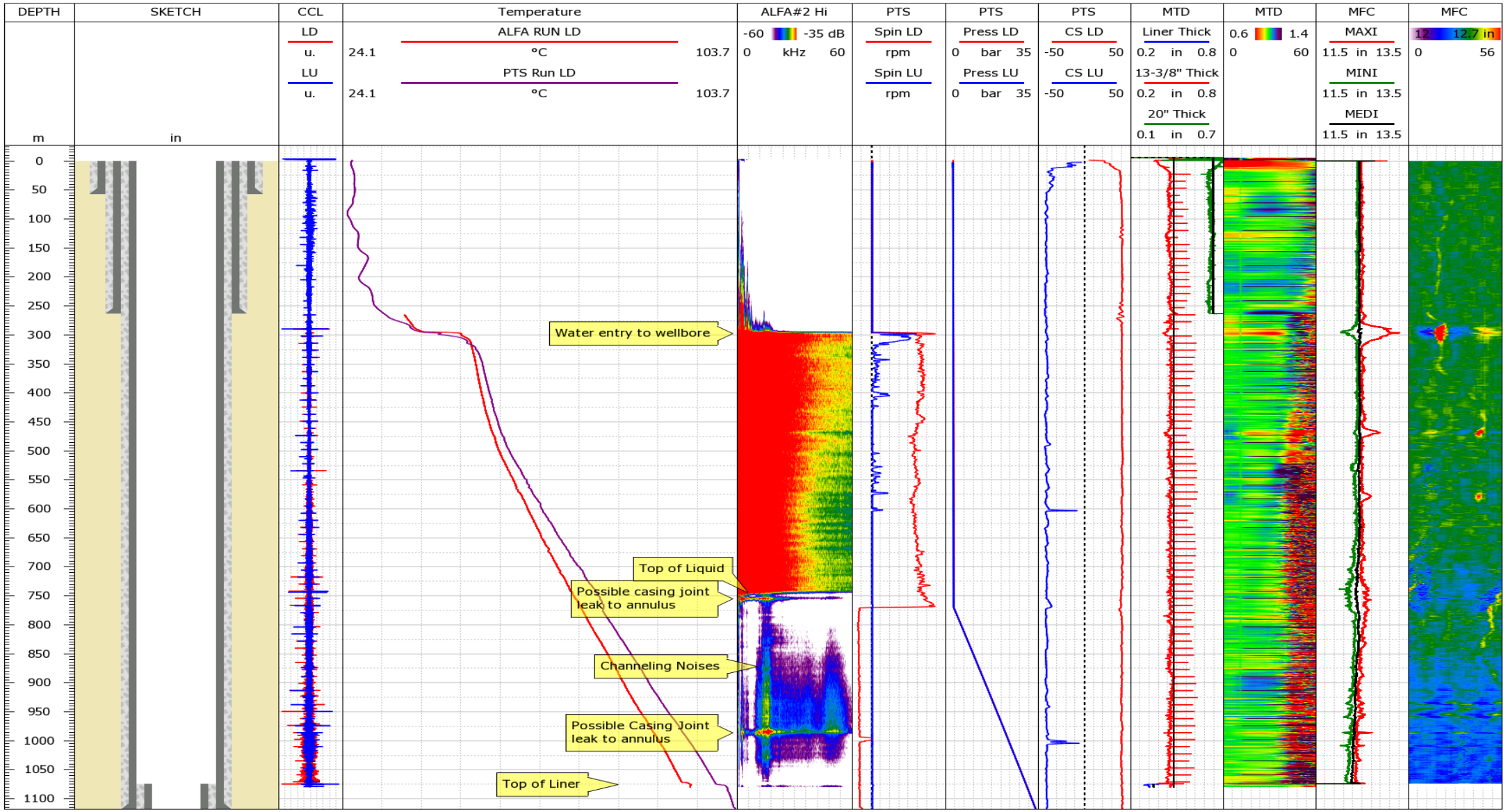




Casing Deformation



MFC, MTD & ALFA ANALYSIS



Magnetic Thickness Detector (MTD) Tool





Introduction – Magnetic Thickness Detector (MTD) Tool

- Evaluate the condition of **multiple completion pipes** in the same run without having to retrieve tubing
- Get **quantitative** assessment of the defects identified using advanced interpretation and forward modelling techniques
- Monitor the progress of corrosion throughout the lifetime of the well through time lapse monitoring
- Locate Well Completion elements (ex:CSG shoe) for multiple barriers



Specification of MTD tool

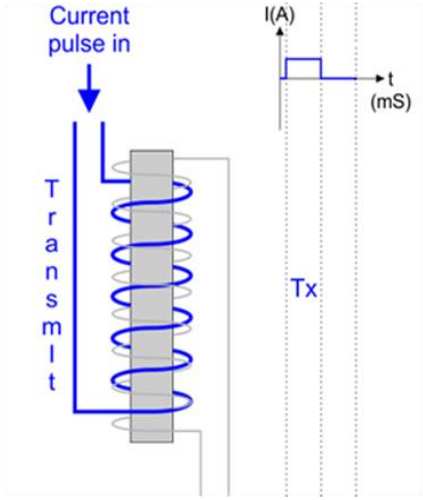
Maximum Pressure	15,000PSI (103MPa)
Temperature Range	−4°F ~ 350°F (-20°C ~ 175°C)
Diameter	1-11/16" (43 mm)
Length	44.3 in. (1.125 m)
Weight	12 lbs (5 kg)
Recommended Logging Speed	30ft/min (single pipe), 8ft/min (double pipe) and 6ft/min (triple pipe)
Thru-wired or Bottom Only	Thru Wired
Measuring Range	2.362 in. ~ 18.625 in. (60 mm ~ 473.1 mm)
Metallurgy	17-4 SST, Titanium & Al-Bronze
Total Pipe Wall Thickness	1.75 in. (44.4 mm)
Combinability	13pin PegasusStar
WALL THICKNESS MEASUREMENT	
First Pipe Measurement	
Maximum Pipe Wall Thickness	0.9 in. (22.86 mm)
Thickness Accuracy	0.0075 in. (0.190 mm)
First Pipe (2-7/8") minimum aperture **	0.5 in. (12.7 mm)
Second Pipe Measurement	
Maximum Pipe Wall Thickness	1.2 in. (30.48 mm)
Thickness Accuracy	0.01 in. (0.254 mm)
Second Pipe (2-7/8" + 7") minimum aperture **	1.5 in. (38.1 mm)
Third Pipe Measurement	
Maximum Pipe Wall Thickness	1.5 in. @ 0.06 in. Accuracy (38.1 mm @ 1.52 mm Accuracy)
Thickness Accuracy	0.06 in. (1.52 mm)
Third Pipe (2-7/8" + 13-3/8") Minimum Aperture **	3 in. (76.05 mm)
RELATIVE BEARING MEASUREMENT	
Measuring Range	0° ~ 360°
Accuracy	±5° (Deviation ≥5°)
Sensitivity	±0.1° (Deviation ≥5°)
DEVIATION MEASUREMENT	
Measuring Range	0° ~ 180°
Accuracy	±5°

Applications

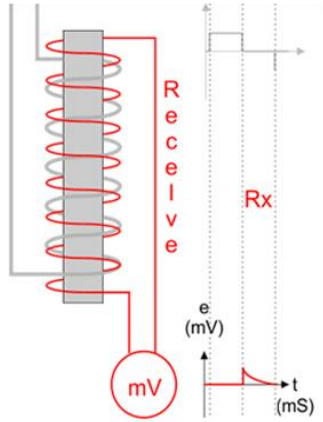
- Wall thickness calculations of pipes (single completion).
- Separate wall thickness calculations of pipes (Dual Completion).
- Determination of steel pipe local damaged sections such as cracks (vertical and transversal), ruptures, collars leak.
- Locating perforation interval and collars depths in both pipes.
- Locating and identifying all types of completion components (packers, various valves, centralizers, crossovers, etc.)

Measurement Principle

Current pulse sent to the transmit coil which emits electromagnetic pulse:



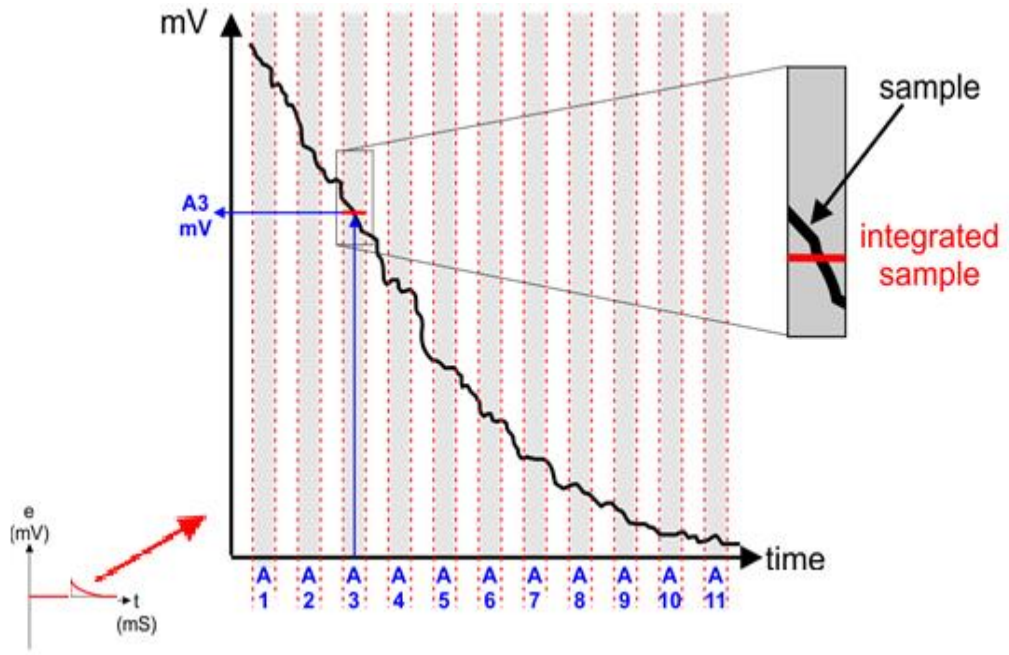
Once pulse transmission complete, sensors switch to receive mode to measure the decaying magnetic field:

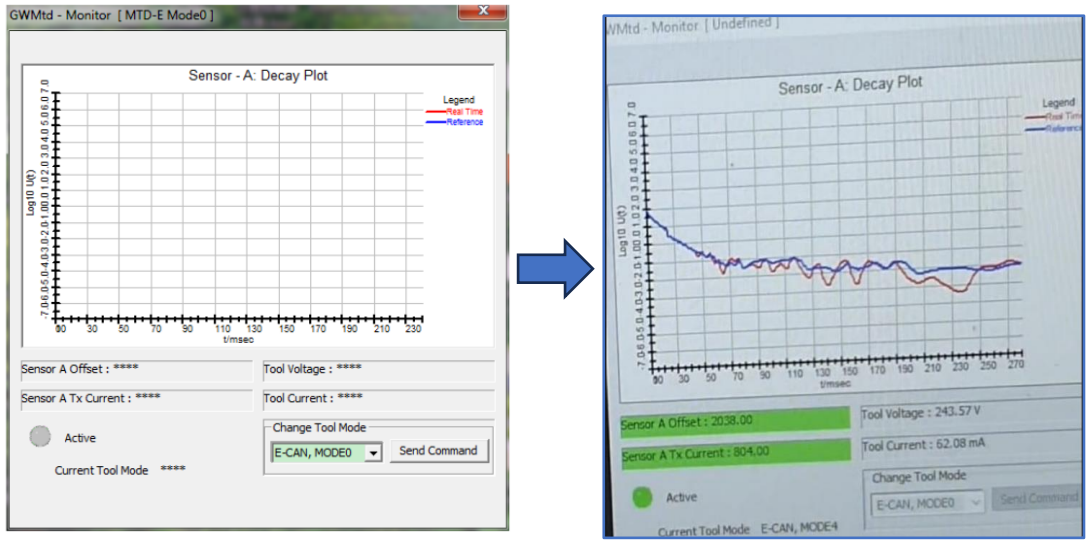


- Same depth Tx and Rx coils
- Short High Energy Pulsed Transmission
- Inherently Multi-Frequency
- Time Domain Received signal
- Measures multiple pipes

Measurement Principle

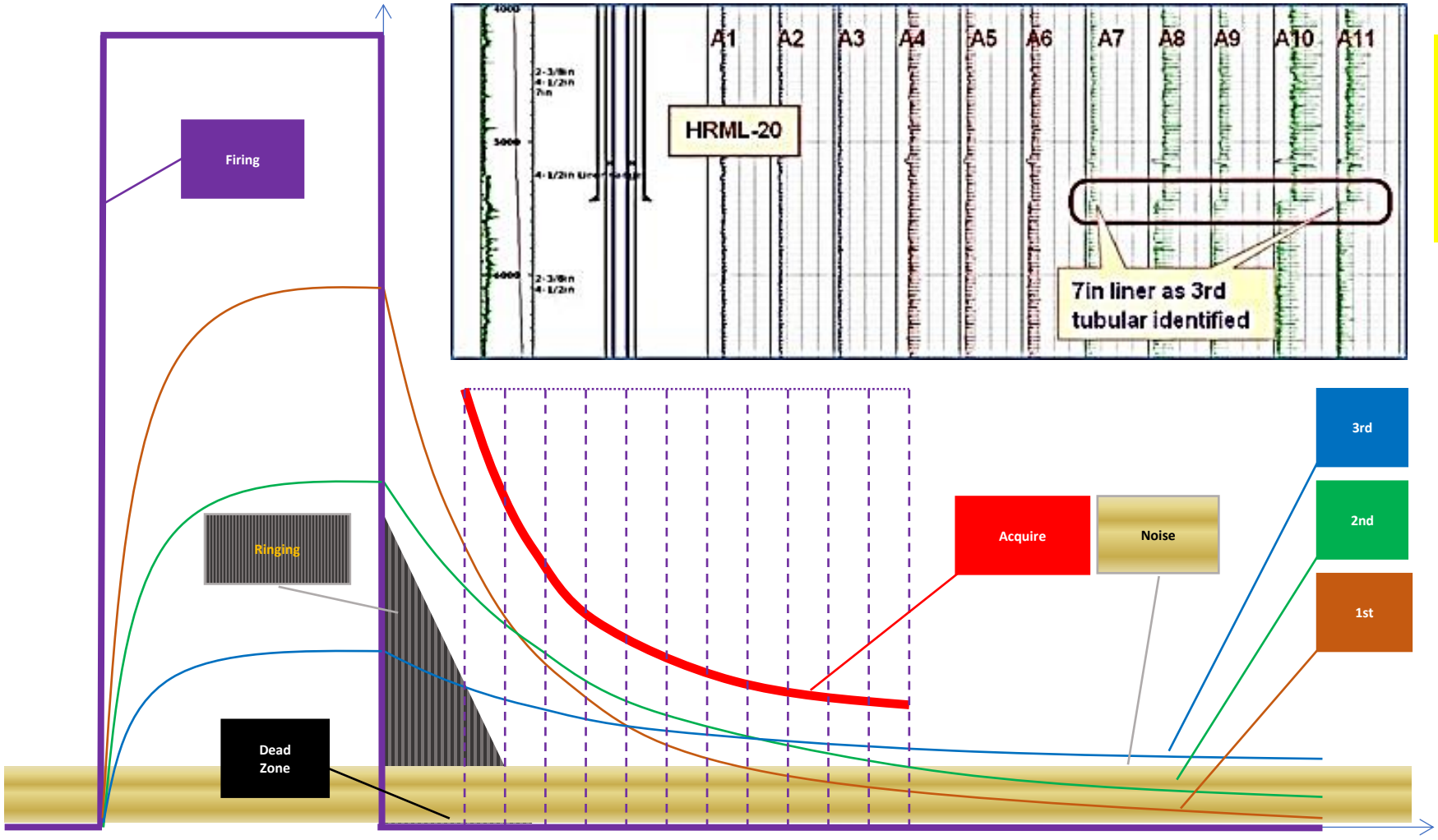
The rate of decay of the signal is related to metal thickness around the transmitter and receiver. **The greater the metal thickness the slower the decay.**





Logging system process

Early time points relate to thickness of the inner tubular, and later time is related to the thickness of outer tubulars.



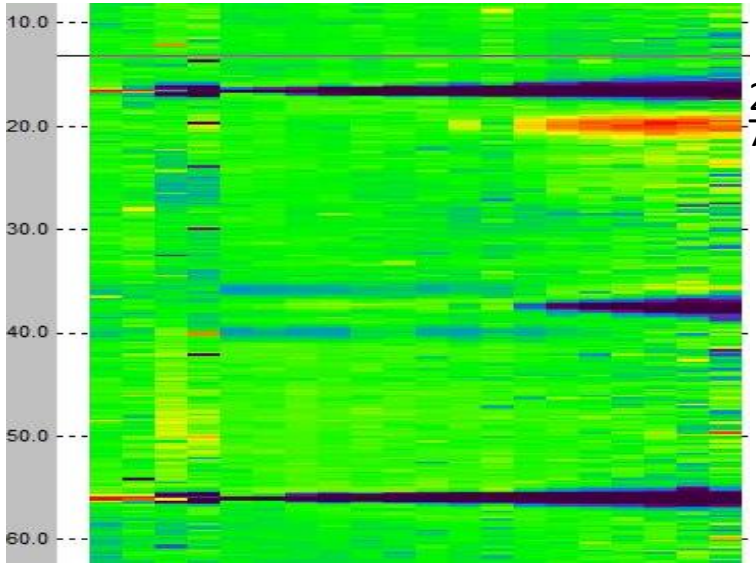
Early time points relate to thickness of the inner tubular, and later time is related to the thickness of outer tubulars.

The rate of decay of the signal is related to metal thickness around the transmitter and receiver. **The greater the metal thickness the slower the decay.**

Data Visualization (VDL)

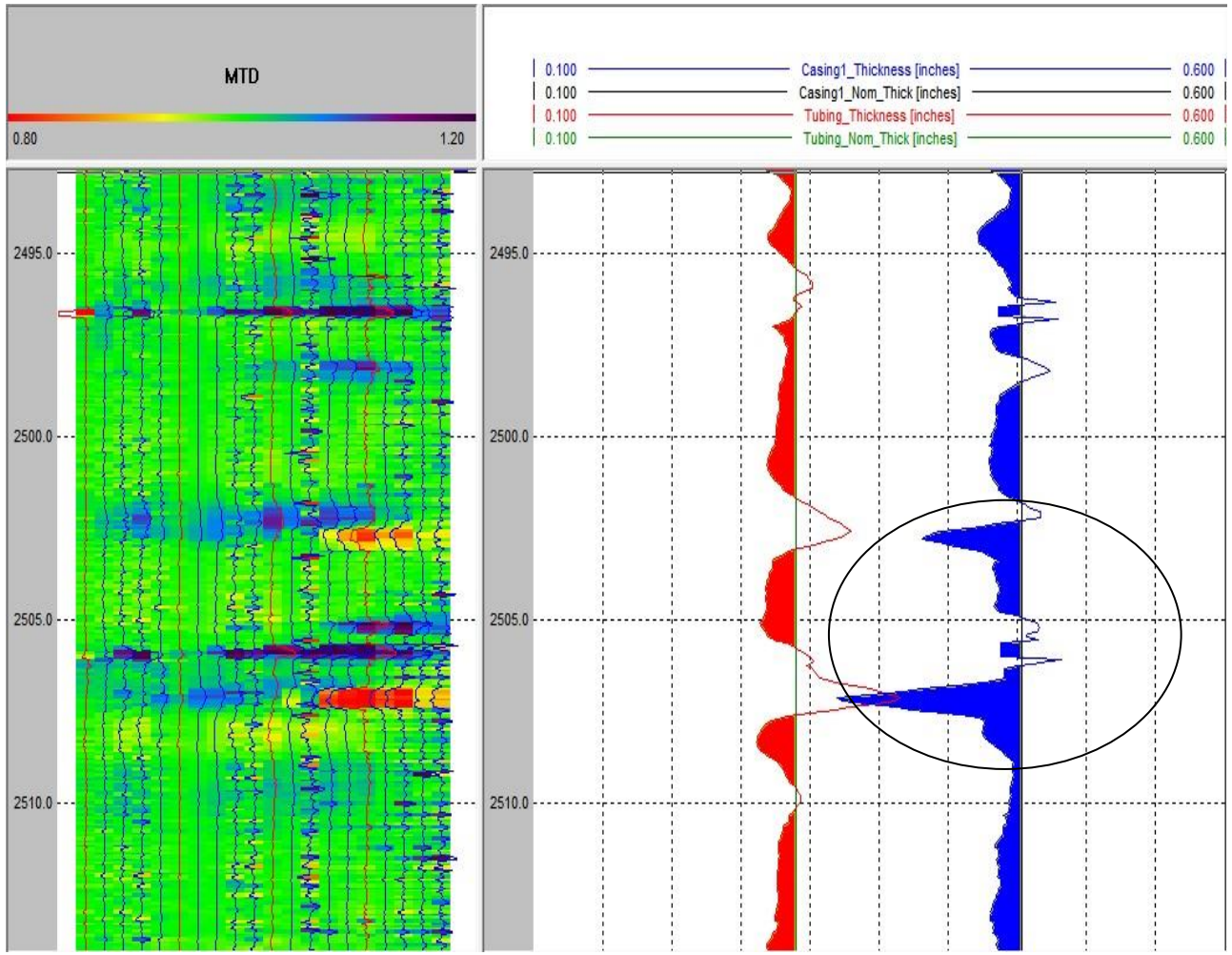
The PEC channel data is normalized such that:

- Green color represents pipe with nominal thickness. i.e. values are one.
- Blue color represents increased thickness
- Red color represents thickness reduction



1. Log display showing two 2-7/8inch tubing collars (x17ft and x57ft),
2. 7inch casing collar (x37ft)
3. Pipe damage in the 7inch casing (x20ft)

MTD Case Study



Producer Well

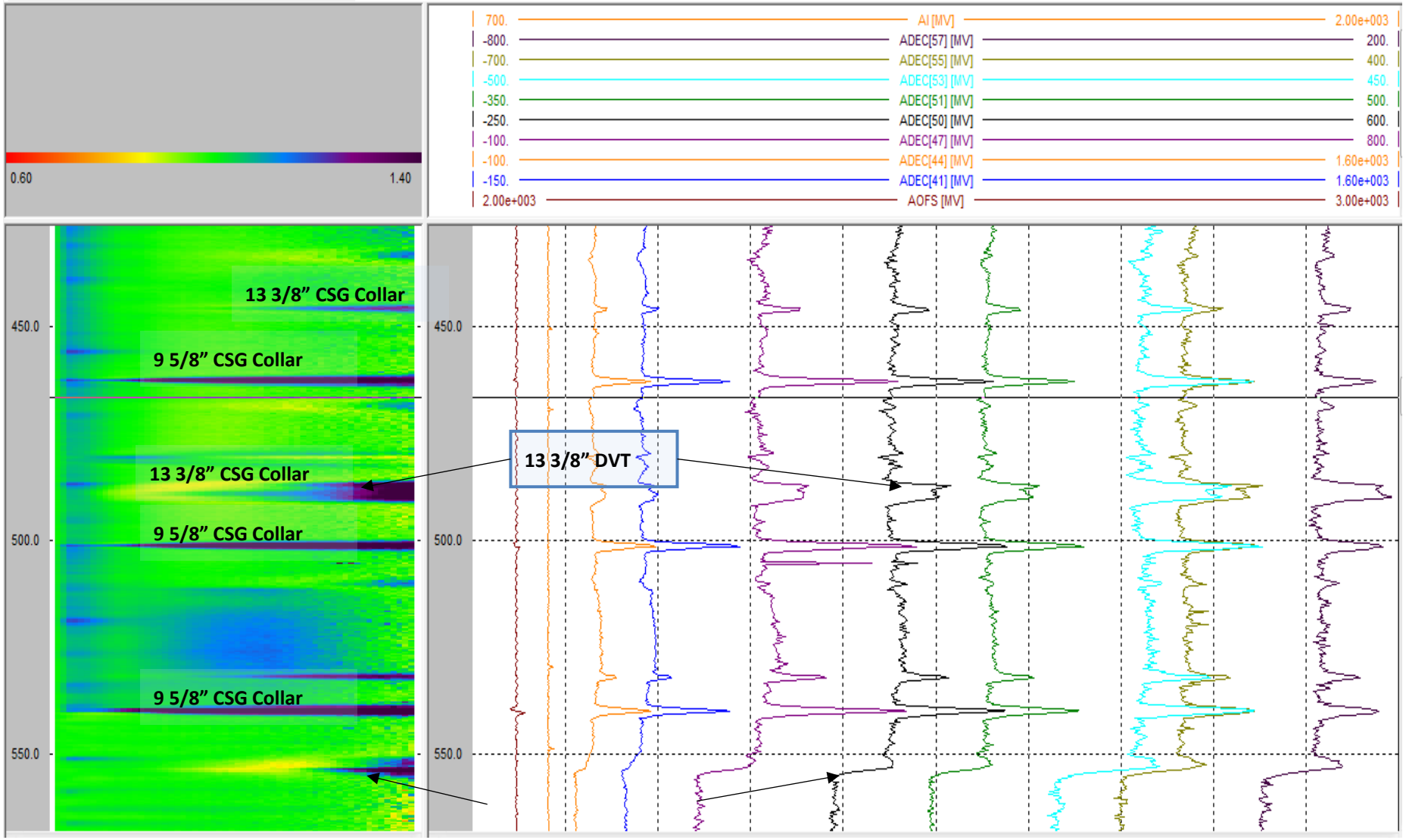
Well integrity problem: **Pressure observed in A annulus**.

MTD ran
in 4-1/2” Tubing

Objective was to evaluate the condition of 7”

Severe casing damage/Well loss is evident in 7” Casing in the interval 2502.3 – 2507.7 m

MTD-E Response in combination 9-5/8”+13-3/8”



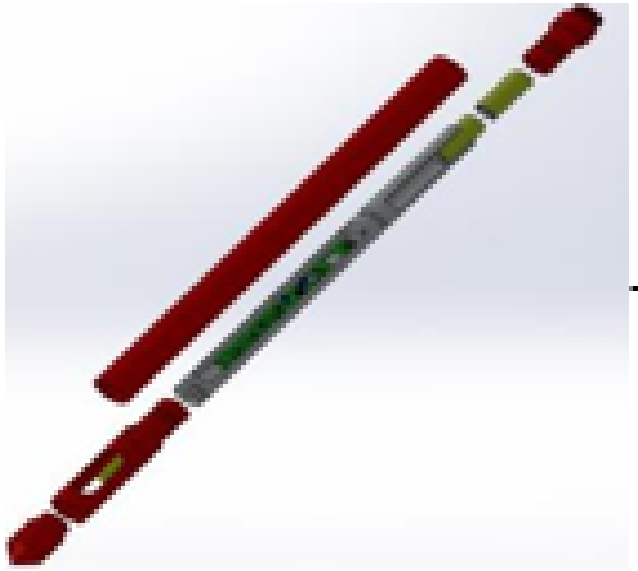
MTD Final Report

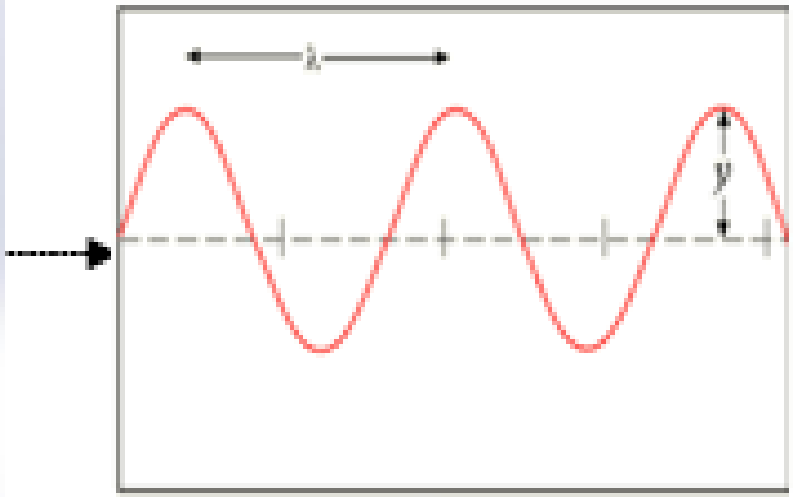
Grade	Max _ML	Corrosion Level	Diagnosis & Recommendations
A	0-5%	Negligible	Within tool accuracy calculation. Generally not a matter of significant concern.
B	5-10%	Light	This is on the threshold of accurate detection and may indicate onset of localized corrosion activity.
C	10-20%	Moderate	This grade may be associated with higher level of corrosion activity and/or with possibility of critical damage across a localized area. Further evaluation and if required, remedial action may be considered if means and access to this tubular is available.
D	>=20%	Intensive	Generally, this is associated with integrity issues in that part of the tubular. Peak corrosion activity in one depth has a higher probability of penetration damage of the tubular. Remedial action after further evaluation is recommended.
E	<0%	Special Joint	Associated with joints that have a higher weight or electro-magnetic permeability.

#	Top Body (m)	Bottom Body (m)	Body Length (m)	Nom Thk (in)	Min Thk (in)	Max Loss Depth (m)	Max Loss (%)	Grade	Damage Profile (% wall loss)
1	-5.7	-0.5	5.2	0.480	Well Header				
2	-0.2	10.8	10.9	0.480	Affected by tool ex-centralization				
3	11.0	22.2	11.3	0.480					
4	22.4	34.4	12.0	0.480	0.450	32.9	6.3	B	
5	34.9	46.5	11.6	0.480	0.440	46.4	8.3	B	
6	47.0	58.7	11.7	0.480	0.437	51.5	8.9	B	
7	59.3	71.0	11.7	0.480	0.439	60.0	8.6	B	
8	71.6	82.8	11.2	0.480	0.444	79.0	7.6	B	
9	83.4	94.9	11.6	0.480	0.443	94.8	7.7	B	
10	95.5	107.1	11.6	0.480	0.444	95.8	7.4	B	
11	107.7	119.2	11.5	0.480	0.445	109.4	7.3	B	
12	119.8	131.5	11.7	0.480	0.426	129.3	11.2	C	
13	132.1	143.7	11.7	0.480	0.431	137.8	10.2	C	
14	144.3	155.5	11.2	0.480	0.429	145.9	10.6	C	
15	156.1	167.6	11.5	0.480	0.449	157.5	6.4	B	
16	168.2	179.8	11.6	0.480	0.447	179.7	6.9	B	
17	180.4	191.8	11.4	0.480	0.454	191.1	5.4	B	
18	192.4	204.1	11.6	0.480	0.446	195.9	7.2	B	
19	204.7	216.2	11.6	0.480	0.458	205.0	4.6	A	
20	216.8	228.4	11.6	0.480	0.456	227.0	5.0	A	
21	229.0	240.5	11.5	0.480	0.437	236.4	8.9	B	
22	241.1	252.6	11.5	0.480	0.440	243.5	8.3	B	
23	253.2	264.8	11.6	0.480	0.453	254.8	5.6	B	
24	265.4	276.9	11.4	0.480	0.435	276.8	9.3	B	
25	277.5	289.2	11.8	0.480	0.416	279.2	13.2	C	
26	289.8	301.5	11.7	0.480	0.410	296.8	14.5	C	
27	302.1	313.6	11.5	0.480	0.437	313.4	8.9	B	

Acoustic Leak Flow Analyzer

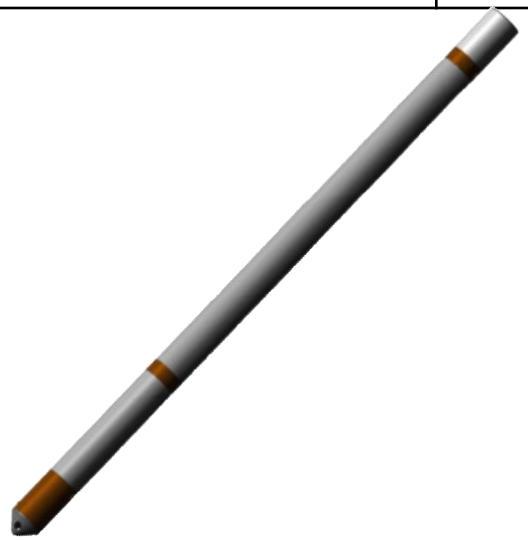
Acoustic Leak Flow Analyzer (ALFA)





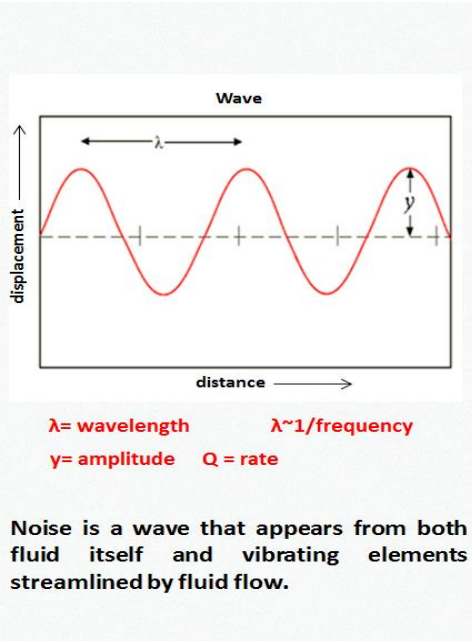
Acoustic Leak Flow Analyzer

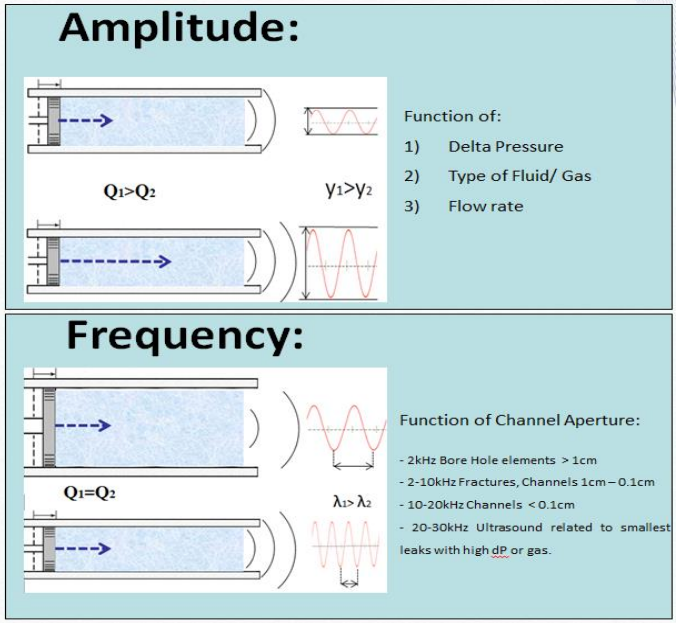
Parameter	Value
Operating frequency range, kHz	8 – 60
Frequency resolution, kHz	0.06
Number of frequency channels	512
Dynamic Range, dB	80
Maximum ambient operating pressure, psi	14,500
H2S resistance, %	10
Operating temperature range, F	307 degF / 150 degC



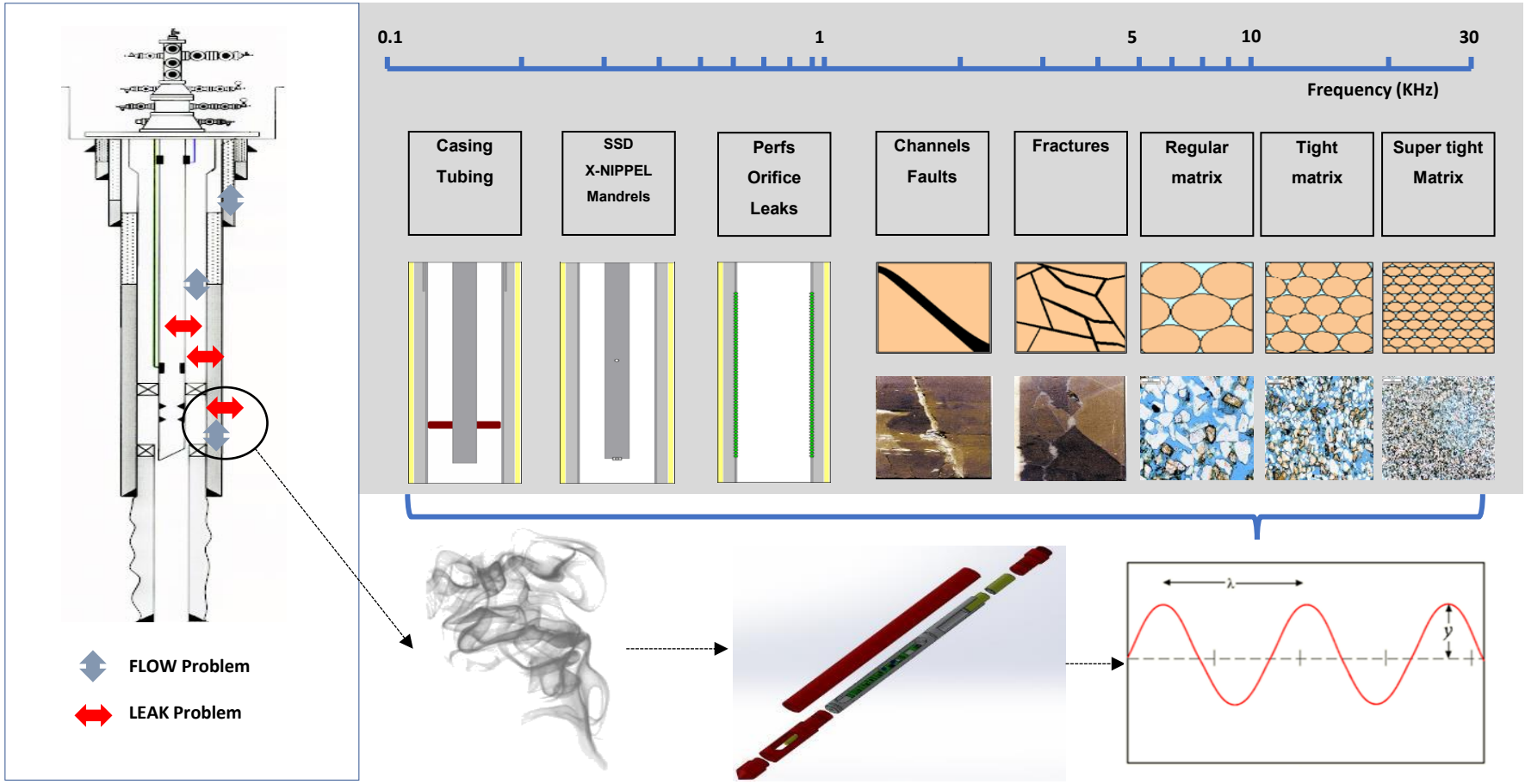
- ✓ **Deep scanning radius**
- ✓ **Scanning through the multiple barriers**

  (Ex: behind both tubing and casing)
- ✓ **Evaluation of any cross-flows behind casing and leakages**
- ✓ **Direct measurement allow to capture any flow**

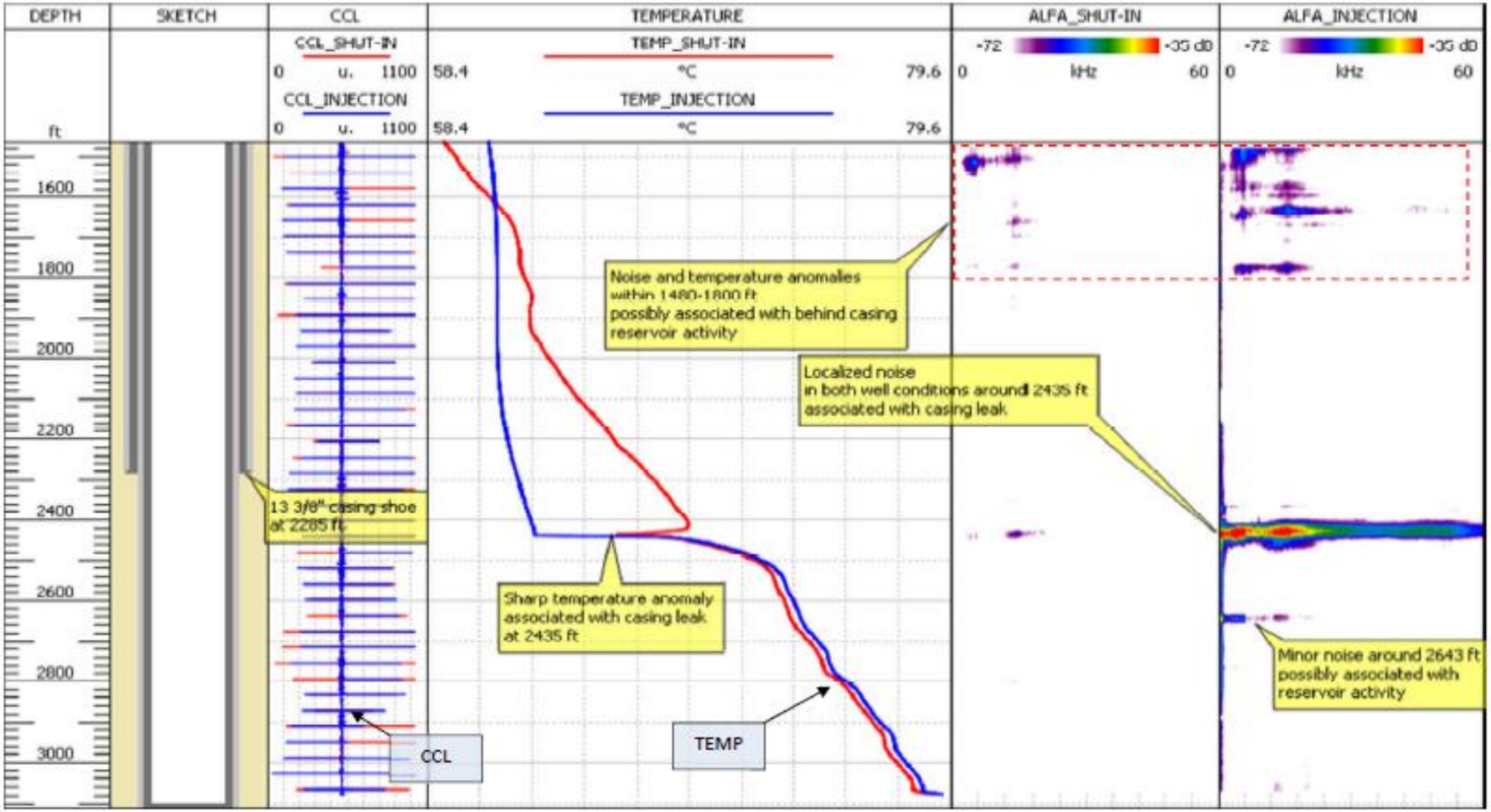


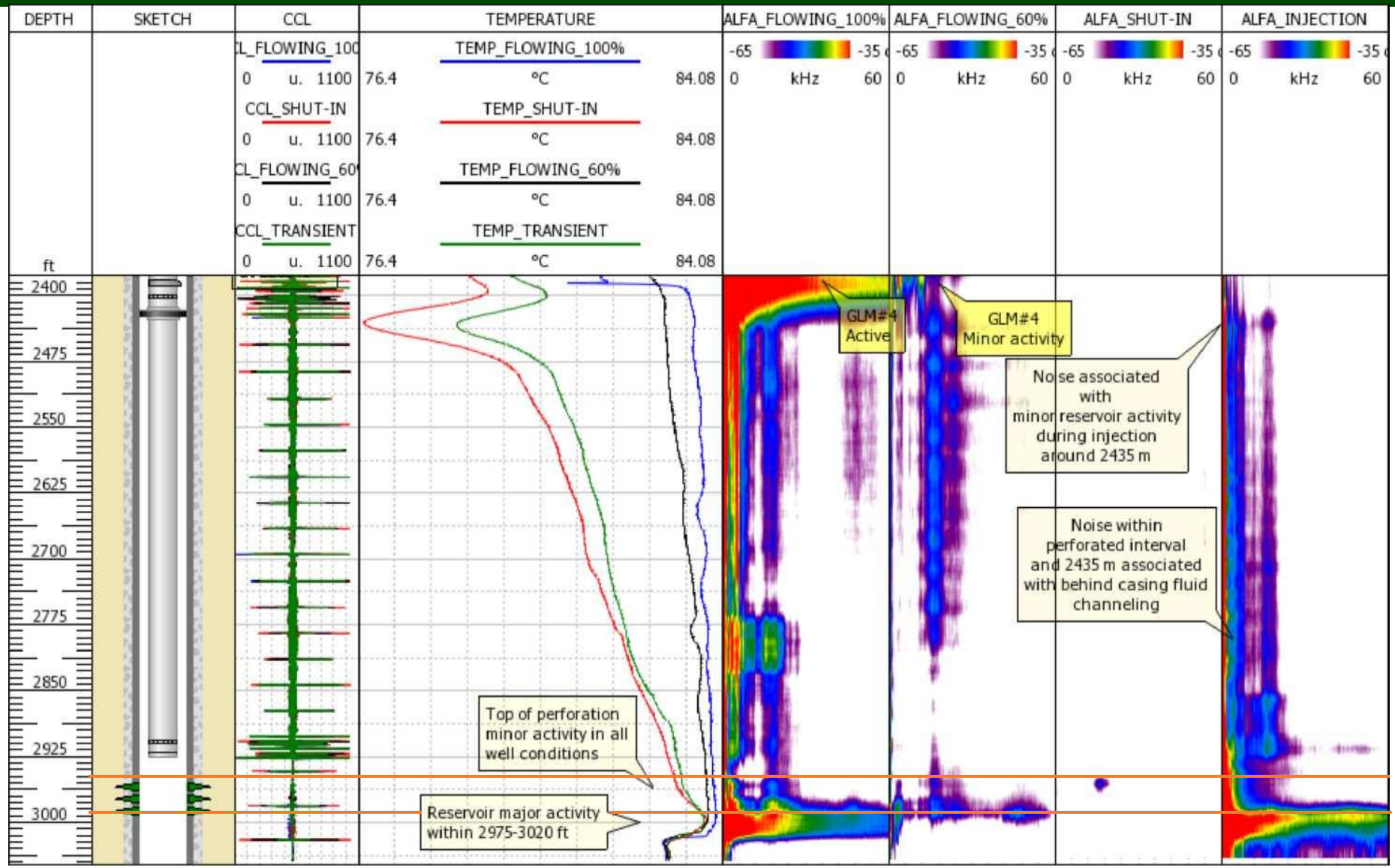


Matrix Indication

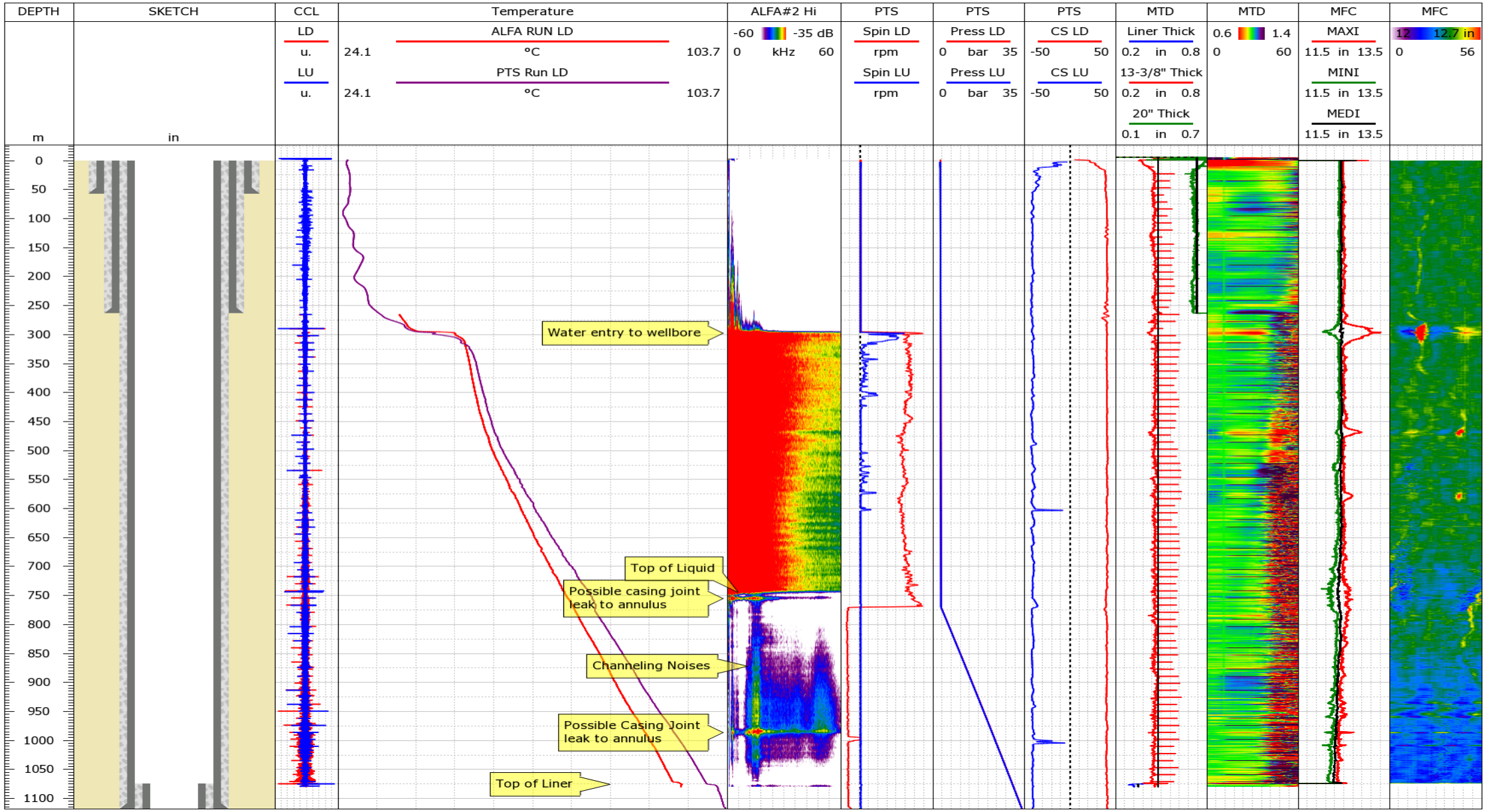


Casing Leak


DEPTH
SKETCH
CCL
TEMPERATURE
ALFA_SHUT-IN
ALFA_INJECTION
CCL_SHUT-IN
0 u. 1100
CCL_INJECTION
0 u. 1100
TEMP_SHUT-IN
58.4 °C 79.6
TEMP_INJECTION
58.4 °C 79.6
-72 -35 dB
0 kHz 60
ft
1600
1800
2000
2200
2400
2600
2800
3000
Noise and temperature anomalies within 1480-1800 ft possibly associated with behind casing reservoir activity
Localized noise in both well conditions around 2435 ft associated with casing leak
13 3/8" casing shoe at 2285 ft
Sharp temperature anomaly associated with casing leak at 2435 ft
Minor noise around 2643 ft possibly associated with reservoir activity
CCL
TEMP


DEPTH
SKETCH
CCL
TEMPERATURE
ALFA_FLOWING_100%
ALFA_FLOWING_60%
ALFA_SHUT-IN
ALFA_INJECTION
CCL_FLOWING_100
0 u. 1100
CCL_SHUT-IN
0 u. 1100
CCL_FLOWING_60
0 u. 1100
CCL_TRANSIENT
0 u. 1100
TEMP_FLOWING_100%
76.4 °C 84.08
TEMP_SHUT-IN
76.4 °C 84.08
TEMP_FLOWING_60%
76.4 °C 84.08
TEMP_TRANSIENT
76.4 °C 84.08
-65 -35
0 kHz 60
ft
2400
2475
2550
2625
2700
2775
2850
2925
3000
GLM#4 Active
GLM#4 Minor activity
Noise associated with minor reservoir activity during injection around 2435 m
Noise within perforated interval and 2435 m associated with behind casing fluid channeling
Top of perforation minor activity in all well conditions
Reservoir major activity within 2975-3020 ft

MFC, MTD & ALFA ANALYSIS



Downhole Camera/Video (DHV)

1. Detect safety issues :

- Casing or tubing leaks
- Mineral deposition
- Scale and corrosion
- Bacterial buildup

2. General inspection :

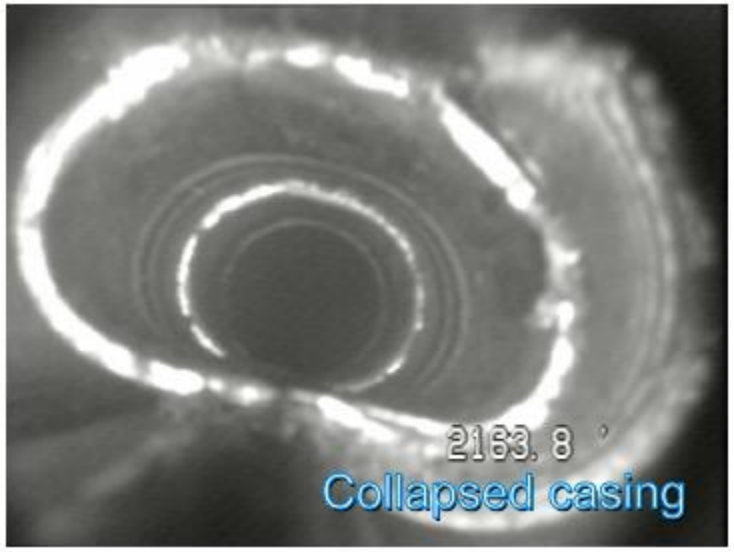
- Tubing hangers
- Plugged perforations
- Casing and tubing spli
- Gas lift valves



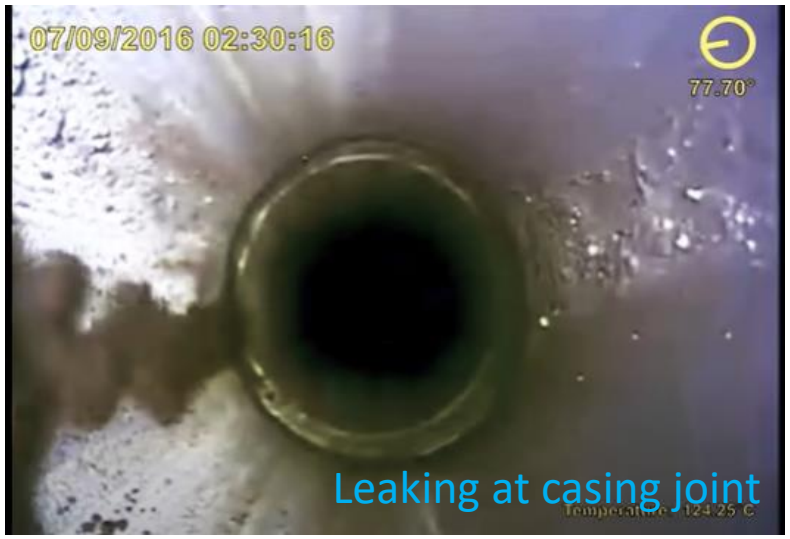
3. Determine the condition of downhole equipment :

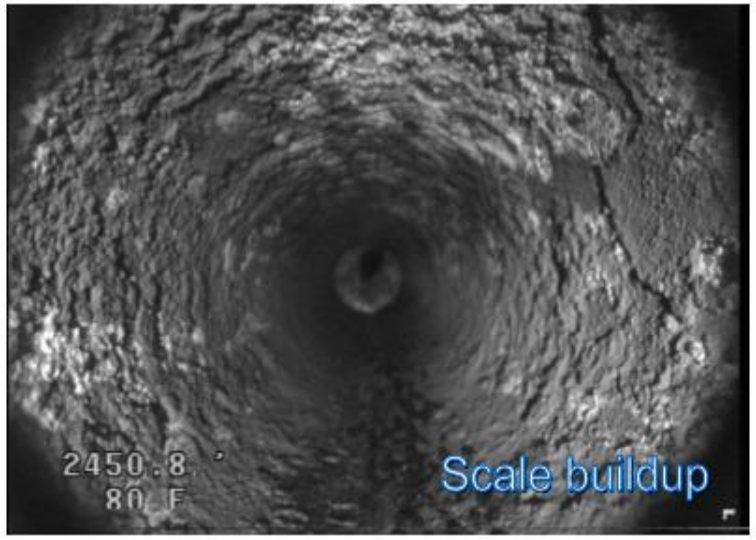
- Nipple profiles
- Chokes
- Sand screens and slotted liners (inspect for wear, plugging, corrosion, and scale buildup)

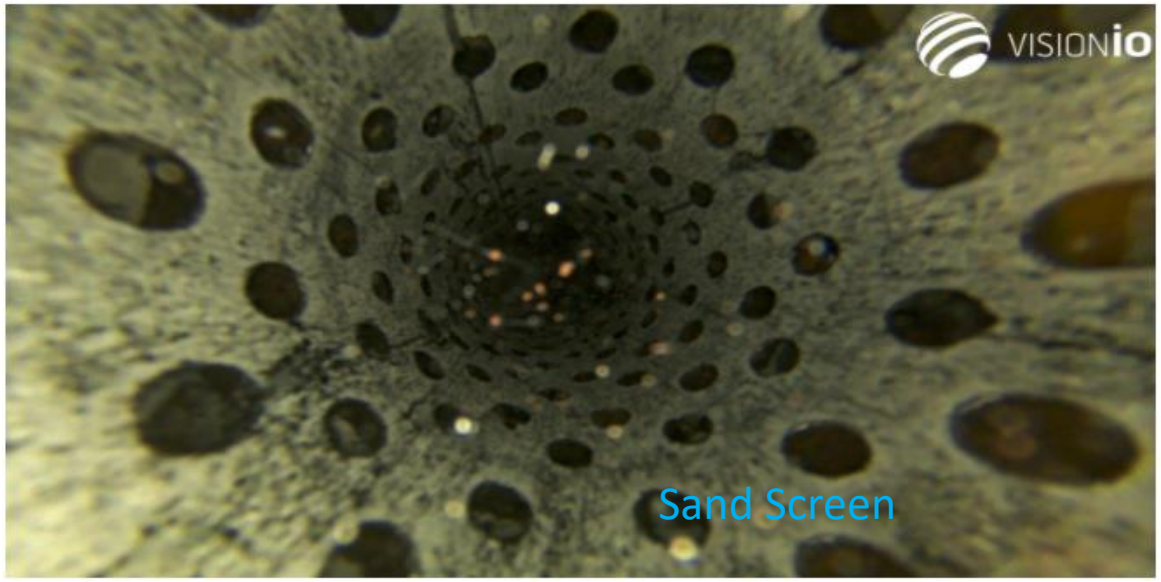
Downhole Vision


2183.8
Collapsed casing

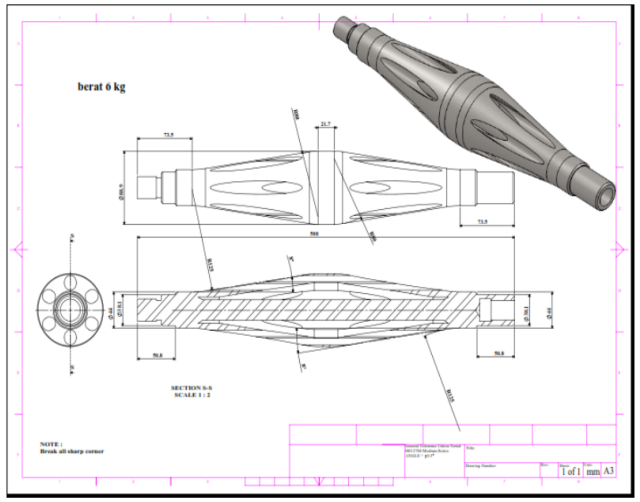

07/09/2016 23:01:40
89.42


07/09/2016 02:30:16
77.70
Leaking at casing joint


2450.8
80 F
Scale buildup

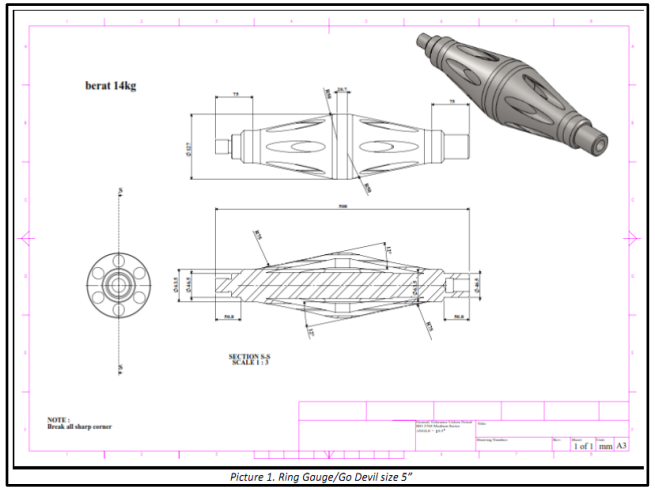

VISIONio
Sand Screen

Geothermal Ring Gauge/Go Devil

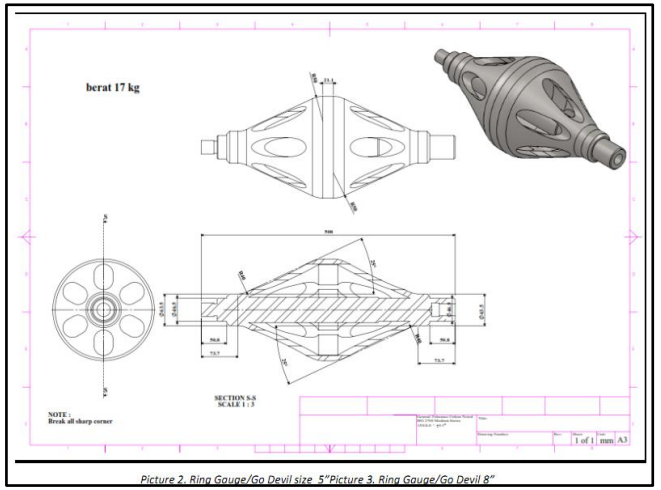


Picture 1. Ring Gauge/Go Devil size 3.5"

Ring Gauge using to check the condition of inside casing and can figure how big the size of the clearance of obstruction.



Picture 1. Ring Gauge/Go Devil size 5"



Picture 2. Ring Gauge/Go Devil size 5"Picture 3. Ring Gauge/Go Devil 8"

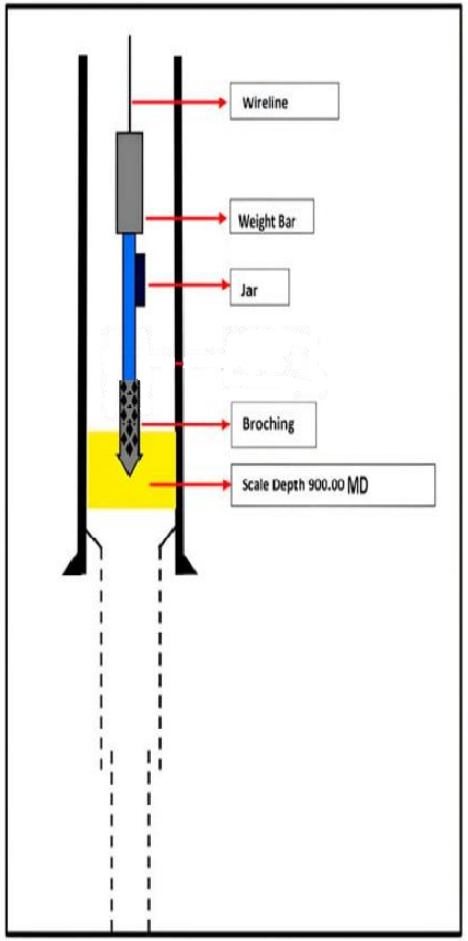


Broaching Tool



Broaching tools are heavy duty tools used to scale breaking combination with jar to get impact to broach out casing restrictions inside the wellbore.





Impression Block (IB)





Impression Block tools using for taking high-definition impressions of object, known or unknown, within the wellbore.
A single jar down is all that required to obtain a high-definition impression.

3.5" Impression Block



5" Impression Block



Geothermal Scale Catcher





Scale Catcher used to collect samples of scale from inside geothermal wells

Down Hole Sampler

DHS used to collect fluid samples inside geothermal wells.
Chamber and tool will be running in open condition until target depth. Closing the valve can be using timer or electric power from surface.







Thank You

NESR

Kadek Agus Dedyantara,
Service Leader Wireline Department